Odd prime-factor parity in shifted primes

A proof of Green's Problem 64

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Abstract

We prove that infinitely many primes p satisfy $\Omega(p-2) \equiv 1 \pmod 2$, where prime factors are counted with multiplicity. The proof constructs a nonnegative weight of prescribed parity and positive mass. Correlation estimates justify a weighted sieve, while a bound uniform in the width of a balanced factor range controls the remaining composites. Subtracting these composites leaves positive prime mass and supplies a prime above every bound. Detailed proofs of the weight, correlation, congruence and rectangle estimates are included; appendices identify their formal counterparts.

Mathematical sources. During the final stage of the Lean formalization, the agent found Jizhou Guo's manuscript [3] on GitHub and used its exposition as a guide. Guo did not participate in the project or provide personal assistance. The final development follows the specialization of its affine-prime-form argument to $N(u) = u + 2$, including the choice $r_0 = 20$, $\sigma = 1/44$, the arithmetic side conditions and the order of analytic parameter choices. The parity-weight, correlation and sieve strategy is based on the two OpenAI manuscripts [4, 5]. Guo's broader statements concerning other shifts, both parities and squarefree values are outside the scope of the two public Lean theorems here.

The OpenAI collection's README at the cited revision describes results at different stages of verification and warns that some unformalized results may contain errors. No independent human verification of these source manuscripts is claimed here. These citations record the origin of the strategy. The supporting inputs needed for the two Green64 statements are proved in the included Lean development. Its machine checks do not constitute independent human review of the source manuscripts or establish all of their claims.

Formalization and production. The proof and formalization were developed with gpt-6 astra in Codex. This article is derived from the mathematical Markdown exposition of the Lean development. The statement is the fixed-shift question in Green's collection [2, Problem 64]. No claim of priority or a comprehensive literature survey is made. The accompanying verification records specify the checked Lean version, statements and axiom audit. Those checks and the mathematical exposition are separate parts of the verification; this article does not assert an independent human review.

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1 Green's Problem 64

Theorem. There are infinitely many primes p such that $\Omega(p-2)$ is odd, where $\Omega(n)$ counts prime factors with multiplicity.

This answers Problem 64 in Ben Green's list, p. 31. We prove the assertion for the fixed shift $p-2$. Throughout the argument, $\lambda(n) = (-1)^{\Omega(n)}$ for positive integers n.

The proof constructs a nonnegative weight supported on integers u with $\lambda(u) = -1$, and shows that its sum over prime values of $u+2$ is positive in every sufficiently large multiplicative interval. The central analytic input is the shifted-correlation estimate (B18), which gives the Type II estimate (B3) uniformly in its bounded companion coefficient. Its proof transfers cancellation on independent prime coordinates to arithmetic histories, retaining the signed comparison and accounting for repeated labels. A weighted sieve then removes most composites; a separate estimate bounds those whose two prime factors are close in size. The constants in the bound for the remaining composites are chosen before the width of the balanced range; this permits that last contribution to be subtracted.

The argument is divided into weights and their mass, correlation and bilinear estimates, and the sieve estimates. The present file defines the objects used in the conclusion and derives the theorem from the two quantitative sieve estimates.

1.1 Weights with a prescribed parity

Write $x = e^L$, $T = \log L$, and suppose L is sufficiently large. Set

$$r = 20, \qquad \sigma = \frac{1}{44}, \qquad s_j = \sigma 2^{-j} L, \qquad d = \max\{j \ge 0 : s_j \ge 20T\}.$$

For $0 \le j \le d$ let

$$Q_j = \{q \text{ prime} : s_j \le \log q \le \tfrac{6}{5} s_j\}.$$

Also take every prime interval

$$P_j = \{p \text{ prime} : \tfrac{3}{2} s_j \le \log p \le \tfrac{9}{5} s_j\}$$

whose logarithmic interval is contained in either $[L^{1/10}, L^{1/5}]$ or $[L^{3/10}, L^{2/5}]$. Let $\mathcal{G}$ be this set of indices and $\mathcal{P} = \bigcup_{g \in \mathcal{G}} P_g$. All the Q_j and P_g are pairwise disjoint. Put

$$V_j = \sum_{q \in Q_j} \frac{1}{q}, \qquad M_g = \sum_{p \in P_g} \frac{1}{p}.$$

The bounds used below hold uniformly for large L:

$$\frac{1}{12} \le V_j \le 4, \qquad \nu \le M_g \le 4, \qquad \nu = \frac{1}{4} \log \frac{12}{11} > 0.$$

For a positive integer u, separate its prime factors as $u = u_* u_{\mathcal{P}}$, where u_* has no prime factor in $\mathcal{P}$ and $u_{\mathcal{P}}$ has no prime factor outside $\mathcal{P}$. Define

$$\omega_g(u) = \#\{p \in P_g : p \mid u\}, \qquad W(u) = \prod_{g \in \mathcal{G}} \frac{\omega_g(u) 2^{-(\omega_g(u)-1)}}{M_g},$$

with a zero factor when $\omega_g(u) = 0$. Notice the distinction between ω_g, which counts distinct primes, and Ω, which counts multiplicities.

Let $a(v)$ be zero unless all prime factors of v lie in the Q_j and v has exactly 20 factors, with multiplicity, in each Q_j. On that support set

$$a(v) = \prod_{q \mid v} \frac{1}{v_q(v)!}.$$

Equivalently, $a(v)$ counts ordered choices of twenty primes in each Q_j with product v, divided by $(20!)^{d+1}$. Repeated primes are allowed. Define

$$E(u) = (-1)^{\sum_{p \in \mathcal{P}} v_p(u)}, \qquad A_\rho(u) = W(u)a(u_*)\frac{1 + \rho E(u)}{2} \quad (\rho \in \{-1, 1\}).$$

We write $A_0(u) = W(u)a(u_*)$ for the unsigned weight; the subscript zero is a separate notation, outside the two-sign definition above.

Since $20(d+1)$ is even, $a(u_*) > 0$ implies $\lambda(u_*) = 1$. Consequently

$$A_\rho(u) > 0 \quad \Longrightarrow \quad \lambda(u) = \rho. \tag{1}$$

This identity uses prime multiplicities throughout; it does not require u to be squarefree.

Fix a smooth function $\Psi : \mathbb{R} \to [0, 1]$, supported in $[1, 2]$ and positive at some point of $(1, 2)$. Such a function is obtained by translating and rescaling a smooth bump function. Set

$$w_\rho(u) = A_\rho(u)\Psi(u/x), \qquad X_\rho = \sum_{u \geq 1} w_\rho(u).$$

Set $w_\rho(0) = 0$, consistently with $\Psi(0) = 0$. All these sums are finite because $w_\rho(u)$ vanishes outside $[x, 2x]$. The mass estimate proved in Section 2.3 gives a constant $c_0 > 0$, depending on Ψ but not on L or ρ, such that

$$X_\rho \geq c_0 x L^{-221} > 0 \tag{2}$$

for both signs and all sufficiently large L.

1.2 The two sieve estimates

Let

$$\mathfrak{S} = 2 \prod_{p \geq 3,\ p \text{ prime}} \left(1 - \frac{1}{(p-1)^2}\right).$$

The product converges, and $\mathfrak{S} \geq 1$; the elementary product bound is proved in Section 12.1. Write

$$M_\rho(L) = \sum_{u+2 \text{ prime}} w_\rho(u)$$

and, for $0 < \kappa < 1/50$, write

$$\mathcal{C}_\rho(L,\kappa) = \sum_{\substack{u+2 \text{ composite} \\ P^-(u+2)>x^{1/2-\kappa}}} w_\rho(u),$$

where $P^-(n)$ is the least prime factor of $n > 1$.

The weighted sieve and the subtraction of composites outside the balanced range give, for every fixed $0 < \kappa < 1/50$ and all sufficiently large L,

$$\mathcal{M}_{-1}(L) + \mathcal{C}_{-1}(L,\kappa) \geq \frac{\mathfrak{S}}{2}\frac{X_{-1}}{L}. \tag{3}$$

The negative sign is the one needed here. The complete subtraction, including the order of the approximation parameters, is given in Section 12.4.

The bound for the remaining composites has the following uniform form: there exists $C_{\mathrm{bal}} > 0$ such that, for every $0 < \kappa < 1/50$ and every $\varepsilon > 0$, both signs satisfy

$$\mathcal{C}_\rho(L,\kappa) \leq (C_{\mathrm{bal}}\kappa + \varepsilon)\frac{X_\rho}{L} \tag{4}$$

once L is sufficiently large. The threshold may depend on κ and ε; C_{bal} does not. Its proof in Section 12.5 uses an upper bound for positive factor pairs in a rectangle, followed by a count of compatible dyadic rectangles. It includes pairs with equal prime factors.

1.3 Positive prime mass and infinitude

Choose

$$\kappa = \min\left\{\frac{1}{100}, \frac{\mathfrak{S}}{8C_{\mathrm{bal}}}\right\}, \qquad \varepsilon = \frac{\mathfrak{S}}{8}.$$

Then $0 < \kappa < 1/50$ and $C_{\mathrm{bal}}\kappa + \varepsilon \leq \mathfrak{S}/4$. For this fixed choice, intersect the eventual ranges in (2), (3), and (4). Subtracting (4) from (3) gives

$$\mathcal{M}_{-1}(L) \geq \frac{\mathfrak{S}}{4}\frac{X_{-1}}{L} > 0. \tag{5}$$

Every summand in $\mathcal{M}_{-1}$ is nonnegative. Therefore some positive integer u satisfies

$$u + 2 \text{ is prime}, \qquad A_{-1}(u) > 0, \qquad \Psi(u/x) > 0.$$

The support of Ψ gives $x \leq u \leq 2x$. By (1), $\lambda(u) = -1$, so $\Omega(u)$ is odd. For $p = u + 2$ we have $p - 2 = u > 0$ and hence $\Omega(p-2)$ odd.

Given any natural number M, choose L in the eventual range of (5) with $e^L > M$. The resulting prime satisfies $p = u+2 > M$. Thus the set of primes in the theorem is unbounded, and therefore infinite. □

2 The weights and their mass

We use the notation and fixed prime bands from the main proof. This section proves nonnegativity, the parity implication, and the uniform lower bound $X_\rho \gg xL^{-221}$.

2.1 Geometry and exact factorization

The inequalities

$$1 < \frac{6}{5} < \frac{3}{2} < \frac{9}{5} < 2, \qquad \frac{9}{10} < 1$$

separate the two intervals at a given dyadic scale from each other and from the intervals at the next scale. Thus no prime occurs in two bands. Every Q_j prime is at least $e^{20T} = L^{20}$. Every P_g prime is at least $\exp(L^{1/10})$, which exceeds L^{20} eventually.

The logarithm of the ratio of the endpoints of any one band, measured in the logarithmic variable, is $\log(6/5)$. The prime number theorem and partial summation therefore give

$$\sum_{e^a \le p \le e^{6a/5}} \frac{1}{p} = \log(6/5) + o(1) \qquad (a \to \infty).$$

The error is uniform when $a \ge 20T$. We use the weaker uniform bounds

$$\frac{1}{12} \le V_j \le 4, \qquad \nu \le M_g \le 4, \qquad \nu = \frac{1}{4}\log\frac{12}{11} > 0.$$

The containment conditions give $O(T)$ marked groups and at least one group for all sufficiently large L. The moving ordinary depth satisfies $d + 1 \le 2T$. The containment conditions retain both endpoints of each whole prime interval.

For $u > 0$ the decomposition $u = u_* u_{\mathcal{P}}$ is unique. Its factors are coprime because their prime supports are disjoint. If $a(u_*) > 0$, then $\Omega(u_*) = 20(d+1)$. Complete additivity of Ω on positive integers gives

$$\lambda(u) = (-1)^{20(d+1)}(-1)^{\sum_{p\in\mathcal{P}} v_p(u)} = E(u).$$

The projection $(1 + \rho E(u))/2$ is either zero or one. This proves $A_\rho \ge 0$ and the implication $A_\rho(u) > 0 \Rightarrow \lambda(u) = \rho$. In particular, repeated ordinary primes contribute their full multiplicity; replacing Ω by the number of distinct prime divisors would invalidate this step.

For $k \ge 1$, $k2^{-(k-1)} \le 1$. Since $a(v) \le 1$ and $M_g \ge \nu$,

$$0 \le w_\rho(u) \le A_\rho(u) \le W(u)a(u_*) \le \nu^{-|\mathcal{G}|} \le L^{C_0}. \tag{W1}$$

2.2 Harmonic probability laws

Put

$$H_Q = \prod_{j=0}^{d} \frac{V_j^{20}}{20!}.$$

Expanding the product over the ordered ordinary slots gives the exact identity

$$\sum_{v\ge 1}\frac{a(v)}{v}=H_Q. \tag{W2}$$

For one marked group define

$$Z_g(q)=\prod_{p\in P_g}\left(1+\frac{q}{p-1}\right),\qquad H_g=\frac{Z_g'(1/2)}{M_g},\qquad H_P=\prod_g H_g.$$

A nonempty set of distinct group primes of size k receives the factor $k(1/2)^{k-1}/M_g$. Summing all positive multiplicities of a prime gives $\sum_{a\ge 1}p^{-a}=1/(p-1)$. Differentiating the finite product Z_g therefore proves

$$H_g=\sum_{\substack{n\ge 1\\ p\mid n\Rightarrow p\in P_g}}\frac{\omega_g(n)2^{-(\omega_g(n)-1)}}{M_g n}=\frac{Z_g(1/2)}{M_g}\sum_{p\in P_g}\frac{1}{p-1/2}. \tag{W3}$$

The zero term for $n=1$ is understood. All series here are absolutely convergent. Since the prime sets are disjoint, multiplying (W3) gives

$$\sum_{n:n_*=1}\frac{W(n)}{n}=H_P. \tag{W4}$$

The harmonic bounds and $\log(1+t)\le t$ give $Z_g(1/2)\le C$ and $\sum_{p\in P_g}(p-1/2)^{-1}\le CM_g$ for an absolute C. The reverse inequality $(p-1/2)^{-1}\ge p^{-1}$ gives

$$1\le Z_g(1/2)\le H_g\le C,\qquad 1\le H_P\le L^{C_1}. \tag{W5}$$

Define independent random positive integers V and R by

$$\Pr(V=v)=\frac{a(v)}{vH_Q},\qquad \Pr(R=n)=\frac{W(n)}{nH_P}\quad (n_*=1).$$

The law of V can be realized by independent choices in the twenty slots of each Q_j, each prime $q\in Q_j$ having probability $1/(qV_j)$. The prime-group components of R are independent as well.

For a single group, total multiplicity one has unnormalized harmonic mass $M_g/M_g=1$. Total multiplicity two arising from two distinct primes has mass

$$\frac{1}{M_g}\sum_{p<q\in P_g}\frac{1}{pq}=\frac{M_g^2-\sum_{p\in P_g}p^{-2}}{2M_g}.$$

The second expression is bounded below by a positive absolute constant: $M_g\ge\nu$ and $\sum p^{-2}=o(1)$ uniformly. By (W5), division by H_g leaves a positive uniform probability for both parities. Condition on every group except one. Whatever the parity of their product, the remaining group can supply either total parity. Hence there is $\delta>0$ such that

$$\Pr(E(R)=\rho)\ge\delta\qquad(\rho=\pm 1). \tag{W6}$$

We also need a tail estimate. Take $t = L^{-2/5}$. Replace each factor $1/(p-1)$ in the Euler-product calculation by $p^{-1+t}/(1-p^{-1+t})$. Since $\log p \le L^{2/5}$, we have $p^t \le e$. The harmonic sums of these replacement factors are bounded uniformly, as are the corresponding differentiated finite products. There are $O(T)$ groups, so

$$\sum_{n:n_*=1} \frac{W(n)}{n^{1-t}} \le L^{C_2}.$$

For every fixed $\eta > 0$, multiplication by $x^{-\eta t}$ then gives

$$\sum_{\substack{n>x^\eta \\ n_*=1}} \frac{W(n)}{n} \le L^{C_2} \exp(-\eta L^{3/5}). \tag{W7}$$

Because $H_P \ge 1$, (W7) also bounds $\Pr(R > x^\eta)$. This estimate retains all positive prime multiplicities in R.

2.3 Localization

Set $U = VR$. The disjoint prime supports make the factorization into V and R unique. Combining the two probability laws gives

$$X_\rho = H_Q H_P \, \mathbb{E}[U\Psi(U/x)\, 1_{E(R)=\rho}]. \tag{W8}$$

We prove a lower bound of order x/L for the expectation, uniformly in ρ.

Choose $1 < \alpha < \beta < 2$ and $c_\Psi > 0$ with $\Psi \ge c_\Psi$ on $[\alpha, \beta]$. Write

$$m_j = \frac{11}{10} s_j, \qquad \Delta = \frac{1}{5}\sigma L.$$

The midpoint sum over all ordinary scales is exactly $20 \sum_{j\ge 0} m_j = L$. Keep one Q_0 slot unconditioned. Set $k = 13$ and $\epsilon = 1/62400$. For each nonreserved slot in a band $j < k$, restrict the selected prime by

$$(11/10 - \epsilon)s_j < \log q \le (11/10 + \epsilon)s_j.$$

The numerical conditions for this corridor are

$$288r = 5760 \le 2^{13}, \qquad kr\epsilon = \frac{1}{240} \le \frac{1}{120}, \qquad 0 < \epsilon \le \frac{1}{10}.$$

There are finitely many restrictions, and the prime reciprocal-sum estimate gives each a positive harmonic probability bounded below independently of L. Their joint probability has the same property.

For large L we have $d \ge 12$. The largest possible sum of logarithms in the remaining bands is at most

$$20\frac{6}{5}\sigma L \sum_{j\ge 13} 2^{-j} = \frac{6}{11} 2^{-12} L < \frac{\Delta}{24}.$$

The infinite midpoint sum over those bands is also less than $\Delta/24$. Their difference consequently has absolute value less than $\Delta/24$.

Independently require $E(R) = \rho$ and $\log R \le \Delta/12$. By (W6) and (W7), this event has probability at least $\delta/2$ for large L. On the intersection, if B is the logarithm of the product of all variables except the reserved prime, then

$$|B - (L - m_0)| \le \frac{\Delta}{6}.$$

The interval

$$[L + \log \alpha - B,\ L + \log \beta - B]$$

lies inside the logarithmic interval of Q_0, at distance comparable to L from its endpoints. Its length is the fixed positive number $\log(\beta/\alpha)$. The prime number theorem and partial summation give harmonic probability at least c/L for the reserved prime to lie there, uniformly in the conditioned variables. On this event $U\Psi(U/x) \ge \alpha c_\Psi x$.

The probabilities of the restrictions on the other variables have a positive product independent of L. Applying (W8) proves

$$X_\rho \ge c\frac{xH_Q H_P}{L} \qquad (\rho = \pm 1). \tag{W9}$$

Finally, use $200^{-20} \le (1/12)^{20}/20!$, together with $\log 200 \le 11/2$. Thus $V_j \ge 1/12$ and $d + 1 \le 2T$ give

$$H_Q \ge \left(\frac{(1/12)^{20}}{20!}\right)^{d+1} \ge 200^{-20(d+1)} \ge L^{-220}.$$

Together with $H_P \ge 1$ and (W9), this yields $X_\rho \ge c_0 x L^{-221}$, as required.

3 Replacing prime and roughness tests

We replace prime and roughness indicators by their complete-cell averages. The weights A_ρ, $x = e^L$, and X_ρ are those of the main proof. All constants are fixed before taking $L \to \infty$. In particular, a constant attached to a finite partition is not allowed to vary with L.

The estimates concern the particular prime and roughness discrepancies defined below. The other coefficient in a bilinear sum may be an arbitrary bounded complex function. This is the property needed when replacing the second of two factors.

3.1 Complete logarithmic cells

Put $W = \exp(\sqrt{L})$ and $\Delta = L^{-S}$, where $S > 0$ will be chosen. For $j \geq 0$ define

$$C_j = (e^{j\Delta}, e^{(j+1)\Delta}] \cap \mathbb{N}.$$

Every integer $n \geq 2$ belongs to the unique cell with $j = \lceil \log n/\Delta \rceil - 1$. Define the roughness indicator

$$R_y(n) = 1_{n\geq 2,\ P^-(n)>y}.$$

For either $I(n) = R_y(n)$, with $y \geq W$, or $I(n) = 1_{n \text{ prime}}$, define its approximation by

$$B_I(n) = R_W(n)\,\frac{\sum_{m\in C_j} I(m)}{\sum_{m\in C_j} R_W(m)} \quad (n \in C_j), \qquad \alpha_I(n) = I(n) - B_I(n). \tag{C1}$$

A zero denominator is assigned the value zero. On a cell where I is supported on the W-rough integers, a zero denominator also forces the numerator to vanish. Consequently

$$\sum_{n\in C_j} \alpha_I(n) = 0. \tag{C2}$$

Restricting a coefficient to a factor interval means multiplying it by that interval's indicator. The ratio in (C1) continues to use the whole cell, including when the interval cuts one or both endpoint cells. This convention makes restriction to a smaller interval an exact identity.

For the principal character, (C2) handles complete cells. On each such cell, subtract the value of n^{-1+it} at one endpoint. The derivative bound

$$\left|\frac{d}{dn} n^{-1+it}\right| \leq \frac{1+|t|}{n^2}$$

bounds the variation. At most two endpoint cells remain. Since $q \leq W$, the principal character is one on every supported integer. The resulting bound is

$$\left(8(e^\Delta - 1) + \frac{2}{a}\right)(2 + |t|).$$

For a nonprincipal character, write each y-rough integer uniquely as $n = pd$, where $p = P^-(n)$. For fixed d, the prime p lies in the interval obtained from $a \leq pd \leq b$, $p > y$, and $p \leq P^-(d)$; when $d = 1$ the last condition is omitted. This is a single clipped interval,

even when p divides d. Siegel–Walfisz and partial summation bound its prime character sum with weight p^{-1+it} by $K(2+|t|)(\log y)^{-D'}$ for any requested D', uniformly when the modulus is at most a fixed power of $\log y$. The outer cofactor costs only

$$\sum_{d\le b}\frac{1}{d}\le 1+\log b.$$

Because $\log y\ge\sqrt{L}$, $b\le e^{2L}$, and $q,|t|\le L^B$, choosing D' sufficiently large gives any fixed inverse power of L for the rough character sum. On each cell slice, the proxy is a scalar in $[0,1]$ times the W-rough indicator, so the same estimate applies. There are at most $2L^{S+1}+2$ slices. Requesting saving $L^{-(B+S+4)}$ for the rough sums makes the total test-minus-proxy sum at most L^{-B}.

Combining the two character cases gives the following precise statement: for fixed $B>0$, $S>2B$, and $\beta>0$, eventually, uniformly for

$$2\le a\le b\le 2a,\quad e^{\beta L}\le a,\quad b\le e^{2L},\quad 1\le q\le L^B,\quad |t|\le L^B,\quad y\ge W,$$

and every Dirichlet character χ modulo q,

$$\left|\sum_{a\le n\le b}\alpha_{R_y}(n)\chi(n)n^{-1+it}\right|\le L^{-B}. \tag{C3}$$

For the prime discrepancy, use $y=2\sqrt{a}$: every complete cell meeting $[a,b]$ lies in a range where y-roughness is equivalent to primality. The two proxies therefore agree on the complete cells, including the two cells cut by the interval endpoints.

3.2 The bilinear estimate used in the proof

For finite sets of positive integers K,J, set

$$\mathcal{J}_\rho(K,J;\alpha,\beta)=\sum_{0\le u\le Y}w_\rho(u)\sum_{\substack{m\in K,\ n\in J\\ mn=u+2}}\alpha(m)\beta(n),\qquad Y=\lfloor 2x\rfloor. \tag{C4}$$

All pairs are ordered, and $m=n$ is retained. In applications α is a restricted discrepancy (C1), while $|\beta|\le 1$. The bilinear chapter states the estimate with all factor ranges and quantifiers: on an admissible dyadic rectangle it bounds (C4) by CxL^{-D} for any prescribed D, with the cell mesh and C chosen before the interval endpoints and the bounded companion coefficient.

Here is the parameter choice used to apply it. Fix $0<b_1<1/100$ and $0<\kappa<1/50$, and put

$$\beta_P=b_1/12,\qquad \beta=b_1/3,\qquad b_*=b_1/3.$$

Reserve one ordinary prime slot in a fixed band Q_{j_Q} so small that

$$\frac{6}{5}\sigma 2^{-j_Q}\le\frac{\beta_P}{4}.$$

Such an index exists because $2^{-j}\to 0$; the moving depth eventually contains it. Given an error exponent D, choose the analytic precision A and the localization exponent K_0 with

$$D + 2 + C_E + C_F \le A, \qquad 2A + 2C_W + 344 \le K_0. \tag{C5}$$

Here C_W, C_E, C_F are the fixed polynomial exponents respectively bounding the marked weight, the selected-factor coefficients, and the Fourier integrals of the ordinary-slot cutoffs; their values are specified in the bilinear reduction. Choose A first, then K_0.

The derivation in Section 4 expands the ordinary slots, allocates a selected divisor $e \in [EL^{-88}, E]$ with $E = UL^{-K_0}$, and applies Cauchy–Schwarz. For the explicitly defined sum $\mathcal{B}'$ in (B8), with $G = L^{C_E} \prod_i |\hat{f}_{\delta_i}(\vartheta_i)|$, it gives

$$|\mathcal{B}'|^2 \le 6EVG^2 \left(2xL^{2C_W+344} + (C_{\mathrm{phys}} + 24C_1)xL^{-A}\right). \tag{C6}$$

The first term is the diagonal divisor-moment bound. The second follows from the marked correlation and its Fourier tail. Since $EV \le C_{\mathrm{box}} xL^{-K_0}$, (C5) implies $|\mathcal{B}'| \le CGxL^{-A}$. The remaining slot integrals cost $L^{C_E+C_F}$. Fourier inversion of $\psi(t) = \Psi(e^t)$ restores the localization; its zeroth and Hth absolute Fourier moments are finite, so the resulting constant is independent of L and of the bounded companion coefficient.

Apply the unsigned and signed estimates to the exact projection $A_\rho = (A_0 + \rho A_0 E)/2$. The triangle inequality and $|\rho| = 1$ give the same bound for $\mathcal{T}_\rho$.

3.3 Two successive replacements

Write $\mathcal{T}_{ij}$ for (C4) with the first coefficient sharp when $i = 0$ and approximated when $i = 1$, and similarly for the second coefficient. Then

$$\mathcal{T}_{00} - \mathcal{T}_{11} = (\mathcal{T}_{00} - \mathcal{T}_{10}) + (\mathcal{T}_{10} - \mathcal{T}_{11}). \tag{C7}$$

On the cells under consideration, (C1) gives $0 \le B_I \le 1$. Both companions are therefore bounded, so (B3) applies in either order. We use bounds Q and $2Q$, respectively, for the two differences.

A rectangle is partitioned into its attained binary dyadic rectangles. The finite carrier consists of the scales of actual positive factor pairs $mn = u + 2$ with $\Psi(u/x) \ne 0$. There are at most

$$D_Y = (\lfloor \log_2(Y + 2) \rfloor + 1)^2 \tag{C8}$$

such pairs of scales. All full-cell ratios remain those of (C1). The internal subdivision cost is absorbed by the extra 2 in (C5). The external rectangle count (C8) is then paid explicitly. In the special balanced positive estimate, a sharper linear count is available; that separate count is used in Section 12.3.

3.4 The finite error sum and its quantifiers

Fix an integer N and a nondecreasing sequence

$$t_0 = b_1, \qquad t_N = 1/2 - \kappa.$$

For $i < N$, the unbalanced region consists of positive pairs $mn = u + 2$ with $x^{t_i} < n \le x^{t_{i+1}}$. Its first sharp coefficient is $R_{x^{t_i}}(m)$ and its second is the prime indicator at

n. For the balanced region both coefficients are prime indicators, and both factors lie in $[x^{1/2-2\kappa}, x^{1/2+2\kappa}]$.

Let $U^{(i)}_{ab,\rho}$ and $B_{ab,\rho}$ be the corresponding sums with replacement flags $a, b \in \{0, 1\}$. The total error actually used is

$$\mathcal{E}(L) = \sum_{\rho=\pm 1} \left(\sum_{i<N} |U^{(i)}_{00,\rho} - U^{(i)}_{10,\rho}| + |B_{00,\rho} - B_{10,\rho}| + |B_{10,\rho} - B_{11,\rho}| \right). \tag{C9}$$

A second quantity $\mathcal{E}_{\text{all}}$ sums all four successive differences, in both replacement orders, for each of the $N + 1$ regions and both signs.

The bounds have the following quantifier order:

$$\begin{gathered} \text{fix } N, b_1, \kappa, D \text{ and the cutoff } \Psi; \\ \text{choose } H, J, B, S, C \text{ with } S > 2(B+1) + 3, C > 0; \\ \text{then for every admissible sequence } (t_i), \text{ eventually in } L, \\ \mathcal{E}(L) \le 2(N+3)Q(L), \qquad \mathcal{E}_{\text{all}}(L) \le 12(N+1)Q(L), \\ Q(L) = CD_Y x L^{-D}. \end{gathered} \tag{C10}$$

The threshold may depend on the partition. The approximation parameters and C precede its endpoints. To check the factors in (C10), for one sign there are N unbalanced first steps and two balanced steps with bounds Q and $2Q$. This gives $(N+3)Q$. For each region and sign the four steps in both orders cost $Q + 2Q + Q + 2Q = 6Q$.

The conclusion follows from the rectangle estimates by finite partition and the triangle inequality.

3.5 Normalization by the actual mass

Take $D = 230$. Since $Y = \lfloor 2e^L \rfloor$, (C8) gives $D_Y = O(L^2)$. For fixed N and the fixed parameters in (C10),

$$\mathcal{E}(L), \mathcal{E}_{\text{all}}(L) = O(e^L L^{-228}) = o(e^L L^{-222}). \tag{C11}$$

The lower bound from Section 2 is uniform in the sign:

$$\frac{X_\rho}{L} \ge c_0 e^L L^{-222}.$$

Given $\varepsilon > 0$, apply (C11) with the absolute error tolerance εc_0 and intersect its eventual range with the mass lower bound. This proves, for either sign,

$$\mathcal{E}(L) = o(X_\rho/L), \qquad \mathcal{E}_{\text{all}}(L) = o(X_\rho/L). \tag{C12}$$

In particular, (C7) gives the balanced two-step error required in the final sieve argument.

4 The bilinear reduction

This chapter gives the finite sums and estimates behind the bilinear replacement in Section 3. We use its complete-cell discrepancies and the weights from Section 1. To distinguish parity from Fourier frequency, $\rho \in \{\pm 1\}$ denotes parity and $\tau \in \mathbb{R}$ denotes a Mellin frequency.

4.1 The statement on one factor rectangle

Fix $D > 0$, $\eta > 0$, and $C_{\mathrm{box}} \geq 1$. Reserve a slot in a fixed ordinary band Q_{j_Q} with

$$\frac{6}{5}\sigma 2^{-j_Q} \leq \eta/4.$$

Choose fixed exponents A, K_0 satisfying

$$A \geq D + C_E + C_F, \qquad K_0 \geq 2A + 2C_W + 344. \tag{B1}$$

Here $C_W \geq 0$ bounds $W(u)$ by L^{C_W}, $C_E = 2\log(20!)$, and C_F is the fixed Fourier cost defined below. There are integers H, J, B, a mesh exponent $S > 2(B+1)+3$, and $C > 0$ with the following property. For each fixed $\beta, \gamma > 0$, eventually in L, the conclusion holds uniformly in all the data below:

$$\begin{gathered} U, V > 0, \quad x \leq 4UV, \quad UV \leq C_{\mathrm{box}} x, \quad V \geq 1, \\ E = UL^{-K_0}, \quad 1 \leq E \leq x^{1-\eta}, \\ 2 \leq a \leq b \leq 2a, \quad x^{\beta} \leq a, \quad b \leq x^2, \\ K = [a,b] \cap \mathbb{N} \subset [U, 2U), \qquad N \subset [V, 2V] \cap \mathbb{N}, \qquad |b_n| \leq 1. \end{gathered} \tag{B2}$$

Let α be the restriction to K of either the prime discrepancy or the x^{γ}-rough discrepancy on the complete cells of width L^{-S}. Then

$$|\widehat{\mathcal{J}_\rho}(K, N; \alpha, b)| \leq CxL^{-D} \qquad (\rho = \pm 1). \tag{B3}$$

The parameters H, J, B, S, C are fixed before β, γ and the rectangle endpoints; the eventual threshold may depend on β, γ. There is no regularity hypothesis on b_n. An empty first interval gives zero and may be omitted. Here $\widehat{\mathcal{J}_\rho}$ is (C4), with $w_\rho(u) = \Psi(u/x)A_\rho(u)$ and $x = e^L$; thus the shift is $mn = u + 2$. The constants also depend on the fixed smooth cutoff Ψ and the reserved band j_Q. The integer H is a Fourier-tail order, J is the number of additional shared marks, B is the requested complete-cell discrepancy order, and S is its mesh exponent.

The range allows a companion factor of size x^{η}: if $U \asymp x^{1-\eta}$ and $V \asymp x^{\eta}$, then $E = UL^{-K_0} \leq x^{1-\eta}$ eventually. More generally (B2) implies $V \geq \frac{1}{4}x^{\eta}L^{-K_0}$. In the sieve application one fixes $\eta = b_1/12$ with $0 < b_1 < 1/100$, so $0 < \eta < 1/1200$. There is no lower bound on this fixed positive exponent independent of b_1, but it cannot tend to zero with L in this statement. Uniformity in b_n concerns the entire sum on scale x. For a singleton $N = \{n\}$, (B3) still has right side CxL^{-D}, not $C(x/n)L^{-D}$; it does not assert a prime-correlation saving at the individual dilation's natural scale x/n.

The proof first establishes the same conclusion for each of the weights $W(u)a(u_*)\varepsilon(u)$, where $\varepsilon = 1$ or $\varepsilon = E(u)$. The identity $A_\rho = (A_0 + \rho A_0 E)/2$ then proves (B3).

4.2 Allocating ordinary slots

Order the nonreserved ordinary slots by decreasing logarithmic scale, breaking ties by their slot labels. Write their number as $n = 20(d+1)-1$, their scales as s_i, and their prime choices as q_i. Repeated prime values retain their separate labels. An allocation $\delta \in \{0,1\}^n$ assigns the slots with $\delta_i = 1$ to a product e and the remaining slots to a product r.

Use two fixed smooth functions f_0, f_1, with values in $[0,1]$, supported in $[-85,85]$. On $[-84,84]$ they are $f_1(t) = h(t)$ and $f_0(t) = 1 - h(t)$, where h is a smooth transition equal to zero on $(-\infty, 0]$ and one on $[1,\infty)$. At slot i, set

$$R_i = \log E - \sum_{j<i} \delta_j \log q_j, \qquad \prod_{i=1}^{n} f_{\delta_i}\left(\frac{R_i - \log q_i}{s_i}\right) \tag{B4}$$

for the allocation weight. The slot geometry makes these weights sum to one on the relevant tuples and places every contributing selected product in the interval

$$EL^{-88} \le e \le E. \tag{B5}$$

Here is the numerical reason for the lower endpoint. A skipped slot has $R_i < \log q_i + s_i$. In every nonterminal band, the next band contains at least nineteen nonreserved slots; their total logarithmic mass is at least $(19/2)s_i > (11/5)s_i \ge \log q_i + s_i$. In the last band, $s_d < 40 \log L$, so $\log q_i + s_i \le 88 \log L$. Induction on the remaining slots bounds the final residual by $88 \log L$. A selected slot has $R_i > \log q_i$ and therefore keeps the residual nonnegative. The same prefix bounds keep each used argument inside the region where $f_0 + f_1 = 1$; summing the two choices successively proves the partition identity in (B4).

The initial supply condition is ensured after excluding a large marked part. On $u \in [x, 2x]$, a marked part at most $x^{\eta/4}$ and the reserved prime at most $x^{\eta/4}$ leave at least $x^{1-\eta/2}$ in the remaining ordinary slots, whereas $E \le x^{1-\eta}$. The Rankin estimate (W7) bounds the excluded tuples by any prescribed logarithmic saving. The finite source comparison also retains the Fourier truncation error; its total, with coefficients of modulus at most one, is bounded by $4xL^{-D}$ after the required precision is fixed.

For clarity, the Fourier-separated coefficients can be written explicitly. Use $\mathrm{e}(t) = e^{2\pi i t}$ and $\hat{f}(\xi) = \int f(t)\mathrm{e}(-t\xi)\,dt$. For fixed $\delta, \vartheta = (\vartheta_i)$, put

$$\omega_i = -\frac{\vartheta_i}{s_i} - \delta_i \sum_{j>i} \frac{\vartheta_j}{s_j}.$$

The exponent in Fourier inversion of (B4) is $\log E \sum_i \vartheta_i/s_i + \sum_i \omega_i \log q_i$. Define the two finite coefficient sums

$$\begin{aligned} b_e &= \sum_{\substack{(q_i)_{\delta_i=1} \\ \prod q_i = e}} \prod_{\delta_i=1} \mathrm{e}(\omega_i \log q_i), \\ c_r &= (20!)^{-(d+1)} \sum_{\substack{(q_i)_{\delta_i=0} \\ \prod q_i = r}} \prod_{\delta_i=0} \mathrm{e}(\omega_i \log q_i), \\ H(v_*) &= \sum_{\substack{p \in Q_{j_Q} \\ p \mid v_*}} c_{v_*/p}. \end{aligned} \tag{B6}$$

Each prime variable lies in its assigned band. In particular H retains the reserved prime and all repeated-prime assignments. For $v > 0$, define

$$a_v(v) = v^{i\tau} W(v)\varepsilon(v)H(v_*),$$

and

$$a_e(e) = 1_{EL^{-88}\le e\le E}\, \exp(i\tau \log e)\, \mathrm{e}\!\left(\log E \sum_i \vartheta_i/s_i\right) \prod_i \hat{f}_{\delta_i}(\vartheta_i)\, b_e. \tag{B7}$$

Here e in $\log e$ is the selected positive integer. With this notation the finite sum to be estimated is

$$\mathcal{B}' = \sum_e a_e(e) \sum_{n\in N} b_n \sum_{\substack{m\in K\\ mn>2,\ e\mid mn-2}} \alpha(m) a_v((mn-2)/e). \tag{B8}$$

This formula uses the actual integer $u = mn - 2$. When a containing interval is needed after Cauchy–Schwarz, $u \le \lceil 6UV \rceil$ suffices.

At most $(20!)^{d+1}$ selected assignments can have a prescribed product. Since $d + 1 \le 2\log L$,

$$|a_e(e)| \le G, \qquad G = L^{C_E} \prod_i |\hat{f}_{\delta_i}(\vartheta_i)|. \tag{B9}$$

For the complementary sum, the full factorial normalization in (B6) gives $|c_r| \le 1$. Thus $|H(v_*)| \le d_2(v_*)$, where d_2 is the divisor-counting function, and $|a_v(v)| \le L^{C_W} d_2(v)$.

4.3 The positive square and the diagonal

Cauchy–Schwarz in (e, n) gives

$$|\mathcal{B}'|^2 \le \left(\sum_{e,n} |a_e(e)b_n|^2\right) \sum_{e,n} \left|\sum_m \alpha(m) 1_{e\mid mn-2} a_v((mn-2)/e)\right|^2. \tag{B10}$$

The first factor is at most $6EVG^2$. Enlarge the second, nonnegative sum to all odd $e \le 2E$ and all $n \le 3V$, using a nonnegative smooth weight $\eta_e(e/E)\eta_n(n/V)$ that is at least one on (B5) and $n \in [V, 2V]$. Its supremum is at most $2L^{88}$. Here the supports are part of the construction:

$$\eta_e(t) \ne 0 \Longrightarrow \tfrac{1}{2}L^{-88} < t < 2, \qquad \eta_n(t) \ne 0 \Longrightarrow \tfrac{1}{2} < t < 3. \tag{B10a}$$

For completeness, put $a = L^{-88}$ and fix a smooth function b with $0 \le b \le 1$, supported in $(1/2, 3/2)$ and equal to one on $[2/3, 4/3]$. The actual majorant is $\eta_e(t) = b(t) + \sum_{j\ge 0,\, 2^j a\le 1} b(t/(2^j a))$. The plateaux of successive scales meet, so this is at least one on $[a, 1]$. Every scale lies in $[a, 1]$, which proves (B10a). The number of summands is at most $2L^{88}$ for $L \ge 8$, and, for $K_j = \|b^{(j)}\|_\infty$, $\|\eta_e^{(j)}\|_\infty \le 2K_j L^{88(j+1)}$. The fixed bump η_n, equal to one on $[1, 2]$ and supported in $(1/2, 3)$, has derivative bounds independent of L. Only after this enlargement do we expand the square into its diagonal and off-diagonal parts.

Write

$$F_{e,n}(m) = \alpha(m)1_{mn>2,\ e|mn-2}a_v((mn-2)/e).$$

The off-diagonal is the concrete finite sum

$$O = \sum_{\substack{e\le 2E\\ e\ \mathrm{odd}}} \sum_{n\le 3V} \eta_e(e/E)\eta_n(n/V) \sum_{\substack{m,m'\in K\\ m\neq m'}} F_{e,n}(m)\overline{F_{e,n}(m')}. \tag{B11}$$

No positivity is used for (B11). Positivity was used only for the unexpanded square in (B10).

For the diagonal, fix $u = mn-2$. There are at most $d_2(u+2)$ choices of (m,n) and at most $d_2(u)$ divisors e. The bound after (B9) therefore gives an unweighted diagonal at most

$$L^{2C_W} \sum_{1\le u\le \lceil 6UV\rceil} d_2(u)^3 d_2(u+2).$$

The elementary divisor-moment estimate is

$$\sum_{u\le Y} d_2(u)^3 d_2(u+2) \le (Y+2)(1+\log(Y+2))^{255}. \tag{B12}$$

Indeed, $d_2(u)^3 d_2(u+2) \le d_2(u)^4 d_2(u+2)^4$. Cauchy–Schwarz bounds the sum by the eighth divisor moment up to $Y+2$. The pointwise comparison $d_2(n)^8 \le d_{256}(n)$ and successive harmonic summation give $\sum_{n\le N} d_{256}(n) \le N(1+\log N)^{255}$. Since $UV \le C_{\mathrm{box}}x$, the right side of (B12) is at most xL^{256} eventually. Including the smoothing weight gives

$$\mathcal{D} \le 2xL^{2C_W+344}. \tag{B13}$$

Thus the remaining estimate needed for (B3) is an arbitrary logarithmic saving for (B11), with all signed marked factors retained.

4.4 The shifted correlation

For a nonzero summand of (B11), both $mn-2$ and $m'n-2$ are divisible by the odd integer e. The congruence $mn \equiv 2 \pmod e$ makes n invertible modulo e, so $e \mid m'-m$. Put

$$k = \frac{m'-m}{e} \neq 0, \qquad z = m'\frac{mn-2}{e}, \qquad z+2k = m\frac{m'n-2}{e}. \tag{B14}$$

Both physical positions are positive. The support of the smoothing gives $|k| \le L^{K_0+89}$ and places the dyadic scales of z between $xL^{-(K_0+90)}$ and xL^{K_0+90}, with a fixed enlargement for the smooth dyadic partition. These bounds follow by substituting $E = UL^{-K_0}$, $m, m' \asymp U$, and the support bounds on e, n. Indeed, (B10a) gives $e > EL^{-88}/2$ and $n > V/2$, while $|m'-m| \le U$. Thus $|k| < 2L^{K_0+88} \le L^{K_0+89}$. Also $m, m' \in [U, 2U]$, $e < 2E$ and $n < 3V$ give $cU^2V/E \le z \le CU^2VL^{88}/E$ for fixed $c, C > 0$; the lower bound uses $mn-2 \ge mn/2$ eventually. Together with $UV \asymp x$, these are the stated polynomial logarithmic bounds, also for negative k.

The coefficient α is supported on integers with no prime factor at most $W = e^{\sqrt{L}}$. Hence m, m' are units modulo $2|k|$ eventually. The odd quotient condition on e is the following exact residue mask, with $d_k = |k|$:

$$T_{d_k} = \begin{cases} \{d_k + 1\} \pmod{2d_k}, & d_k \text{ even}, \\ \emptyset, & d_k \text{ odd}. \end{cases}$$

For characters modulo $2d_k$, define

$$c_\chi = \frac{1}{\varphi(2d_k)} \sum_{r \in T_{d_k}} \overline{\chi(r)}.$$

Character orthogonality gives

$$1_{m'm^{-1} \in T_{d_k}} = \sum_\chi c_\chi \chi(m') \overline{\chi(m)}, \qquad \sum_\chi |c_\chi| \le 1. \tag{B15}$$

The smoothing factors have explicit expressions in the three logarithmic coordinates $t_1 = \log(m/U)$, $t_2 = \log(m'/U)$ and $t_3 = \log(z/Z)$:

$$\begin{aligned} g_e(t) &= \frac{U}{kE}(e^{t_2} - e^{t_1}), \\ g_n(t) &= \frac{Z}{kUV} e^{t_3}(e^{-t_1} - e^{-t_2}) + \frac{2}{UV} e^{-t_1}. \end{aligned} \tag{B15a}$$

They equal e/E and n/V respectively by (B14). Multiply $\eta_e(g_e(t))\eta_n(g_n(t))$ by fixed smooth buffers in the three coordinates, each equal to one on the relevant containing interval. This gives a compactly supported function on a fixed logarithmic box. Since $|k| \ge 1$, $U/E = L^{K_0}$ and $Z/(UV)$ is bounded by a fixed power of L, the chain rule and the derivative bounds after (B10a) bound each of its fixed-order derivatives by a fixed power of L. Fourier inversion of this three-variable function separates m, m', z; the quantitative truncation is given in the frequency summation.

This reduces (B11) to correlations at the shift $2k$. Here are their actual coefficient functions. For the residual c in (B6), write

$$H_\sigma(v) = \sum_{\substack{p \in Q_{j_Q} \\ p \mid v}} p^{i\sigma} c_{v/p},$$

and, for $n > 0$, put

$$\begin{aligned} F(n) &= \sum_{m \mid n_*} \overline{\alpha(m)} \chi(m) m^{i(\xi_2 - \tau)} H_\sigma(n_*/m), \\ G(n) &= \sum_{m \mid n_*} \alpha(m) \overline{\chi(m)} m^{i(\xi_1 + \tau)} \overline{H_\sigma(n_*/m)}. \end{aligned} \tag{B16}$$

The same original α occurs at both endpoints, with the displayed conjugation. If m supports α, dividing by m removes no marked prime. Thus $W(n/m) = W(n)$ and $\varepsilon(n/m) = \varepsilon(n)$ whenever $m \mid n_*$. Multiplicativity of imaginary powers then turns the full cofactor expansion exactly into

$$\begin{aligned} C_{k,Z} = \sum_{\substack{z \ge 1 \\ z + 2k > 0}} & \psi(z/Z) z^{i(\tau + \xi_3)} (z + 2k)^{-i\tau} F(z) G(z + 2k) \\ & \cdot W(z) W(z + 2k) \varepsilon(z) \varepsilon(z + 2k). \end{aligned} \tag{B17}$$

The smooth function ψ is supported in $[1/64, 64]$, so the sum is finite. The marked-correlation estimate says that, for any fixed $D_1, C_Z, C_k, C_f > 0$, the mesh can be chosen so that

$$|\mathcal{C}_{k,Z}| \le ZL^{-D_1} \tag{B18}$$

uniformly for $0 < |k| \le L^{C_k}$, $xL^{-C_Z} \le Z \le xL^{C_Z}$, characters modulo $2|k|$, both choices of ε, and $|\tau|, |\sigma|, |\xi_i| \le L^{C_f}$. The cutoff is smooth with fixed support and prescribed bounds on its logarithmic derivatives. The approximation parameters precede the fine interval's positive lower exponent and the roughness exponent.

The proof of (B18) is given in Section 5. It combines alternating coordinate averages, their arithmetic character approximation, a physical moment estimate, and a separate discrepancy bound for the comparison term. The scope of this assertion matters. α is exactly the restricted discrepancy (C1), and c_r is exactly the ordinary-slot coefficient (B6), for an arbitrary allocation and arbitrary slot Fourier frequencies. Only the external frequencies $\tau, \sigma, \xi_1, \xi_2, \xi_3$ have the displayed polynomial restriction. For every fixed j_Q, D_1, C_Z, C_k, C_f and cutoff derivative budget, there are common J, B, S; for each fixed $\beta, \gamma > 0$ there is then one threshold, uniform in the fine interval, its further restriction, allocation, slot frequencies, character, external frequencies, shift and scale. The result is not claimed for an arbitrary bounded first coefficient α.

4.5 Summing frequencies and returning to the bilinear sum

For completeness, the losses in passing from (B18) to (B11) can be paid before choosing the cell mesh. Fix the desired off-diagonal accuracy A and let

$$D_{\mathrm{tail}} = A + K_0 + 200, \qquad D_{\mathrm{main}} = K_0 + 315 + D_{\mathrm{tail}}.$$

Fix an integer $P > 3$, so $(1 + \|t\|)^{-P}$ is integrable on $\mathbb{R}^3$, and set

$$\begin{aligned} &p_0 = 5(K_0 + 90)(P + 1), \quad p_1 = 5(K_0 + 90)(P + 2), \\ &D_1 = D_{\mathrm{main}} + p_0, \qquad\quad C_f = p_1 + D_{\mathrm{main}} + 2C_W + 30, \qquad T_f = L^{C_f}. \end{aligned} \tag{B19}$$

The five factors defining the cutoff $f = f_{k,Z,L}$ are the two compositions after (B15a) and the three fixed buffers. They have common compact support. The chain and product rules give, for each fixed derivative order j, a bound $\|\partial^j f\|_\infty \le C_j L^{5(K_0+90)(j+1)}$. Integration by parts through orders P and $P + 1$, respectively, on this fixed support, therefore gives the two uniform moment bounds

$$\int_{\mathbb{R}^3} |\hat{f}(\zeta)|\, d\zeta \le C_0' L^{p_0}, \qquad \int_{\mathbb{R}^3} \|\zeta\| |\hat{f}(\zeta)|\, d\zeta \le C_1 L^{p_1}. \tag{B19a}$$

The constants C_0', C_1 depend on the fixed buffers, not on L, k, Z or the fine cells. In particular the zeroth Fourier moment itself need not be bounded independently of L.

Split Fourier inversion at the ball of radius T_f. On the ball, apply (B18) with saving D_1, after shifting ξ_i by 2π times the Fourier variables and increasing the fixed frequency exponent. The character coefficients have total modulus at most one by (B15). Thus the integrated principal term at each k, Z is at most

$$ZL^{-D_1} C_0' L^{p_0} = C_0' Z L^{-D_{\mathrm{main}}}. \tag{B19b}$$

There are at most $O(L^{K_0+89})$ shifts and $O(L)$ dyadic scales; their largest scale is $O(xL^{K_0+90})$. Hence the sum of (B19b) is $O(xL^{2K_0+180-D_{\mathrm{main}}})$, which is $O(xL^{-A})$. The extra p_0 in the requested correlation precision is essential and is chosen before J, B, S.

Outside the ball, use

$$\int_{\|\zeta\|>T_f} |\hat{f}(\zeta)|\, d\zeta \le T_f^{-1} \int \|\zeta\| |\hat{f}(\zeta)|\, d\zeta. \tag{B20}$$

Finite sums of absolute coefficient values and the derivative bounds for the scaled cutoffs give a polynomial factor in L; the additional powers in (B19) make the total tail $O(xL^{-D_{\mathrm{tail}}})$. In this comparison, a physical pair where a cutoff vanishes is still included in the finite Fourier integral. Its full Fourier integral is zero, so its truncated contribution equals minus its tail and is bounded by (B20). This accounts for the two endpoint cutoff errors without discarding a signed subfamily.

The resulting bounds are

$$|O_{\mathrm{ball}}| \le C_{\mathrm{phys}} x L^{-A}, \qquad |O - O_{\mathrm{ball}}| \le 24 C_1 x L^{-D_{\mathrm{tail}}},$$

where C_1 is the fixed coefficient in (B19a), and $C_{\mathrm{phys}} = 250 C_{\mathrm{box}}(1+2/\log 2)(C_0' + 132 C_1)$ with a fixed zeroth-moment coefficient C_0'. Hence

$$|O| \le (C_{\mathrm{phys}} + 24 C_1) x L^{-A}. \tag{B21}$$

Together, (B10), (B13), and (B21) give

$$|\mathcal{B}'|^2 \le 6 EVG^2 \left(2xL^{2C_W+344} + (C_{\mathrm{phys}} + 24C_1) x L^{-A}\right).$$

Since $EV \le C_{\mathrm{box}} x L^{-K_0}$, (B1) implies $|\mathcal{B}'| \le C' G x L^{-A}$ for a fixed C'.

Let $M_f \ge 1$ bound the L^1 norms of $\hat{f}_0, \hat{f}_1$. There are at most $40 \log L$ nonreserved slots. Integrating all slot frequencies and summing the two choices at each slot costs at most

$$(2M_f)^n \le L^{C_F}, \qquad C_F = 40 \log(2M_f). \tag{B22}$$

Thus (B9) and (B1) absorb the total loss $L^{C_E + C_F}$. Finally write $\psi_0(t) = \Psi(e^t)$ and restore the localization by Fourier inversion. The moments

$$M_0 = \int |\widehat{\psi}_0(s)|\, ds, \qquad M_H = \int |s|^H |\widehat{\psi}_0(s)|\, ds$$

are finite. The first controls bounded Mellin frequencies; for the tail choose H large enough and use the second. With the source-comparison error from the slot allocation, the resulting constant is bounded by $4 + C' M_0 + 20(2\pi)^H M_H$. This proves (B3) once (B18) has been established.

5 Cancellation in the marked correlation

We prove the estimate (B18) of Section 4. All the sums and operators below are finite. We use prime Siegel–Walfisz, the complete-cell discrepancy (C3), the high-frequency prime estimate (U1), smooth Fourier inversion, and the displayed finite energy estimates. The bilinear exponential-sum bound comes from the row sums of an additive Gram matrix and Cauchy–Schwarz.

Write $L = \log x$ and $\mathrm{e}(t) = e^{2\pi i t}$. We retain the actual marked groups and the multiplicity parity from Section 1. The BIG groups have $\log p \geq L^{3/10}$; the SMALL groups used for omissions have $\log p \leq L^{1/5}$. Their harmonic masses lie in $[v, v_{\max}]$, where $v = \frac{1}{4}\log(12/11)$ and $v_{\max} = 4$. We write these masses as V_g in this chapter; they are M_g in the definition of the weight. The fixed damping parameter is $q = 1/2$. The number of groups is $O(\log L)$. Constants implicit in this chapter may depend on the fixed accuracy, frequency bounds, and mark counts, but not on L, the shift, or the endpoints of a fine cell.

The proof estimates a signed difference between the original correlation and an arithmetic resampling term. A separate use of the fine-cell discrepancy then makes the resampling term small. There are three different spaces in this argument: a product of prime alphabets in the orthogonal-average argument, the residue and memory space, and the integer-position space. The local contraction is on the second space. Its relation to the third is a moment identity and an interval-counting estimate, not an identification of their norms. Throughout the operator argument write $s = 2k$ for the integer shift in (B17). A forward frame moves by sD, and a reverse frame by $-sD$.

5.1 Orthogonal averages and the alternating identity

In a BIG group P_g, use the probability law

$$V_g = \sum_{p \in P_g} \frac{1}{p}, \qquad \mu_g(p) = \frac{1}{V_g p}.$$

There are $M = J + \ell$ labelled coordinates in each group. Here J will be fixed later and the application has $\ell = 1$. Precisely, put $\mathcal{X} = \prod_{g \in \mathrm{BIG}} P_g^M$, with probability $\mu(x) = \prod_{g,i} \mu_g(x_{g,i})$, and use $\langle f, h \rangle_\mu = \sum_x \mu(x) \overline{f(x)} h(x)$. The tests before symmetrization are arbitrary functions $F, H : \mathcal{X} \to \mathbb{C}$. Set

$$(\mathcal{S}F)(x) = \frac{1}{(M!)^G} \sum_{\sigma \in \prod_g S_M} F((x_{g,\sigma_g(i)})_{g,i}), \qquad f = \mathcal{S}F, \quad h = \mathcal{S}H, \tag{M0}$$

where $G = \#\mathrm{BIG}$. Relabelling coordinates preserves μ, so Jensen gives $\|\mathcal{S}F\|_\mu \leq \|F\|_\mu$. The actual kernel independently averages its source and target permutations: changing variables by their inverses moves these two averages onto F and H, exactly. This identity, rather than a symmetry assumption on the arithmetic coefficients, licenses (M0). The tests can depend jointly on all groups. The maps (T14)–(T15) identify them as Fourier coefficients of the two physical endpoint tests.

Let E_i average coordinate i. These are commuting orthogonal projections. The Hoeffding components

$$H_S f = \prod_{i \in S} (I - E_i) \prod_{i \notin S} E_i f \quad (S \subseteq [M])$$

are orthogonal and sum to f. Symmetry gives equal squared norms to components of the same degree $|S|$. Choose disjoint sets of m probe coordinates and ℓ trailing coordinates. At degree j, the fraction of components containing every probe and no trailing coordinate is

$$\frac{\binom{M-m-\ell}{j-m}}{\binom{M}{j}} \le \binom{m+\ell}{m}^{-1}.$$

Consequently

$$\Big\| \prod_{i\in\text{probe}} (I - E_i) \prod_{i\in\text{tail}} E_i f \Big\|_2^2 \le \binom{m+\ell}{m}^{-1} \|f\|_2^2. \tag{M1}$$

Apply this successively to different groups, using Fubini. For $\ell = 1$ and s groups the factor is $(m+1)^{-s}$. The actual BIG groups contain two disjoint blocks $\mathcal{A}, \mathcal{B}$, each of size at least $c \log L$, where $c = (120 \log 2)^{-1}$. Given $A_0 > 0$, choose a fixed m so that $A_0 \le c \log(m+1)$. Then each block gives squared-energy saving L^{-A_0}.

The coupled phase cannot be moved through these coordinate averages. To retain it exactly, put

$$R(f,h) = \mathbb{E}[\bar{f}\, \Phi h], \qquad |\Phi| = 1,$$

For a fixed positive SMALL divisor d, orientation $o \in \{1,-1\}$, and real ζ, θ, its precise value is

$$D(x) = d \prod_{g\in\text{BIG}} \prod_{i=0}^{J-1} x_{g,i}, \qquad \Phi(x) = D(x)^{io\zeta} \mathrm{e}(osD(x)\theta). \tag{M0a}$$

Thus $|\Phi| = 1$, and Φ ignores the ℓ trailing coordinates. It need not ignore the probes. Let U average all trailing coordinates, and set

$$E_a = U \prod_{i\in a} E_i, \qquad P_{\mathcal{A}} = \sum_{a\subseteq \mathcal{A}\times[m]} (-1)^{|a|} E_a = U \prod_{i\in \mathcal{A}\times[m]} (I - E_i).$$

Define $P_{\mathcal{B}}$ in the same way. Expanding the two products gives the exact identity

$$\begin{aligned} R(Uf,Uh) - \sum_{\substack{\emptyset\ne a\subseteq \mathcal{A}\times[m]\\ \emptyset\ne b\subseteq \mathcal{B}\times[m]}} (-1)^{|a|+|b|} R(E_a f, E_b h) \\ = R(P_{\mathcal{A}} f, Uh) + R(Uf, P_{\mathcal{B}} h) - R(P_{\mathcal{A}} f, P_{\mathcal{B}} h). \end{aligned} \tag{M2}$$

Both subsets in the comparison sum are nonempty. Each term on the right has a contracted endpoint; since $|\Phi| = 1$, Cauchy–Schwarz and (M1) give

$$|\text{right side of (M2)}|^2 \le 9 L^{-A_0} \|f\|_2^2 \|h\|_2^2. \tag{M3}$$

The number of comparison patterns is a fixed power of L: m is fixed and there are $O(\log L)$ groups. We will pay this cost in the arithmetic approximation to each pattern. It is not subtracted from the exponent supplied by the already summed identity (M2).

5.2 Implementing the averages by characters

For a nonempty collection of probes, let D be their product under the product probability law. Partition its logarithm into cells $I_j = [e^{jh}, e^{(j+1)h})$, put $\nu_j = \Pr(D \in I_j)$, and retain cells with $\nu_j \geq \xi$. The arithmetic kernel is

$$K(D,X) = \sum_{\nu_j \geq \xi} \frac{\mathbf{1}_{D,X \in I_j}}{\nu_j} \sum_{\substack{\chi \text{ primitive} \\ \operatorname{cond}\chi \leq Q}} \overline{\chi(D)}\chi(X). \tag{M4}$$

The reference operation is exact matching of the product, not the constant kernel. If $\pi(X) = \Pr(D = X)$, put

$$I_\pi(D,X) = \frac{1_{D=X}}{\pi(X)}, \qquad \sum_D \pi(D) I_\pi(D,X)\phi(D) = \phi(X) \quad (\pi(X) > 0). \tag{M4a}$$

Set the kernel to zero when $\pi(X) = 0$; such X do not occur in the product law. Applying (M4a) separately in the two probe blocks sets the auxiliary products equal to the target first-probe and source second-probe products. The other, freed endpoint coordinates are then averaged with their original laws. This is the ordinary resampling form in (M2). In particular, it evaluates the phase at the spliced products; it does not average that phase independently. The estimate below compares this matching form with (M4), including its phase and endpoint tests; it does not assert that their kernels are close in operator norm.

All marked primes exceed Q. Characters in (M4) thus have modulus one on its support. The cell Gram matrix has diagonal ν_j. For two different primitive characters, freeze every prime coordinate but one. The condition on the product clips that remaining prime to an interval. The quotient of the two characters induces a nonprincipal primitive character with conductor at most Q^2. Prime Siegel–Walfisz and partial summation bound the corresponding entry by

$$|G_j(\chi,\chi')| \leq (K_1/V_g)L^{-B_1}.$$

The other coordinates have total probability one. Summing over them does not introduce the number of tuples. There are at most Q^2 characters. Therefore, if $Q \leq L^{C_q}$ and $\xi^{-1} \leq L^{C_\xi}$, requesting accuracy $B_1 = B + C_\xi + 2C_q$ gives

$$\|K\|_{2\to 2} \leq 1 + (K_1/v)L^{-B}. \tag{M5}$$

Different cells give orthogonal blocks of this operator, so their number does not multiply (M5).

To compare (M4) with ordinary resampling, retain the full phase $\mathrm{e}(\theta' XY)(XY)^{i\zeta}$. On a major arc $\theta' = a/r + \beta$, $r \leq Q$, the function $\mathrm{e}(aXY/r)$ on units is a sum of primitive-character modes with coefficient ℓ^1 norm at most r. The two kernels reproduce these modes up to

$$(2\epsilon + \epsilon^2)r, \qquad \epsilon = (K_1/v)L^{-B}.$$

Within two matched logarithmic cells, the remaining phase varies by at most

$$Q^4(|\zeta| + 2\pi|\beta|R_0)(h_A + h_B), \qquad R_0 = e^{(j_A+1)h_A + (j_B+1)h_B}. \tag{M6}$$

Here R_0 bounds the product of the two cell variables. This follows by differentiating the phase in the two logarithmic coordinates. The factor Q^4 bounds the two finite character sums.

On minor arcs, move the two kernels onto the endpoint tests and use (M5), eventually with norm at most 2. The finite integer bilinear exponential-sum inequality on dyads of sizes M_0, N_0 has spectral factor bounded by

$$\frac{1}{r} + \frac{2}{N_0} + \frac{\log(2r)}{M_0} + \frac{r\log(2r)}{M_0N_0}. \tag{M7}$$

For example, for windows of integer lengths $P \le M_0$, $Q' \le N_0$, the bound preceding (M7) is $[P + (1 + Q'/r)(P + r\log(2r))]/(M_0N_0)$. Thus (M7) is at most $5L^{-D}$ provided

$$r, N_0 \ge L^D, \quad M_0 \ge L^D\log(2r), \quad M_0N_0 \ge L^D r\log(2r). \tag{M8}$$

Here is a denominator choice that verifies this alternative. Put $d_* = 2/5$, $S_D = L^{D+d_*+2}$, and apply Dirichlet approximation with cutoff $\lfloor M_0N_0/S_D \rfloor$. It gives a reduced a/r with

$$r \le M_0N_0/S_D, \qquad |\beta|M_0N_0 \le S_D/r, \qquad |\beta| \le r^{-2}.$$

Each nonempty BIG probe product is at least $e^{L^{3/10}}$ and has logarithm $O_m(L^{2/5}\log L)$; in particular $\log(2M_0N_0) \le L^{d_*+2}$ eventually. If $r > Q \ge L^D$, these bounds give all four inequalities in (M8). If $r \le Q$, the matched cells satisfy $R_0 \le 4e^{h_A+h_B}M_0N_0$, so $|\beta|R_0$ has a fixed polynomial bound. This supplies the major-arc variation estimate with explicit control of its denominator.

Passing from labelled prime tuples to their products costs at most

$$\prod_g \frac{k_g!}{V_g^{k_g}}, \tag{M9}$$

because a product in a group has at most $k_g!$ ordered representations. Here $k_g \le m$, so (M9) is another fixed power of L. After including it in the requested minor-arc precision, the source bounds for the raw term and the two-kernel term are, respectively, $125L^{-B}\|f\|_2^2\|h\|_2^2$ and $2000L^{-B}\|f\|_2^2\|h\|_2^2$. Their squared difference is bounded by $4250L^{-B}\|f\|_2^2\|h\|_2^2$.

If the two product ranges meet n_A, n_B cells, the probability lost on discarded cells is at most $2(n_A + n_B)\xi$. To bound their number, each probe product has logarithm at most $B_L = mC_GL^{2/5}\log L$, where $C_G\log L$ bounds the BIG group count. Consequently

$$n_A, n_B \le 2 + B_L/h, \qquad \#\{\text{coarse dyads for either product}\} \le 2 + B_L/\log 2. \tag{M9a}$$

These are polynomial bounds for the mesh (M10), uniform in the conditioned shared coordinates. They apply also when the product range is further clipped. The number of patterns is at most $2^{m(|\mathcal{A}|+|\mathcal{B}|)}$, also polynomial. Shared coordinates are conditioned on and then integrated by Cauchy–Schwarz. For a requested one-rectangle saving B_{save}, let C_{factor} pay for (M9). The following choices suffice:

$$\begin{gathered} D \ge B_{\text{save}} + 2C_{\text{factor}}, \qquad C_s \ge D + d_* + 2, \\ C_4 \ge C_s + 1, \quad C_4 \ge C_{\text{freq}}, \\ H_m \ge B_{\text{save}} + 4C_q + C_4, \qquad C_\xi \ge B_{\text{save}} + H_m + d_* + 4. \end{gathered}$$

Here $|\zeta| \le L^{C_{\mathrm{freq}}}$ and C_q bounds the conductor cutoff. Request precision $B_{\mathrm{save}} + C_q$ in the rational-phase reproduction to pay for $r \le L^{C_q}$; the prime Siegel–Walfisz precision is then increased by $C_\xi + 2C_q$ as in (M5). Any fixed constants can be absorbed by an extra power and a larger threshold. After paying the pattern and dyad counts in B_{save}, (M3)–(M9) give the small local signed-operator estimate with

$$h = \frac{\log 2}{\lceil 4L^{H_m} \rceil}, \qquad Q = \lceil 4L^{D+1} \rceil, \qquad \xi = L^{-C_\xi}. \tag{M10}$$

Choose $C_q > D + 1$ after D and before H_m, C_ξ.

5.3 The arithmetic state space and the local contraction

5.3.1 Residues, active marks and memory

Fix $N = 2\lceil L^{13/25} \rceil$, $M = J + 1$, and finite cutoffs B_g for the BIG memory lengths. The estimates below are uniform in these cutoffs. For a SMALL group put

$$\mathbb{T}_g = \prod_{p \in P_g} \mathbb{Z}/p\mathbb{Z}, \qquad \mathcal{S}_g = \{(y, a) : y \in \mathbb{T}_g,\ a : [M] \hookrightarrow P_g,\ y_{a_i} = 0\ (i \in [M])\}. \tag{T1}$$

Here and below $[M] = \{0, \dots, M-1\}$. Give every element of $\mathcal{S}_g$ mass $|\mathbb{T}_g|^{-1} V_g^{-M}$, and write $\omega_g(y) = \#\{p : y_p = 0\}$. This is Haar probability in y and counting measure in the valid ordered marks, with the displayed factor V_g^{-M}; it is not a probability on the marked fibre. Translation means $y_p \mapsto y_p + \Delta \pmod p$ simultaneously for every p.

For BIG primes set

$$q_i = \begin{cases} \sqrt{q}, & i = 0, N, \\ q, & 0 < i < N, \end{cases} \qquad \eta_i = 1 - q_i, \quad H_N = \sum_{i=0}^{N} \eta_i, \quad b_p = 1 - H_N/p, \quad \tilde{\mu}_g(p) = \frac{1}{V_g p b_p}. \tag{T2}$$

Eventually $b_p \ge 1/2$. A BIG state consists of an active array $a_g \in P_g^M$ and an ordered list $((p_1, r_1), \dots, (p_j, r_j))$, $0 \le j \le B_g$, where $r_i \in \mathbb{Z}/p_i\mathbb{Z}$. Its mass is

$$\left(\prod_{i<M} \tilde{\mu}_g(a_{g,i}) \right) \frac{1}{j!} \prod_{i=1}^{j} \tilde{\mu}_g(p_i). \tag{T3}$$

The residues in this list have **counting** measure: summing the mass of (p, r) over r gives $p\tilde{\mu}_g(p)$, while its zero residue has mass $\tilde{\mu}_g(p)$. Equal primes and equal atoms are allowed here. The active BIG array also has no distinctness condition at this stage. Distinctness is imposed on the appropriate birth lists in the transition and then globally in the forest expansion.

Let $\widetilde{\mathcal{S}}$ be the product of (T1) and these BIG spaces, with mass w. Let π replace each ordered OLD list by its multiset, leaving all SMALL states and active arrays unchanged. On $\mathcal{S} = \pi(\widetilde{\mathcal{S}})$ put

$$\bar{w}(X) = \sum_{\pi(x) = X} w(x), \qquad \mathcal{E}(F) = \sum_{X} \bar{w}(X) |F(X)|^2, \qquad (\Pi u)(X) = \frac{\sum_{\pi(x)=X} w(x) u(x)}{\bar{w}(X)}. \tag{T4}$$

A zero-mass fibre is assigned zero. In our nonempty fibres the masses are positive. Pullback $\pi^* F = F \circ \pi$ is an isometry, $\Pi\pi^* = I$, and Jensen gives $\mathcal{E}(\Pi u) \le \sum_x w(x)|u(x)|^2$. For a multiset with atom multiplicities n_z, the OLD part of its mass is $\prod_z \tilde{\mu}_g(p(z))^{n_z} / \prod_z n_z!$. This follows by counting its $j!/\prod_z n_z!$ ordered representatives in (T3). In particular, passing to multisets never discards collisions.

An OLD atom records a previously introduced prime and its current residue. On a step $\Delta = osD$ all surviving residues become $r + \Delta \pmod p$. A *promotion* supplies an unshared target active slot from an OLD occurrence whose translated residue is zero; distinct promoted slots use distinct OLD indices. A *store* puts an unshared source active occurrence into OLD, at residue $\Delta \pmod p$. Stored and promoted primes must not divide D. The remaining unshared target slots are new births sampled with $\tilde{\mu}_g$. The output length must stay below B_g. A transition is *clean* when it has no stores and no promotions. Clean transitions translate every OLD occurrence, including repetitions.

For definiteness the active operator is the following finite sum. Choose the raw pattern or a nonempty pair (a, b) of probe subsets, auxiliary primes with the original harmonic law μ, and two independent uniform permutations in each SMALL and BIG group. Copy the shared active coordinates, use the divisor D of (M18b), and sum all legal store/promotion choices just described. In each BIG group the coefficient is

$$\rho_0^{z_g + z'_g} V_g^{-t_g} \prod_{i \text{ fresh at the target}} \tilde{\mu}_g(a'_{g,i}), \tag{T5}$$

where z_g, z'_g count zero-residue OLD occurrences at the two endpoints and t_g is the number of promotions. Choose $q^{1/4} < \rho_0 < 1$. In each SMALL group multiply by $V_g^{-1} q^{(\omega_g(y) + \omega_g(y') - 2M)/2}$, imposing shared marks and $y' = y + \Delta$. Finally multiply by the auxiliary mass, the permutation masses, and $K_f D^{i\zeta} \mathrm{e}(sD\theta)$ in the forward orientation; the reverse orientation conjugates $K_f D^{i\zeta}$ and changes θ to $-\theta$ in the return phase. There is only one such complex scalar for the full frame.

The operator E_o is this row sum, pulled back and averaged by (T4), with the two gates $|\mathrm{OLD}| \le T = \lceil L^2 \rceil$ and the condition that the list of fresh target numerical primes is distinct. Only distinctness between different birth times is omitted. These are the operators denoted E_i in (F7), with alternating o. The integer-position matrix (M18c) has a different carrier and will be related to their full path sum in the residue-to-interval transfer.

5.3.2 The frozen BIG form, including its errors

We first identify the harmonic form bounded by (M1)–(M10). Freeze the SMALL shared tuple and its product $d > 0$. For a pattern ν, let $S_g \subset [J]$ be its shared coordinates. For source and target arrays u, v and permutations σ, τ, put

$$B_\mu(u, v) = \prod_g \left[\prod_{i \in S_g} 1_{u_{g,\sigma_g(i)} = v_{g,\tau_g(i)}} \prod_{i \notin S_g} \mu_g(v_{g,\tau_g(i)}) \right]. \tag{T6}$$

In this formula the complement is in $[M]$, so it includes the trailing coordinate. Multiply (T6) by the auxiliary mass, the two permutation masses, the two-endpoint auxiliary ban, and the single oriented scalar from (T5); sum over ν, auxiliaries and permutations to obtain a kernel $\mathsf{B}_\mu(d, \zeta, \theta, o, s)$. Its pairing is $\sum_{u,v} \mu(u) \overline{F(u)} \mathsf{B}_\mu(u, v) H(v)$. This defines the frozen form for arbitrary tests; no factorization of F or H between groups is required.

Here is the identity connecting it to (M2). Temporarily remove the auxiliary ban. In a comparison pattern replace the two character kernels by ordinary probe resampling. Use the shared-copy indicators to identify the unchanged coordinates. At a probe belonging to the first subset use the target value, at a probe belonging to the second use the source value, and integrate the freed counterpart. The trailing coordinates at both endpoints are independently integrated. The divisor in the phase is the head product of this spliced array, so the resulting pairing is exactly $-(-1)^{|a|+|b|}R(E_a\delta F, E_b\delta H)$. The empty pattern has the opposite sign and gives $R(U\delta F, U\delta H)$. Changing variables by the independent endpoint permutations proves

$$\mathsf{B}_\mu(F,H) = \mathsf{R}(F,H) + \mathsf{C}(F,H) - \mathsf{Z}(F,H), \tag{T7}$$

where R is the right side of (M2), C is the sum of (character-resampling minus ordinary-resampling) comparison forms without the ban, and Z is the character form on auxiliaries meeting either endpoint. This is an identity of signed forms, before taking absolute values. In particular, the phase is never commuted through a coordinate average.

Equations (M4)–(M10) bound $|\mathsf{C}|^2$ by $L^{-A}\|F\|_\mu^2\|H\|_\mu^2$ at any fixed accuracy A. The same bound holds for Z eventually: for a fixed endpoint array, the probability that one of at most mG auxiliaries hits its MG marked values is at most $mMG^2/(ve^{L^{3/10}})$. Split the ban into its source and target parts and apply weighted Cauchy–Schwarz to each part. The unrestricted marginal at the other endpoint is the original product law. The pointwise kernel cap and pattern sum are polynomial in L, so this exponential atom saving absorbs them. This argument keeps the tests in their original L^2 norms; it does not bound a test on a rare set by its supremum. Together with (M3) and $|a+b-c|^2 \le 3(|a|^2+|b|^2+|c|^2)$, (T7) gives the explicit squared bound $33L^{-A}$ times the two energies. To obtain exponent $2E$, use $A = 2E+2$. The reduction to $d = s = 1$ replaces θ by $ds\theta$ and introduces the unit scalar $d^{io\zeta}$; hence this bound is uniform in $d > 0$, $s \neq 0$ and $\theta \in \mathbb{R}$.

The density in (T6) must next be changed to (T2). This is done with one multiplier common to every pattern and permutation. Put

$$r_g(p) = b_p^{-1/2}, \qquad (\mathcal{A}F)(u) = \prod_{g,i} r_g(u_{g,i})F(u). \tag{T8}$$

Then $\|\mathcal{A}F\|_\mu = \|F\|_{\tilde\mu}$. The precise ratio of the tilted joint law to the harmonic joint law, after this change at both endpoints, is

$$R_\nu(u,v;\sigma,\tau) = \prod_g \prod_{i\notin S_g} r_g(u_{g,\sigma_g(i)})r_g(v_{g,\tau_g(i)}). \tag{T9}$$

For a shared coordinate the indicator in (T6) makes the two endpoint square roots supply its full ratio; for a free coordinate an additional square root is needed at each end. This verifies (T9) coordinate by coordinate. Auxiliary draws still use μ, not $\tilde\mu$.

Let $\mathsf{B}_{\tilde\mu}^{\mathrm{nd}}$ include the fresh-birth list condition in E_o, and let $\mathsf{B}_\mu^{\mathrm{bad}}$ be the same harmonic form restricted to its failure. The exact identity is

$$\mathsf{B}_{\tilde\mu}^{\mathrm{nd}}(F,H) = \mathsf{B}_\mu(\mathcal{A}F,\mathcal{A}H) + \mathsf{T}(F,H) - \mathsf{B}_\mu^{\mathrm{bad}}(\mathcal{A}F,\mathcal{A}H), \tag{T10}$$

where T uses the multiplier $(R_\nu - 1)$ and retains the fresh-birth condition. Set $P = e^{L^{3/10}}$, $a_0 = 1 - H_N/P$, $I = MG$, and

$$\delta_{\mathrm{tilt}} = a_0^{-I} - 1, \qquad \mathscr{K} = 1 + (2^{m|\mathcal{A}|} - 1)(2^{m|\mathcal{B}|} - 1)Q^4\xi^{-2}. \tag{T11}$$

The raw pattern has cap one and every other pattern has cap $Q^4\xi^{-2}$, so $\mathscr{K}$ is their total cap. From (T9), $1 \le R_\nu \le a_0^{-I}$; normalized auxiliary and permutation sums and the two marginals of (T6) therefore give

$$\begin{aligned} |\mathsf{T}(F,H)|^2 &\le (\mathscr{K}\delta_{\mathrm{tilt}})^2 \|F\|_{\widetilde{\mu}}^2 \|H\|_{\widetilde{\mu}}^2, \\ |\mathsf{B}_\mu^{\mathrm{bad}}(\mathcal{A}F, \mathcal{A}H)|^2 &\le \mathscr{K}^2 \frac{I^2}{vP} \|F\|_{\widetilde{\mu}}^2 \|H\|_{\widetilde{\mu}}^2. \end{aligned} \tag{T12}$$

For the second bound, expose the fresh coordinates in (T6). Any pair coincides with conditional probability at most $1/(vP)$; there are at most I^2 pairs. The restricted row has this saving and the full column has mass one. Weighted Schur gives the displayed squared bound, valid even for tests concentrated on the bad set. Since $H_N \le 5L$, $I = O_J(\log L)$, and $\mathscr{K} \le L^{C(m,C_q,C_{\bar{\xi}})}$, both errors in (T12) are smaller than any inverse power of L. Applying the three-term square inequality to (T10) gives $9L^{-2E}$ times the tilted endpoint energies, after choosing the harmonic precision. This is the full tilted BIG estimate, including the ban and the actual list filter.

5.3.3 The SMALL maps and the translated OLD residues

It remains to transfer this bound to (T4). Fix the OLD lengths j_g and the ordered OLD primes π_g, retaining their residues as variables. On this sector use the finite abelian group

$$\mathbb{T} = \prod_{g \in \mathrm{SMALL}} \mathbb{T}_g \times \prod_{g \in \mathrm{BIG}} \prod_{i=1}^{j_g} \mathbb{Z}/\pi_{g,i}\mathbb{Z}. \tag{T13}$$

Repeated OLD primes give separate coordinates. The translation $t \mapsto t + \Delta$ acts diagonally on every occurrence; this is exactly the clean OLD update. For a SMALL permutation σ and shared tuple $d = (d_{g,i})_{g,i<J}$, define

$$(P_{\sigma,d}U)(a;y,r) = \prod_{g \in \mathrm{SMALL}} V_g^{-1/2} q^{\max(\omega_g(y)-M,0)/2} \sum_{\substack{(y,c) \in \prod_g \mathcal{S}_g \\ c_{g,\sigma_g(i)} = d_{g,i}\ (i<J)}} U((y,c),a,r). \tag{T14}$$

A fibre with no valid marks contributes zero. These are conditional sums, not conditional probabilities. In the clean coefficient the SMALL scalar is exactly the product of the two half factors in (T14). The shared tuple has the same value at the two endpoints, and their residues differ by $\Delta = osD$. Partitioning both endpoint sums by (d,y,r) gives precisely the pairing of $P_{\sigma,d}U$ with the translated $P_{\tau,d}F$, with the frozen BIG kernel between them. The displacement still depends on d, the BIG shared marks and the auxiliaries. It has not been frozen before this disintegration.

Use normalized Fourier coefficients, so that

$$\hat{u}(\chi) = \frac{1}{|\mathbb{T}|} \sum_t u(t)\overline{\chi(t)}, \qquad u(\widehat{\cdot + \Delta})(\chi) = \chi(\Delta)\hat{u}(\chi).$$

Parseval replaces translation by multiplication by a character. Write $\chi(\Delta) = \mathrm{e}(\vartheta_\chi \Delta)$; the frozen BIG block now has frequency $\theta + \vartheta_\chi$. The bound above is uniform in this frequency, however large the OLD sector. For this fixed prime sector put

$$c_\pi = \prod_{g \in \mathrm{SMALL}} V_g^{-M} \prod_{g \in \mathrm{BIG}} \frac{\prod_{i=1}^{j_g} \pi_{g,i} \tilde{\mu}_g(\pi_{g,i})}{j_g!}.$$

The factors $\pi_{g,i}$ convert OLD counting measure to normalized Haar measure on (T13). The exact clean pairing in this sector is

$$c_\pi \, \mathbb{E}_{\sigma,\tau} \sum_d \sum_{\chi \in \widehat{\mathbb{T}}} \mathsf{B}^{\mathrm{nd}}_{\tilde{\mu}}(\widehat{P_{\sigma,d}U}(\chi), \widehat{P_{\tau,d}F}(\chi); d_S, \zeta, \theta + \vartheta_\chi, o, s), \quad d_S = \prod_{g,i<J} d_{g,i}. \tag{T15}$$

The two arguments of B are functions of the whole BIG active array a. After (T8) and (M0) they are the tests f, h of the orthogonal-average argument. Formula (T15) is the required operator identity: it keeps the full joint phase, both endpoint permutations, all shared values, and every translated OLD occurrence.

There is no loss equal to $|\mathbb{T}|$ in its norm estimate. For a fixed y, d, the number of completions in one SMALL group is $(\omega_g(y) - J)_1$; for ℓ omitted marks it is $(\omega_g(y) - J)_\ell$. Cauchy–Schwarz in that fibre, followed by summation over d and Parseval, bounds each lifted endpoint energy by C_{cond} times its original sector energy, where

$$C_{\mathrm{cond}} = \prod_{g \in \mathrm{SMALL}} \frac{\ell!}{(1-q)^{\ell+1} V_g^\ell}, \qquad \ell = 1. \tag{T16}$$

Indeed, on a nonempty fibre $\omega_g = J + \ell + z$, and $q^z (z+\ell)_\ell \le \sum_{z \ge 0} q^z (z+\ell)_\ell = \ell!/(1-q)^{\ell+1}$. Every original ordered mark state occurs once when its shared tuple is summed. Cauchy–Schwarz over (d, χ) and then over sectors yields

$$|\langle U, E^{(1)}_{\mathrm{clean}} F \rangle|^2 \le 9 L^{-2E_0} C^2_{\mathrm{cond}} \, \mathcal{E}(U) \mathcal{E}(F). \tag{T17}$$

Here the superscript means that the OLD endpoint parameter is one. For general $\rho_0 \le 1$, multiply both tests by $\rho_0^{\#\text{zero OLD}}$; these multipliers and both OLD gates are contractions. Pullback and orbit averaging in (T4) are contractions as well. To pass from a pairing bound to an operator bound, set the first test equal to the output of the operator and cancel its norm (the zero output is immediate).

If $C_G \log L$ bounds the number of SMALL groups and $V_g \ge v$, then $C_{\mathrm{cond}} \le L^{c_{\mathrm{cond}}}$, with $c_{\mathrm{cond}} = C_G \log \max(1, \ell!/((1-q)^{\ell+1} v^\ell))$. Choose $E_0 = E + c_{\mathrm{cond}} + 1$ before choosing m or J. For $L \ge 3$, (T17) is at most L^{-2E} times the two energies. This proves the clean part of (F7), with all normalizations specified.

5.3.4 Promotions, stores and the full transition

Fix an OLD BIG prime p, its residue r, the BIG divisor D_B, and an orientation $o \in \{\pm 1\}$. Let F be the product of all SMALL marks and U the product of the omitted ℓ marks in each SMALL group. The retained SMALL divisor is $D_S = F/U$. Promotion requires

$$r + o s D_S D_B \equiv 0 \pmod{p}, \quad \text{or equivalently} \quad rU + o s D_B F \equiv 0 \pmod{p}. \tag{M11}$$

The constant term is nonzero: $p \nmid D_B F$ and $0 < |s| < p$. Furthermore

$$\log U \le \ell\, \#\mathrm{SMALL}\, L^{1/5} = O(\log L\, L^{1/5}) < L^{3/10} \le \log p.$$

Thus $0 < U < p$, and (M11) determines at most one U. Distinct marks within each group and disjointness of the groups determine the omission tuple from U. Uniform coordinate permutations induce uniform omission subsets. Hence

$$\Pr(\text{an OLD promotion}) \le \frac{\#\mathrm{OLD}}{\binom{J+\ell}{\ell}^{\#\mathrm{SMALL}}}. \tag{M12}$$

In the application $\ell = 1$, $\#\mathrm{OLD} \le \lceil L^2 \rceil$, and $\#\mathrm{SMALL} \ge c_0 \log L$ with $c_0 = (40 \log 2)^{-1}$. The right side is at most $2L^{2-c_0 \log(J+1)}$. If L^κ bounds the complete row cost, choose $J \ge m$ so that

$$2H_0 + 2\kappa + 3 \le c_0 \log(J+1). \tag{M13}$$

This makes the promotion row rare enough for the squared-energy saving L^{-2H_0}. Reverse the transition orientation to obtain the same estimate for storing an active label: it is a rare column. The weighted Schur inequality combines each rare row or column bound with the full complementary row bound, then adds the two exceptional operators. Increasing the fixed precision pays their constant sum. This is why J is chosen after the mesh, conductor, and pattern parameters.

The clean and exceptional parts sum to the literal operator E_o. Apply their squared estimates at precision $E+1$. The inequality $\|u+v\|^2 \le 2\|u\|^2 + 2\|v\|^2$ gives

$$\mathcal{E}(E_o F) \le 4L^{-2(E+1)} \mathcal{E}(F) \le L^{-2E} \mathcal{E}(F). \tag{T18}$$

The exceptional estimate did not require cancellation. The same positive row and column bounds, without the rare-event factor, apply if an arbitrary coefficient of modulus at most one is attached to each expanded transition. Thus, for a fixed exponent κ,

$$\mathcal{E}(E_o^{\phi} F) \le L^{2\kappa} \mathcal{E}(F), \qquad |\phi| \le 1. \tag{T19}$$

The exponent is independent of J; its formula is given below. This distinction is needed in the forest proof: a sign multiplier at a collision owner uses (T19), not (T18).

5.3.5 Background transitions and the constants in the moment bound

At visit i, the background operator G_i leaves the SMALL state and all active BIG marks fixed. In one BIG group let Z be the indices of OLD atoms at residue zero. Choose $A \subset Z$, delete those OLD occurrences, and append b new atoms $(p_1, 0), \dots, (p_b, 0)$. The coefficient of this choice is exactly

$$\left(-\frac{\eta_i}{\rho_0}\right)^{|A|} \left(\frac{q_i}{\rho_0^2}\right)^{|Z|-|A|} \frac{1}{b!} \left(-\frac{\eta_i V_g}{\rho_0}\right)^{b} \prod_{j=1}^{b} \tilde{\mu}_g(p_j). \tag{T20}$$

Sum over these choices, with the finite output cap, then take the product over groups and the orbit average (T4). These appended occurrences are *background births*. A later use of an existing occurrence is not a birth. Formula (T20) explains both the signs and the

single birth factorial. The OLD factorial in (T3) is the Hilbert measure and is not inserted a second time in an edge coefficient. The powers of ρ_0 in (T5) and (T20) cancel at successive endpoints; they are an auxiliary norm device, not a change to the original residue density.

Here are explicit constants for (F7)–(F11). They also distinguish the row constant used in removing the OLD cutoff from the smaller conditional-map constant (T16). For $d_0 = \ell + m$ put

$$\begin{aligned} C_{\mathrm{prom}}(\rho_0, d_0) &= \sum_{j=0}^{d_0} \frac{j!}{(1-\rho_0)^{j+1}}, \\ C_{\mathrm{small}} &= \prod_{g \in \mathrm{SMALL}} \frac{\ell!}{V_g^{\ell}(1-\sqrt{q})^{\ell+1}}, \\ S_{\mathrm{small}} &= \sum_{x \in \prod_{g \in \mathrm{SMALL}} \mathcal{S}_g} \prod_{g} \frac{1}{|\mathbb{T}_g| V_g^M}, \\ a &= \mathcal{K} C_{\mathrm{small}} \prod_{g \in \mathrm{BIG}} \left[w_0^{d_0} 4^{d_0} C_{\mathrm{prom}}(\rho_0, d_0) \max(1, V_g^{-1})^{d_0} 2^{d_0} \right]. \end{aligned} \tag{T21}$$

Here $\mathcal{K}$ is (T11), and w_0 is specified in the history restoration. Thus S_{small} is a finite total mass, not a universal constant or a probability normalization. Both C_{small} and a depend on the actual group masses and L; their polynomial **exponents** are fixed. The bounds on V_g and the number of groups make $\mathcal{K} C_{\mathrm{small}} \le L^{C'_E}$ and $a \le L^{C_b}$ eventually. C'_E here is a row-budget exponent, unrelated to the ordinary-slot exponent C_E in (B1).

An explicit exponent for (T19) can be read without choosing J. Choose C_G so that both group counts and $|\mathcal{A}| + |\mathcal{B}|$ are at most $C_G \log L$ and put

$$\begin{aligned} R_v &= \max(1, v_{\max}, v^{-1}), \\ B_{\mathrm{row}} &= \max\{1, 4^{d_0} C_{\mathrm{prom}}(\rho_0, d_0) \max(1, v^{-1})^{d_0} 2^{d_0}\}, \\ \kappa &= m C_G \log 2 + 4 C_q + 2 C_\xi + 1 \\ &\quad + C_G \log \max\left(1, \frac{\ell!}{(1-\sqrt{q})^{\ell+1} v^{\ell}}\right) + C_G \log B_{\mathrm{row}} + \frac{d_0 C_G}{2} \log R_v. \end{aligned} \tag{T22}$$

To see the uniformity, at most d_0 slots per group are unshared. If z OLD occurrences have residue zero, choosing $j \le d_0$ of them costs at most $(z)_j \rho_0^{z-j} \le C_{\mathrm{prom}}$. The SMALL row is bounded by the same falling-factorial calculation as (T16), with $\sqrt{q}$ in place of q. The number of patterns and the character caps give the first line of the exponent in (T22). The comparison of forward and reverse harmonic masses contributes at most $R_v^{d_0 G}$ to the squared bound. Weighted Schur, followed by (T4), gives (T19). Applying the rare-row estimate (M12) to this same calculation gives (M13), so J can indeed be chosen afterwards.

The background estimate is uniform even in unit multipliers on its individual choices. For W_0, w_0, ρ_0 as below, summing (T20) in absolute value with weight $w_0^{|\mathrm{OLD}|}$ uses

$$\frac{1}{\rho_0 w_0} + \frac{\sqrt{q}}{\rho_0^2} \le 1, \qquad \sum_{p \in P_g} \tilde{\mu}_g(p) \le 2.$$

The deletion sum is binomial and the birth sum is exponential. In the transposed sum the factorials in (T3) give the corresponding binomial coefficients; hence the same weighted Schur argument gives $\mathcal{E}(G_i^{\phi} F) \le \Gamma^2 \mathcal{E}(F)$, with $\Gamma = \exp(4 W_0 w_0 G / \rho_0)$. Set

$$l = L^{-S_0}, \qquad h = L^{\kappa}, \qquad b = L^{S_0+\kappa}, \quad K_r = S_0 + \kappa, \qquad P_0 = \left| \prod_{g \in \mathrm{BIG}} \prod_{p \in P_g} b_p \right| \mathcal{E}(v) \Gamma^{N+1}, \tag{T23}$$

where v is one on states with empty OLD memory and zero otherwise. The initial distinct-active mask is retained in the history sum and can only reduce its boundary energy. Equations (T18)–(T23) supply all three estimates in (F7); the forest argument below restores the omitted distinctness between different birth times.

5.4 Restoring all histories

Equalities between prime labels in a history are indexed by forests. The exact signed expansion, its coefficients, and the two bounds below are proved in Section 7. Its random-sign factorization explains why each identification loses the local contraction at at most two visits. For $N = 2\lceil L^{13/25} \rceil$, the number of available labels is bounded by

$$\mathcal{A} = (N+1)\#\mathrm{BIG}(J + 1 + \lceil L^2 \rceil) \le L^4.$$

A forest of rank r requires r identifications and there are at most $\mathcal{A}^{2r}$ such forests. The expansion has two bounds for every forest contribution F_r:

$$|F_r| \le \min\{P_0 (L^{-S_0})^N b^{2r},\ P_0 a^N \delta^r\}, \tag{M15}$$

where

$$P_0 \le L^{(C_v+2)N}, \qquad b \le L^{K_r}, \qquad a \le L^{C_b},$$
$$S_0 = E + C_v + 5, \qquad \delta = \frac{2}{v e^{L^{3/10}}}.$$

The first bound uses the local signed contraction at each step; each identification replaces at most two local factors by their row bounds. The second uses absolute row bounds everywhere and charges one factor δ to each independent equality of BIG prime labels. This is a finite reindexing of the same history expansion.

Split at $r = \lceil NL^{-3/20} \rceil$. Below the split, the exponent cost $(8+2K_r)r$ from the forest count and b^{2r} is at most N eventually. The first bound in (M15) is therefore at most $L^{-(E+2)N}$ after counting forests of that rank. Above the split,

$$(\mathcal{A}^2\delta)^r \le e^{-rL^{3/10}/2} \le e^{-NL^{3/20}/2},$$

which dominates $L^{(C_v+C_b+E+4)N}$. The second bound gives the same $L^{-(E+2)N}$. There are at most $\mathcal{A} + 1 \le L^N$ ranks; summing proves an upper bound L^{-EN} for the signed history sum with the OLD cutoff $\#\mathrm{OLD} \le \lceil L^2 \rceil$.

There is no circular parameter choice here. The fixed volume exponent C_v is chosen from the harmonic mass bounds and q, before m or J. One may first choose $\rho_0 \in (q^{1/4}, 1)$ and then use

$$W_0 = \max(1, v_{\max}), \quad w_0 = \max\left(1, \frac{4W_0}{\rho_0(1 - \sqrt{q}/\rho_0^2)}\right), \quad C_v = \frac{4W_0 w_0 C_0}{\rho_0},$$

where $C_0 \log L$ bounds the group count. Thus S_0 is fixed before the local parameters. The later exponents K_r, C_b affect only the eventual threshold in the rank split.

To remove this cutoff, put $T_0 = \lceil L^2 \rceil$ and $H_b = \lceil L^2/4 \rceil$. The number of active BIG labels encountered at all visits is at most $(N+1)G(J+1)$, where G is the BIG group count. Eventually $H_b + (N+1)G(J+1) \le T_0$. The first crossing of the OLD cutoff therefore requires at least H_b background births. In the positive history expansion, multiply each birth weight by 2. The discarded mass is at most 2^{-H_b} times this single positive root sum; no factor counting histories is needed. Its explicit bound is

$$S_{\text{small}} e^{4(N+1)(1+\sqrt{L})} \left[C_\nu C_{\text{small}} \left(16 \max(1, v^{-1}) e^{\sqrt{L}} \right)^{(1+m)G} \right]^N,$$

Here $C_\nu = \mathscr{K}$, and

$$C_\nu C_{\text{small}} \le L^{C'_E}, \qquad S_{\text{small}} \le \exp((C_0+1) L \log L (\sqrt{L} + \log \max(1, v^{-1}))).$$

The discarded mass is consequently at most $\exp(-\frac{1}{4}(\log 2) L^2 + O_{\text{fixed}}(L^{7/4}))$, hence at most $L^{-(E+1)N}$. Use the forest bound at precision $E+1$ as well. Adding the two errors gives $2L^{-(E+1)N} \le L^{-EN}$ for the full history sum.

5.5 From residue averages to the integer interval

For a fixed history ω, let h_i be its offsets. Its density at an integer n is

$$D_\omega(n) = \prod_p \prod_{i=0}^{N} \begin{cases} \mathbf{1}_{p \mid n+h_i}, & p \text{ is active at visit } i, \\ q_i^{\mathbf{1}_{p \mid n+h_i}}, & \text{otherwise}, \end{cases} \tag{M16}$$

with $q_0 = q_N = \sqrt{q}$ and $q_i = q$ internally. Its model density v_ω is the product of the exact averages over residues modulo each prime. This retains coincidences between offsets modulo a prime. Explicitly, let A_p be the visits at which p is active, and $I_{p,t} = \{i : h_i \equiv t \pmod p\}$. The local factor is

$$v_{\omega,p} = \begin{cases} 0, & \text{the offsets on } A_p \text{ are not all congruent}, \\ \dfrac{1}{p} \displaystyle\prod_{i \in I_{p,t} \setminus A_p} q_i, & A_p \neq \emptyset,\ h_i \equiv t\ (i \in A_p), \\ 1 - \dfrac{1}{p} \displaystyle\sum_{t \bmod p} \left(1 - \prod_{i \in I_{p,t}} q_i \right), & A_p = \emptyset. \end{cases} \tag{M16a}$$

The minus sign between the initial integer residue and t disappears under uniform averaging. There is one residue per **numerical prime** in (M16a), even if it appears in many active lists.

We verify its connection to the memory expansion, which is needed before applying an interval estimate. A prime lifespan is either empty or an interval $[a,z] \subset [0,N]$. For a nonempty lifespan require $h_a \equiv h_z \pmod p$, require every active visit to lie in $[a,z]$ and have offset congruent to h_a, and allow $a = z$ only when a is active. Its weight is

$$\frac{1}{p} \epsilon_a \epsilon_z \prod_{a<i<z,\ i \notin A_p} (1 - \eta_i 1_{h_i \equiv h_a \ (p)}), \qquad \epsilon_i = \begin{cases} 1, & i \in A_p, \\ -\eta_i, & i \notin A_p. \end{cases} \tag{T24}$$

The empty lifespan is allowed only if $A_p = \emptyset$ and has weight $b_p = 1 - \sum_i \eta_i/p$. The sum of all these weights is exactly (M16a). For an absent prime, expand the elementary identity

$$\prod_{i\in I}(1-\eta_i) = 1 - \sum_{i\in I}\eta_i + \sum_{a<z,\ a,z\in I}\eta_a\eta_z \prod_{a<i<z,\ i\in I}(1-\eta_i) \tag{T25}$$

in each residue class $I = I_{p,t}$ and average over t. The identity follows by telescoping twice, or induction on $|I|$. For a represented prime, let a_0, z_0 be its first and last active visits. The inactive factors strictly before a_0 have the expansion $1 - \sum_{a<a_0}\eta_a 1_{h_a\equiv h_{a_0}} \prod_{a<i<a_0}(1 - \eta_i 1_{h_i\equiv h_{a_0}})$; use the reversed expansion after z_0 and leave the middle factors unchanged. Multiplication gives exactly (T24), including the cases $a = a_0$, $z = z_0$, and a single active visit.

To recover the transition description from these lifespans, follow each prime through its active visits and keep it in OLD between separated active runs. Leaving an active run stores it; returning promotes the same occurrence. An inactive left endpoint of a lifespan is a background birth, and an inactive right endpoint a background deletion. The intermediate residue is $h_i - h_a \pmod p$, so all zero tests are the ones in (T24). Conversely, a history with distinct births determines these lifespans and active runs uniquely, apart from the ordering of simultaneous births and OLD representatives. Their permutation counts cancel respectively the birth factorials in (T20) and the sector factorials in (T3). At each prime the ρ_0 powers cancel; $b_p\tilde{\mu}_g(p) = 1/(V_g p)$; the remaining V_g^{-1} factors count active runs. These are exactly the factors obtained from the integer matrix's state and edge masses. Thus multiplication over primes and summation over the frame choices identify the restored signed memory sum with $\sum_\omega c_\omega v_\omega$. The frame coefficient retains its sign, auxiliary ban and complex phase throughout this equality.

Here is the finite approximation used for (M16). If active congruences at some prime are incompatible, both the integer density and its residue average vanish. Otherwise let Q_0 be the product of the distinct active primes. Their congruences specify one class modulo Q_0, with a fixed positive factor from the inactive visits at those primes. For each remaining prime put

$$f_p(r) = 1 - \prod_{i=0}^{N} q_i^{1_{r+h_i\equiv 0 \ (p)}}.$$

Then $0 \le f_p \le 1$, it is supported on at most $N+1$ residue classes, and its average is at most $(N+1)/p \le 1/2$ eventually, since every marked prime is at least $e^{L^{1/10}}$. Put

$$\lambda = \sum_p \frac{1}{p}\sum_{r \bmod p} f_p(r), \qquad V_0 = \prod_p\left(1 - \frac{1}{p}\sum_r f_p(r)\right), \qquad U_0 = 1 + \sum_p\sum_r f_p(r).$$

Bonferroni truncation at orders $2K$ and $2K+1$, followed by CRT on each retained product, gives relative error on an interval of length X at most

$$\frac{e^{4\lambda}}{2^{2K+1}} + \frac{Q_0}{X}\,2(2K+2)e\,U_0^{2K+1}e^{2\lambda}. \tag{M17}$$

Indeed, the omitted elementary symmetric sums are bounded using the sum λ of the local averages; the normalization by V_0 costs at most $e^{2\lambda}$. For each retained residue pattern, CRT counts its interval representatives with error at most one. Summing the absolute coefficients of the retained patterns gives the second term in (M17).

Take $K = \lceil L^{11/20} \rceil$. The actual ranges give

$$\lambda = O(N \log L), \qquad \log Q_0 = O_J(L^{23/25} \log L), \qquad K \log U_0 = O(L^{19/20}),$$

whereas $X \geq e^{\gamma L}$ for a fixed $\gamma > 0$. Both terms in (M17) are smaller than L^{-EN} for every fixed E eventually. For complex history coefficients, the transfer is made before discarding any signs:

$$\left| \sum_{\omega} c_\omega \sum_{n} D_\omega(n) - X \sum_{\omega} c_\omega v_\omega \right| \leq X L^{-EN} \sum_{\omega} |c_\omega| v_\omega. \tag{M18}$$

The absolute mass on the right is at most L^{CN}. Its exponent is fixed after J; increasing the requested CRT accuracy absorbs it. The signed main sum on the left is the one already bounded by (M15).

We spell out the finite physical matrix to which these history estimates apply. Fix a seed interval $I = [a, b) \cap \mathbb{Z}$ of positive integers, of length $X = b - a \geq e^{\gamma L}$. Put $R = \lceil L^{13/25} \rceil$, $r_{\text{path}} = \lceil 2R\Delta_{\max} \rceil$, with $\Delta_{\max}$ as in (M20a), and take the carrier positions to be

$$I^+ = [a - r_{\text{path}}, b + r_{\text{path}}] \cap \mathbb{Z}.$$

Every prefix of every $2R$-step history from I belongs to I^+: each displacement has absolute value at most $\Delta_{\max}$, so the triangle inequality bounds a prefix by $2R\Delta_{\max}$. Thus truncating the matrix to this common carrier introduces no history-dependent restriction on its starting interval. All residue counts below use the same I and the same length X.

Put $M = J + 1$. A state is an integer position $n \in I^+$, together with, for each marked group g, an ordered list p_g of M distinct primes dividing n. The residues of n determine all residue coordinates of the state. All states have the same reference mass

$$m_0 = \prod_g V_g^{-M}; \qquad u_n = \sqrt{m_0} \sum_{\substack{s \text{ a state} \\ \text{pos}(s) = n}} e_s. \tag{M18a}$$

Thus no mark fibre is normalized to have mass one.

A frame consists of a raw pattern or a nonempty pair (a, b) from (M2), an auxiliary prime for each freed BIG probe coordinate, and independent source and target permutations $\sigma_g, \tau_g \in S_M$ in every group. For a comparison pattern, let $C_g \subseteq \{0, \dots, m-1\}$ be its freed coordinates in group g; for a raw pattern and for SMALL groups, $C_g = \emptyset$. The shared coordinates are $S_g = \{0, \dots, J-1\} \setminus C_g$. Write the auxiliary primes as $\pi_{g,i}$ and give them the law

$$\mu_{\text{aux}} = \prod_{g \in \text{BIG}} \prod_{i \in C_g} \frac{1}{V_g \pi_{g,i}}.$$

Auxiliary values may repeat. Each must differ from every source and target mark in its own group. For states s, t, a frame is allowed when their permuted marks agree at each shared coordinate and

$$\begin{aligned} D_f &= \prod_g \prod_{i \in S_g} p^s_{g,\sigma_g(i)} \prod_{g \in \text{BIG}} \prod_{i \in C_g} \pi_{g,i}, \\ \text{pos}(t) &= \text{pos}(s) + 2kD_f. \end{aligned} \tag{M18b}$$

All factors of D_f retain their multiplicities. For the raw pattern let $K_f = 1$. For a comparison pattern put

$$K_f = -(-1)^{|a|+|b|} K_A(D_a, X_a) K_B(D_b, X_b),$$

where D_a, D_b are the corresponding products of auxiliary primes, X_a is the product of the **target** marks in the first probe block, and X_b is the product of the **source** marks in the second block. Both endpoint products use the permuted lists. The two kernels are exactly (M4), with the actual cell masses for these probe products.

For an allowed frame define

$$w_f(s,t) = \mu_{\mathrm{aux}} \prod_g \Big[(M!)^{-2} V_g^{-(M-|S_g|)} q^{(\omega_g(\mathrm{pos}(s))+\omega_g(\mathrm{pos}(t))-2M)/2} \Big].$$

The powers of q are the two endpoint dampings. Their exponents are nonnegative because each endpoint has M distinct marked divisors. The physical matrix is

$$A_{s,t} = \sum_{f \text{ allowed}} w_f(s,t) K_f D_f^{i\zeta}. \tag{M18c}$$

Reversing the two endpoint permutations preserves D_f and the real weight. Reversing the orientation conjugates the whole complex coefficient, including K_f. The backwards matrix is therefore exactly A^*.

Expanding $(AA^*)^R$ gives the alternating histories above. Insert $\mathrm{e}(2kD_f\theta)$ on a forward edge and its conjugate on a backward edge. The product is $\mathrm{e}(\theta h_N)$, so integration for $0 \le \theta \le 1$ keeps precisely the histories returning to the initial integer position. The initial and final mark lists remain free; the vectors u_n in (M18a) perform those sums. The restored history bound and (M18) consequently give

$$T := \sum_{n \in I} \langle u_n, (AA^*)^R u_n \rangle \le X L^{-2RE}, \qquad R = \lceil L^{13/25} \rceil. \tag{M19}$$

5.6 The endpoint bound

The first endpoint has the form $f = \sum_{n \in I} c_n u_n$; its coefficient depends on the integer position but not on the individual marks. The second endpoint g may depend on all marks. The finite spectral estimate used here is

$$|\langle f, Ag \rangle| \le \sqrt{3}\, \|g\|\, \|f\|^{1-1/R} (H_{\mathrm{loc}} C_{\mathrm{sup}}^2)^{1/(2R)} T^{1/(2R)}. \tag{M20}$$

Here $|c_n| \le C_{\mathrm{sup}}$ and a nonzero matrix entry changes the integer position by at most the positive integer H_{loc}. To prove the estimate, partition the positions into disjoint intervals I_j of H_{loc} integers. Let f_j be the restriction of f to I_j, and let g_j be the restriction of g to $I_{j-1} \cup I_j \cup I_{j+1}$. Locality gives $\langle f, Ag \rangle = \sum_j \langle f_j, Ag_j \rangle$ and $\sum_j \|g_j\|^2 \le 3\|g\|^2$. Put $P = AA^*$. Spectral interpolation for the nonnegative matrix P, followed by Cauchy–Schwarz, gives

$$|\langle f_j, Ag_j \rangle| \le \|g_j\|\, \|f_j\|^{1-1/R} \langle f_j, P^R f_j \rangle^{1/(2R)}.$$

Indeed, the spectral measure of P at f_j satisfies $\langle f_j, Pf_j \rangle \le \|f_j\|^{2-2/R} \langle f_j, P^R f_j \rangle^{1/R}$. Writing $f_j = \sum_{n \in I \cap I_j} c_n u_n$ and applying Cauchy–Schwarz to the vectors $P^{R/2} u_n$ yields

$$\langle f_j, P^R f_j\rangle \le H_{\mathrm{loc}} C_{\mathrm{sup}}^2 \sum_{n\in I\cap I_j} \langle u_n, P^R u_n\rangle.$$

Hölder's inequality in j, with exponents 2, $2R/(R-1)$ and $2R$, proves (M20), since $\sum_j \|f_j\|^2 = \|f\|^2$. For $R = 1$, omit the middle factor and use Cauchy–Schwarz. In the application $R > 1$.

In the application we may take $C_{\mathrm{norm}} = 9$ and $C_{\mathrm{sup}} = e^{5\sqrt{L}}$, with $\|f\|, \|g\| \le \sqrt{X}L^9$ eventually. To specify the window, put

$$\begin{aligned} C_D &= 4(J+1)(4 + 1/\log 2),\\ \Delta_{\max} &= 2L^{C_k}\exp(C_D \log L\, L^{2/5}),\\ H_{\mathrm{loc}} &= 2\lceil 2R\Delta_{\max}\rceil + 1. \end{aligned} \tag{M20a}$$

Every frame satisfies $D_f \le \exp(C_D \log L\, L^{2/5})$, including repeated auxiliaries, and every $2R$-step path from n stays within $\lceil 2R\Delta_{\max}\rceil$ of n. The reference measure in (M20) is ordinary counting measure on the finite physical states; $\sqrt{m_0}$ is in the vectors, not the norm. Since $\log H_{\mathrm{loc}} = O_J(\log L\, L^{2/5})$ and $R \ge L^{13/25}$, the local factor in (M20) is bounded. Apply (M19) with $E' = E + 2C_{\mathrm{norm}} + 1$. Then (M20) is at most XL^{-E} eventually. The physical-lift argument gives the exact endpoint vectors, the Fourier pairing identity (L8), and the shared-prime restoration error (L4). It obtains the $J+1$ marked physical states by multiplying both original integers by a common product of J primes per group; it does not require the original integers in (B17) to have that many distinct marked divisors.

This proves the saving for the original term minus its arithmetic comparison term. It remains to estimate that comparison term itself.

5.7 The comparison term and the fine discrepancy

We give the endpoint used in the comparison explicitly. If G is the union of the marked prime groups, let n_* be the part of n with all prime factors in G removed. For bounded coefficients α, c, a prime slot Q_* disjoint from G, a character χ, and real frequencies σ_0, σ_1, set

$$\mathrm{Core}(n_*) = \sum_{\substack{mp\mid n_*\\ p\in Q_*}} \alpha(m)\chi(m) m^{i\sigma_0} p^{i\sigma_1} c(n_*/mp). \tag{M21}$$

For each marked group P_g, let $\mathcal{L}_g(n)$ consist of its ordered lists of t_g distinct prime divisors of n, and let $\omega_g(n)$ count those distinct divisors. For a function b of the BIG lists with $|b| \le 1$, the literal marked weight is

$$W_t^b(n) = \prod_g \frac{q^{\omega_g(n)-t_g}}{V_g^{t_g}} \sum_{\substack{p_g\in\mathcal{L}_g(n)\\ \text{all } g}} b((p_g)_{g\in\mathrm{BIG}}). \tag{M22}$$

This includes the distinguished SMALL group; in the source its number of marks is $k+1$. Multiplicities in n are not restricted by the distinctness of the marks. With a smooth cutoff $\Lambda(n)$, the endpoint Fourier polynomial is

$$\hat{E}(\theta) = \sum_{1\le n\le N_{\mathrm{cap}}} \Lambda(n)\varepsilon(n)\,\mathrm{Core}(n_*)W_t^{b}(n)\mathrm{e}(n\theta). \tag{M23}$$

The cell/character kernel (M4) has a concrete rank decomposition: each (j,χ) contributes the product of $1_{D\in I_j}\overline{\chi(D)}$ and $1_{X\in I_j}\chi(X)$, with coefficient ν_j^{-1}. There are at most n_AQ^2 such terms in the first kernel and n_BQ^2 in the second. Thus a comparison pattern has at most $n_An_BQ^4$ pairs, each coefficient at most ξ^{-2}. The support counts (M9a) bound this rank by a fixed power of L. For retained cells there is a sharper bound: their disjoint probabilities sum to at most one, so their number is at most ξ^{-1} in either block. Including all patterns gives

$$\begin{aligned} \#\{\text{ranks}\} &\le L^{C_{\mathrm{pat}}}Q^4\xi^{-2}, \\ \sum_{\text{ranks}} \nu_{j_A}^{-1}\nu_{j_B}^{-1} &\le L^{C_{\mathrm{pat}}}Q^4\xi^{-4}, \qquad C_{\mathrm{pat}} = m(3+1/\log 2)\log 2. \end{aligned} \tag{M23a}$$

Here $3+1/\log 2$ is an upper group-count constant for the actual BIG family. The two inverse cell masses stay outside the endpoint tests. The remaining endpoint cell indicators and characters are the bounded list coefficient b in (M22).

The auxiliary sum retains its dependence on the shift and cutoff. More explicitly, let $\nu_A(u),\nu_B(v)$ be the probability laws of the two auxiliary products. Independence of the two disjoint probe blocks gives the exact finite identity

$$\begin{aligned} &\sum_{\text{aux}} \mu_{\text{aux}} 1_{D_A\in I_{j_A},D_B\in I_{j_B}} \overline{\chi_A(D_A)\chi_B(D_B)}\,H(D_AD_B) \\ &\quad = \sum_{u\in I_{j_A}}\sum_{v\in I_{j_B}} \nu_A(u)\nu_B(v)\overline{\chi_A(u)\chi_B(v)}\,H(uv). \end{aligned} \tag{M23b}$$

The inverse cell masses and the pattern sign multiply both sides. After the shared-divisor restoration described below, $H(c)$ retains the shift $2kc$ and the cutoff $\Psi(n/(Zc))$. Thus (M23b) does not replace this dependence by a mass-one average. Fourier inversion of the cutoff as in (L8) separates its Mellin factor; additive Fourier inversion separates the two endpoint integers. This yields the pair of polynomials (M23), with an auxiliary bilinear phase $\mathrm{e}(-2kuv\theta)$ on the right of (M23b). Partition u,v into the coarse dyads of (M9a) when applying (M7). The scales of a retained cell pair are $H' = e^{j_Ah_A+j_Bh_B}$ and $Y = ZH'$; its actual product lies between H' and $e^{h_A+h_B}H'$. A fixed smooth cutoff equal to one on this ratio interval gives the endpoint support used in (M25). All rank, coefficient and Fourier costs are fixed before requesting the discrepancy precision below.

Here is the resulting comparison identity with the endpoint factors displayed. For a pattern (a,b) let $t_g = M - |S_g|$ be its number of free marks; thus $t_g = 1$ in SMALL and $t_g = 1+|C_g|$ in BIG. For rank $r = (j_A,\chi_A,j_B,\chi_B)$ define the two list functions

$$b_{1,r} = \overline{1_{X_B\in I_{j_B}}\chi_B(X_B)}, \qquad b_{2,r} = 1_{X_A\in I_{j_A}}\chi_A(X_A). \tag{M23c}$$

Here X_B is the source probe product and X_A the target probe product, with the shared entries deleted and the free entries kept in their original coordinate order. Both functions have modulus at most one and contain no auxiliary variables. With the two cores of (L7), define

$$U_r(n) = \varepsilon(n)n^{it_1}\,\mathrm{Core}_1(n_*)W_t^{b_{1,r}}(n), \qquad V_r(n) = \varepsilon(n)n^{it_2}\,\mathrm{Core}_2(n_*)W_t^{b_{2,r}}(n). \tag{M23d}$$

The first core is conjugated when U_r is conjugated in the pairing. For (B17) the phases are $t_1 = -(\tau + \xi_3)$, $t_2 = -\tau$, $\zeta = -\xi_3$, as in (L7)–(L8). Write $c = uv$ and $s_{a,b} = -(-1)^{|a|+|b|}$. The independent-rank main contribution of this pattern and rank to the signed comparison part of A (hence to $-Q_{\mathrm{cmp}}$ in (L8)) is exactly

$$\begin{aligned}\frac{s_{a,b}}{\nu_{j_A}\nu_{j_B}} \sum_{\substack{u\in I_{j_A}\\ v\in I_{j_B}}} \nu_A(u)\nu_B(v)\overline{\chi_A(u)\chi_B(v)}\, c^{i\zeta} \\ \cdot \sum_{\substack{n>0\\ n+2kc>0}} \frac{\Psi(n/(Zc))}{n}\overline{U_r(n)}V_r(n+2kc).\end{aligned} \tag{M23e}$$

To check the removal of the shared divisor d, substitute the original positions dn, dn' and $D_f = dc$. The displacement becomes $n' = n + 2kc$ and the cutoff becomes $\Psi(n/(Zc))$. The ordinary parts of the two integers are unchanged, since all prime factors of d are marked. The common parity is squared. The phase in d is $d^{i(-t_1+t_2+\zeta)} = 1$. The shared harmonic factors then sum to one on the unrestricted shared family; the exclusions and repeated-mark cases produce the weighted restoration error (M28), not an exact mass-one assertion on the restricted family.

There is one further explicit difference in (M23e): the original auxiliary ban forbids a free endpoint mark from equalling an auxiliary prime. Partitioning the complete pair of free mark lists into the allowed and forbidden sets gives, exactly, the independent product of the two weights minus its collision contribution. A forbidden pair forces $p \mid n$ for an auxiliary prime $p \ge e^{L^{3/10}}$; if it divides the other endpoint, it also divides n, since $p \mid c$. Reindex $n = pr$. The factor $1/n$ and the divisor-moment bound for the two cores give a factor $1/p$ with a fixed logarithmic cost. If $L^{E_{\mathrm{mark}}}$ bounds either full marked weight, the error for one pattern and one fixed auxiliary assignment is at most

$$C_J K_\Psi Q^4 \xi^{-2} L^{2E_{\mathrm{mark}}+16+C_\psi} e^{-L^{3/10}}. \tag{M23f}$$

Here $C_J = 2J(3 + 1/\log 2)\, 64 \cdot 66 \cdot 3^{15}$; E_{mark} depends only on J. Summing auxiliary probabilities and at most $L^{C_{\mathrm{pat}}}$ patterns preserves an error smaller than every fixed inverse power of L. Thus (M23e), summed over ranks and patterns, equals that signed comparison up to the two specified errors (M28) and (M23f).

For completeness, the Fourier pairing of (M23e) also has an exact form. Put $\psi_{\log}(t) = \Psi(e^t)$. Choose a fixed real smooth cutoff η, equal to one at both n/Y and $(n+2kc)/Y$ whenever the summand in (M23e) is nonzero, and supported in a fixed compact subinterval of $(0,\infty)$. Such a choice is uniform because $1 \le c/H' \le e^{h_A+h_B}$ and $|k|/Z \to 0$ uniformly. Set

$$A_{r,s,Y}(n) = \frac{\eta(n/Y)}{n} n^{-2\pi i s} U_r(n), \qquad B_{r,Y}(n) = \eta(n/Y)V_r(n), \tag{M23g}$$

and use $\widehat{A}(\theta) = \sum_n A(n)\mathrm{e}(n\theta)$, likewise for B. Then (M23e) is

$$\int_{\mathbb{R}} \widehat{\psi}_{\log}(s) \int_0^1 \mathscr{K}_{r,s}(\theta)\overline{\widehat{A}_{r,s,Y}(\theta)}\widehat{B}_{r,Y}(\theta)\, d\theta\, ds, \tag{M23h}$$

where the full coupled kernel is

$$\mathcal{K}_{r,s}(\theta)=\frac{s_{a,b}Z^{-2\pi is}}{\nu_{j_A}\nu_{j_B}}\sum_{\substack{u\in I_{j_A}\\ v\in I_{j_B}}}\nu_A(u)\nu_B(v)\overline{\chi_A(u)\chi_B(v)}(uv)^{i\zeta-2\pi is}\mathrm{e}(-2kuv\theta). \tag{M23i}$$

The circle integral imposes $n'=n+2kuv$, and the Mellin integral restores $\Psi(n/(Zuv))$, proving (M23h) term by term. The cell masses in the denominator cancel the total restricted product probability, so $|\mathcal{K}_{r,s}(\theta)|\le 1$ for every s,θ. On minor arcs (M7)–(M9) give the stronger saving, retaining both inverse cell masses in the coefficient budget (M23a). The endpoints (M23g) are now independent of u,v, which is the precise interface needed for the arc-energy estimate.

The coefficient α has the complete-cell discrepancy (C3): for any prescribed B, the fine mesh can be chosen so that

$$\left|\sum_{a\le m\le b}\frac{\alpha(m)\chi(m)m^{it}}{m}\right|\le L^{-B},\qquad q,|t|\le L^{B},\quad [a,b]\subset[1,x^2]. \tag{M24}$$

Here α already includes its restriction to the factor interval $K\subset[x^{\beta},x^2]$ of multiplicative width at most two, with fixed $\beta>0$. Thus the displayed sum means its intersection with K. If nonempty, that intersection satisfies the lower endpoint and width conditions of (C3). The modulus q in this display is an integer conductor variable, distinct from the fixed damping $q=1/2$.

Let the two nonempty probe-product scales be U_1,U_2, put $H'=U_1U_2$ and $Y=ZH'$, and suppose $Z\ge xL^{-C_Z}$. The marked arc estimate obtained from (M24) is

$$\int_{|\beta|\le L^{C_{\mathrm{arc}}}/H'}|\hat{E}(h/d+\beta)|^2\,d\beta\le K_{\mathrm{arc}}YL^{-T}. \tag{M25}$$

Here Q_* is the complete prime interval used by the comparison, contained in $[x^{\tau},x^{\eta}]$ with $0<\tau<\eta<1/4$; the support of α is $e^{\sqrt{L}}$-rough. The cutoff has the prescribed polynomial bounds on its logarithmic derivatives, and $2\le H'/L^{C_{\mathrm{arc}}}\le Y/(8A_{\psi})$. These conditions hold for the actual data from (B16). Under them, (M25) is uniform for the polynomial conductor, frequency, and coefficient budgets in the comparison expansion. The support has $e^{3L/4}\le Y\le e^{3L/2}$; take $N_{\mathrm{cap}}=\lceil A_{\psi}Y\rceil\le x^2$, where A_{ψ} bounds the cutoff support.

Here are the estimates establishing (M25). Peeling one distinguished SMALL mark leaves the core and parity at the original integer and evaluates only the remaining marked weight at n/p. Separate the smooth logarithmic factors by Fourier inversion, and write $H=H'/L^{C_{\mathrm{arc}}}$. The finite local-energy inequality (E1) and the exact separation (E9) in Section 8 give the following four terms. With $B_{\mathrm{amp}}\ll L^{E_{\mathrm{amp}}}$ and $B_s\ll L^{E_s}$ the resulting amplitude bounds, those terms, up to fixed constants, are

$$\begin{aligned}
&YV_s^{-2}d^2(1+\log_2 N_{\mathrm{cap}})^8L^{-(T+2c_m+8)},\\
&Y^{-1}L^{-(j-2)}(LY/H+16N_{\mathrm{cap}}(1+\log N_{\mathrm{cap}}))B_{\mathrm{amp}}^2N_{\mathrm{cap}}(1+\log N_{\mathrm{cap}})^{63},\\
&(H/Y)^2B_{\mathrm{amp}}^2N_{\mathrm{cap}}(1+\log N_{\mathrm{cap}})^{63},\\
&3LB_s^2N_{\mathrm{cap}}(1+\log N_{\mathrm{cap}})^{16}e^{-L^{a_s}}.
\end{aligned} \tag{M26}$$

The first is the finite Mellin-window contribution (E13), obtained from (M24) and the small- and long-prime estimates, with $d\le L^{c_m}$. The second is the weighted Mellin tail;

choose $j = \lceil 2E_{\mathrm{amp}} + 64 + T \rceil + 2$. The third is the error (E3) in replacing linear windows by logarithmic windows. Since $Y = ZH'$, $H/Y = 1/(ZL^{C_{\mathrm{arc}}})$, so its exponential decay pays for every displayed logarithmic factor. The fourth accounts for squares of a peeled SMALL prime, whose size is at least $e^{L^{a_s}}$; here $a_s = 1/10$ is available. Divisor-moment estimates give the displayed powers $8, 63, 16$. Each line of (M26) is $O(YL^{-T})$ with the stated choices, proving (M25). The window replacement follows from the Taylor estimate (E2) and finite overlap, with no prime distribution estimate on an interval of length H.

The first Fourier polynomial in the actual comparison has an additional factor $1/n$. With its cutoff written as $\eta(n/Y)$,

$$\frac{\eta(n/Y)}{n} = Y^{-1}\frac{\eta(n/Y)}{n/Y}.$$

The cutoff is supported away from zero. Applying (M25) to $\eta(u)/u$ therefore gives energy $K_{\mathrm{arc}}Y^{-1}L^{-T}$ for this first polynomial; the second has energy $K_{\mathrm{arc}}YL^{-T}$. Cauchy–Schwarz cancels the two powers of Y. Summing the rational arcs of the comparison gives $2K_{\mathrm{arc}}L^{2c_m-T}$. Off those arcs, the auxiliary-product bilinear exponential-sum estimate gives L^{-A_1} against the global polynomial endpoint energy. Choose T, A_1 to pay for the finite rank and pattern budgets. This makes the whole comparison term smaller than any requested logarithmic power.

Subtracting this separately bounded term from the signed-operator estimate (M20) bounds the original correlation.

5.8 Caps, shared divisors, and the parameter order

The finite comparison first imposes a cap on the marked part and allows marks shared between the two endpoints. Both modifications are removed quantitatively. The Rankin estimate from the weight chapter gives

$$\sum_{\substack{u \le Y \\ u_P > \lfloor e^{L/88} \rfloor}} W_1(u) \le YL^{(3+1/\log 2)\log(6e^{12})}e^{-L^{3/5}/88}. \tag{M27}$$

Stripping shared divisors has error

$$O\Big(L^{8(J+1)\log 2 + 2E_{\mathrm{mark}} + 17 + C_\psi}e^{-L^{1/10}}\Big). \tag{M28}$$

The quotient cap is transported through the exact finite identity before using (M27); its final specialization has shared divisor one and zero extra marks. Thus no assertion that the marked part is squarefree, or that a cap is unchanged by arbitrary division, is needed. Both (M27) and (M28) are negligible at every fixed logarithmic accuracy.

The choices can now be made in order. Fix the desired saving and all external frequency, shift, and cutoff bounds. Choose the local contraction precision and m using (M1)–(M3), then the arithmetic mesh, conductor, and retained-mass parameters using (M4)–(M10). Choose J using (M13). The forest row bounds, total path mass, comparison rank, Fourier-tail orders, and CRT precision are chosen after these fixed parameters. Finally choose the discrepancy order B and the fine-cell exponent $S = 2(B+1) + 4$.

These choices precede the positive lower exponent β of the fine interval and its positive roughness exponent γ. Only the eventual threshold in L depends on β, γ. Combining the

endpoint bound (M20), the comparison estimate (M25), and the two restoration errors (M27)–(M28) gives, with precisely the coefficients of (B16)–(B17),

$$|C_{k,Z}| \le ZL^{-D_1}$$

uniformly for $0 < |k| \le L^{C_k}$, $xL^{-C_Z} \le Z \le xL^{C_Z}$, the stated polynomial frequency bounds, and either multiplicity-parity choice. This is (B18).

6 From one mark to the physical matrix

This chapter supplies the finite change of variables between (B17) and the matrix (M18c). Extra marks are introduced by a common dilation of the two integers. They are then removed by an exact finite reindexing and a quantitative estimate for collisions.

6.1 The raw shared-prime sum

Put $q = 1/2$, $M = J + 1$, and let G denote the set of all marked groups. For each $g \in G$ draw J labelled primes $s_{g,i} \in P_g$ independently with probabilities $1/(V_g s_{g,i})$. Write

$$D_s = \prod_{g,i} s_{g,i}, \qquad \mu(s) = D_s^{-1} \prod_g V_g^{-J}, \qquad \sum_s \mu(s) = 1.$$

Repeated draws are included. For positive w, let $\mathcal{A}_s(w)$ be the set of complete ordered lists of M distinct primes in each group, all dividing $D_s w$, whose first J coordinates are the given $s_{g,i}$. In particular this set is empty if any shared list repeats a prime. Put

$$H(n) = \prod_g q^{\max(\omega_g(n)-M,0)/2}, \qquad R_s(w,w') = \frac{H(D_s w)^2 H(D_s w')^2}{D_s \prod_g V_g^{J+2}} |\mathcal{A}_s(w)||\mathcal{A}_s(w')|. \quad \text{(L1)}$$

On a nonempty completion the maximum in H can be omitted. Formula (L1) is exactly the raw physical slice: the state mass contributes $\prod V_g^{-M}$, the raw row contributes $\prod V_g^{-1}$, and its two half dampings combine with the two endpoint half dampings. Averaging the endpoint permutations gives the same fixed-slot completion sum; each relabelling has the inverse permutation as its unique inverse. The factor D_s^{-1} comes from the harmonic factor $1/n$ after putting $n = D_s w$.

Call a draw clean if its primes are distinct in each group and none divides ww'. Then $\omega_g(D_s w) = J + \omega_g(w)$ and each completion is specified by one prime divisor of w in each group. Substitution in (L1) gives

$$R_s(w,w') = \mu(s) W(w) W(w') \quad \text{on clean draws.} \qquad \text{(L2)}$$

This explains why an original integer with only one marked divisor in a group is allowed: its physical integer is $D_s w$, which has the additional J shared marked divisors on a clean draw.

Let $I = J|G|$, $P_{\min} \le \min_{p \in \cup P_g} p$, and $v \le \min_g V_g$. The probability of a collision or a prime dividing ww' is at most

$$\frac{I(\log_2 w + \log_2 w') + I^2}{v P_{\min}}. \qquad \text{(L3)}$$

Indeed every atom has mass at most $1/(vP_{\min})$; a positive integer has at most its base-two logarithm distinct prime divisors. Sum these bounds over the draws and over pairs of draws. On any draw, completions can only lose free choices compared with the one-mark lists of w, w'. At most J shared primes per group already divide either integer, so removing the associated damping costs at most $q^{-2M|G|}$. Therefore

$$\left|\sum_s R_s(w,w') - W(w)W(w')\right| \le (q^{-2M|G|}+1)W(w)W(w')\frac{I(\log_2 w+\log_2 w')+I^2}{vP_{\min}}. \quad \text{(L4)}$$

This is obtained by summing $R_s - \mu(s)W(w)W(w')$, which vanishes on clean draws by (L2). Thus no conditioning or renormalization of the prime law occurs. Since J is fixed, $|G| = O(\log L)$ and $P_{\min} \ge e^{L^{1/10}}$, the factor in (L4) is a fixed power of L times $e^{-L^{1/10}}$ when $w, w' \le e^{2L}$.

6.2 The arithmetic factors and the exact reindexing

Write $a_1 = \tau + \xi_3$, $a_2 = -\tau$ for the phases in (B17), and set $E = \overline{F}$. Then the product of the two cores is $\overline{E(w)}G(w')$. Let $\Psi(t) = t\psi(t)$ and define

$$K(w,w') = \frac{\Psi(w/Z)}{w}\, w^{ia_1}(w')^{ia_2}\overline{E(w)}G(w')\varepsilon(w)\varepsilon(w').$$

Consequently $C_{k,Z}/Z = \sum_{w>0,\, w+2k>0} K(w, w+2k)W(w)W(w+2k)$. The sum is finite by the support of ψ.

For a raw frame set $n = D_s w$, $n' = D_s w'$. Its equation $n' - n = 2kD_s$ is equivalent to $w' - w = 2k$; the change of variables is bijective on the positive integers divisible by D_s. Moreover

$$(D_s w)_* = w_*, \qquad \varepsilon(D_s w)\varepsilon(D_s w') = \varepsilon(w)\varepsilon(w'). \quad \text{(L5)}$$

The first identity uses only that D_s is supported on marked primes; the second uses complete multiplicativity and $\varepsilon(D_s)^2 = 1$. They hold even on dirty draws and preserve the original multiplicities. Taking the frame phase $D_s^{i\zeta}$ with $\zeta = -(a_1 + a_2) = -\xi_3$ also cancels the common dilation in the two imaginary powers. The cutoff is evaluated at $n/(ZD_s) = w/Z$. Expanding the raw physical pairing and reindexing therefore gives exactly

$$Q_{\text{raw}} = \sum_{w>0,\, w+2k>0} K(w, w+2k) \sum_s R_s(w, w+2k). \quad \text{(L6)}$$

Choose the finite physical window to contain these dilated positions and their edge neighbours. This is possible uniformly: D_s is at most $\exp(O_J(L^{2/5}\log L))$, while $w \asymp Z = xL^{O(1)}$. No contributing term is added or lost when the common finite carrier replaces the individual quotient carriers.

Applying (L4) in (L6), the pointwise logarithmic bounds for W, and the divisor-moment bounds for the cores shows that $Q_{\text{raw}} - C_{k,Z}/Z$ is smaller than any fixed inverse power of L. For example, Cauchy–Schwarz bounds the sum of the two core moduli over $w \asymp Z$ by $ZL^{O(1)}$, and the factor $1/w$ cancels Z. The remaining factor is $L^{O_J(1)}e^{-L^{1/10}}$. Physical marked-part caps are removed separately by (M27); one must not identify a cap at $D_s w$ with the same cap at w.

6.3 The endpoint vectors and the signed difference

Partition the first integer position into dyadic intervals $[X, 2X)$. On a physical state s at position n, put

$$\begin{aligned} f_{X,\nu}(s) &= \sqrt{m_0}\, 1_{X\le n<2X} \frac{X}{n} (n/Z)^{-2\pi i\nu} n^{-ia_1} E(n)\varepsilon(n)H(n), \\ g_X(s) &= \sqrt{m_0}\, n^{ia_2} G(n)\varepsilon(n)H(n). \end{aligned} \tag{L7}$$

Restrict g_X to the finite neighbourhood reachable from the first interval; this does not change the pairing. These formulas specify the extra endpoint damping, separately from the damping already in the matrix. The first vector depends only on the integer position.

Let $f_\Psi(t) = \Psi(e^t)$ and use $\hat{f}_\Psi(\nu) = \int_{\mathbb{R}} f_\Psi(t) e^{-2\pi i\nu t}\, dt$. Fourier inversion of the actual frame cutoff, followed by finite summation, gives

$$Q_{\mathrm{raw}} - Q_{\mathrm{cmp}} = \sum_X \frac{1}{X} \int_{\mathbb{R}} \hat{f}_\Psi(\nu) \langle f_{X,\nu}, A(\zeta - 2\pi\nu) g_X\rangle\, d\nu. \tag{L8}$$

Here $A(v)$ is precisely (M18c) with frame phase D_f^{iv}, and Q_{cmp} is the sum of its nonempty comparison patterns with the signs in (M2). The conjugation of the first vector supplies $(n/Z)^{2\pi i\nu}$; its product with $D_f^{-2\pi i\nu}$ is $e^{2\pi i\nu \log(n/(ZD_f))}$. This proves (L8) without replacing D_f by a representative cell value. Compact support and smoothness make $\hat{f}_\Psi$ integrable. Fourier tails are bounded with the fixed derivative budgets before the polynomial frequency restriction in the correlation estimate is applied.

The vector norms in (M20) follow by first fixing the complete distinct mark product P_0. It divides n, and the core has size at most a fixed logarithmic factor times $d_2(n/P_0)^2$. Thus its squared sum over $n \le 3X$ is at most $(3X/P_0)L^{O(1)}$, using the fourth divisor moment. Finally $m_0 \sum_{\text{distinct lists}} 1/P_0 \le \prod_g V_g^{-M} (\sum_{p\in P_g} 1/p)^M = 1$. This gives $\|f_{X,\nu}\|, \|g_X\| \le \sqrt{X} L^{C_{\mathrm{norm}}}$ with exponent independent of J. The rough-factor counting bound for the core gives the supremum bound in (M20). Applying that estimate in (L8), paying for the logarithmic number of dyads and the Fourier norm, and then using (L4)–(L6), proves the claimed saving for the original correlation minus its comparison term.

7 The forest expansion of the history sum

We give the finite identity and the two estimates used in (M15). The distinction between a new prime label and a later use of the same label is essential: the forest expansion applies to the list of births, not to the list of all visits.

7.1 A linear functional on birth lists

Fix the moment length $N = 2R$, the OLD cutoff T, and all SMALL labels. Expand the product of the N signed transitions, including their comparison patterns and independent endpoint permutations. At the initial state there are $G(J + \ell)$ potential BIG birth slots. At each of the $N + 1$ background steps there are GT slots, and each of the N active transitions has $G(J + \ell)$ slots. Here G is the number of BIG groups. Thus the list has fixed length

$$A = (N + 1)G(J + \ell + T).$$

An unused slot has value $\emptyset$. Retaining a label already present in the state creates no new birth. All local distinctness conditions, transition legality conditions, OLD gates and the initial distinctness condition remain in the coefficients of this expansion. Only the additional condition that different births have different numerical prime labels is set aside temporarily.

Let $\mathcal{H}$ be this finite collection of expanded histories, let $b(H) = (b_0(H), \dots, b_{A-1}(H))$ be its birth list, and let c_H be the product of its transition and boundary coefficients, with the common baseline and tilted prime masses described below. In particular c_H includes the sign of every comparison pattern, all normalized prime and permutation masses, the initial and terminal boundary factors, and the damping factors. Define the linear functional

$$\Lambda(f) = \sum_{H \in \mathcal{H}} c_H f(b(H)). \tag{F3}$$

The projected history sum is $\Lambda(1_{\text{nonempty births distinct}})$. This construction does not replace the coefficients by their absolute values. On histories with distinct births its density agrees with the prime residue averages in (M16). Factoring out the contribution of primes not selected as births gives the common baseline $B_{\text{base}} = \prod_p b_p$ and the tilted laws $\tilde{\mu}_g(p) = 1/(V_g p b_p)$ from the state-space construction.

For a birth list b, replace an unused entry at address j by a symbol $\star_j$ used at no other address. Denote the resulting list by $x_0, \dots, x_{A-1}$. A forest is a choice, for each j, of either no parent or one parent $p(j) < j$. Its rank $r(p)$ is the number of chosen parents. Define

$$T_p(b) = (-1)^{r(p)} \prod_{j:\, p(j) \text{ chosen}} 1_{x_j = x_{p(j)}}, \qquad F_p = \Lambda(T_p). \tag{F4}$$

The exact identity is

$$1_{x_0,\dots,x_{A-1} \text{ distinct}} = \prod_{j=0}^{A-1} \left(1 - \sum_{i<j} 1_{x_j = x_i} \right) = \sum_p T_p(b). \tag{F5}$$

For a distinct list every factor in the product is one. Otherwise, at the first repeated entry the previous entries are distinct, so exactly one summand is one and the product vanishes. Expanding the product gives the second equality. The parent is always earlier, so cycles

are impossible. There are at most A^{2r} forests of rank r: choose at most A^r lists of children and at most A^r lists of parents. Applying Λ to (F5) proves that the original signed history sum is exactly $\sum_p F_p$.

7.2 The bound preserving signed cancellation

The equality constraints in (F4) cannot be inserted into the local contraction by conditioning. Instead use independent random sign fields. For each selected edge $e = (j, p(j))$, let $\epsilon_e(a)$ be independent uniform signs, indexed by the finite alphabet of primes and private symbols. Then

$$1_{x_j = x_{p(j)}} = \mathbb{E}[\epsilon_e(x_j)\epsilon_e(x_{p(j)})]. \tag{F6}$$

Independence between the edge fields gives the product of the equality indicators. For a fixed field assignment, group its signs by address. To make the multiplier at an unused slot equal to one, divide the multiplier at address j by its value at $\star_j$, and keep the product of those values as an overall scalar. That scalar and every slot multiplier have modulus one. Consequently F_p is exactly $(-1)^{r(p)}$ times an average of the same history functional with unit slot multipliers and a unit scalar outside it.

Assign the active birth slots in transition i to that transition; assign background and initial slots to adjacent visits. Only the owners of selected children and their parents can acquire a nontrivial slot multiplier. There are at most $2r(p)$ such owners. The background operator bound is uniform in all unit multipliers. At an unmarked active transition the original signed contraction still applies; at a marked transition use the bound for an arbitrary unit multiplier.

Use exactly the orbit norm $\mathcal{E}$ defined in (T1)–(T4) of Section 5.3. Its OLD residues have counting measure and its ordered memory sectors have the single factorial in (T3). The local estimates, proved in (T6)–(T23), are

$$\mathcal{E}(E_i f) \le l^2 \mathcal{E}(f), \qquad \mathcal{E}(E_i^{\phi} f) \le h^2 \mathcal{E}(f) \quad (|\phi| \le 1), \qquad \mathcal{E}(G_i^{\phi} f) \le \Gamma^2 \mathcal{E}(f). \tag{F7}$$

Here E_i is the active signed transition, G_i is the background transition, and ϕ multiplies the actual transition coefficient. In the notation (T21)–(T23), $l = L^{-S_0}$, $h = L^{\kappa}$, $b = L^{S_0+\kappa}$ and $K_r = S_0 + \kappa$. With the constants W_0, w_0, ρ_0 specified after (M15), the background bound is $\Gamma = \exp(4W_0 w_0 G/\rho_0)$.

Iterating (F7) and applying Cauchy–Schwarz to the initial and terminal vectors bounds each fixed-field scalar by

$$|B_{\text{base}}|\, \mathcal{E}(v)\, \Gamma^{N+1} \prod_{i=0}^{N-1} \begin{cases} h, & i \text{ has a marked owner,} \\ l, & \text{otherwise.} \end{cases}$$

The initial vector includes the retained initial distinctness mask and the conjugated initial slot phase; its energy is at most that of the original boundary vector v. Thus, putting $P_0 = |B_{\text{base}}|\mathcal{E}(v)\Gamma^{N+1}$ and averaging the unit scalars, we obtain

$$|F_p| \le P_0 l^N b^{2r(p)}. \tag{F8}$$

The initial boundary energy occurs once. In particular, (F8) does not pay an additional factor for the number of histories or sign fields.

7.3 The bound from prescribed prime values

For the second bound keep the equality constraints and use absolute coefficients. The tilted law satisfies

$$\sum_{p} \tilde{\mu}_g(p) \le 2, \qquad \sum_{p:\, p=t} \tilde{\mu}_g(p) \le \delta \sum_{p} \tilde{\mu}_g(p), \qquad \delta = \frac{2}{ve^{L^{3/10}}}. \tag{F9}$$

Expose births in address order. A selected child must equal a label whose value has already been exposed. In an active transition with s selected new coordinates the pinned absolute row bound is

$$\sum_{\text{legal transitions with the prescribed values}} |c(x,y)|\, w_0^{|\mathrm{OLD}(y)|} \le \delta^s a\, w_0^{|\mathrm{OLD}(x)|}. \tag{F10}$$

To obtain this bound, expand the comparison pattern, the auxiliary prime choices and the endpoint permutations before summing the new coordinates. Their masses are normalized. Each prescribed new value contributes the atom bound in (F9); cap the character kernel pointwise as in (M4). The remaining SMALL row sums and promotion damping give the same absolute row constant a independently of the prescribed values. For fixed m it satisfies $a \le L^{C_b}$. This argument retains multiplicities of transitions with the same output.

The corresponding background bound is $\delta^s \Gamma$ with the same weight $w_0^{|\mathrm{OLD}|}$. Its geometric sum converges because $1/(\rho_0 w_0) + \sqrt{q}/\rho_0^2 \le 1$. Apply these two bounds backwards to successive suffixes of the history. Each selected child is charged exactly once; a child forced to equal an incompatible group or a private unused symbol gives zero. A selected child in the initial block also gives zero, because initial distinctness was retained. Every surviving forest therefore receives all $r(p)$ atom factors, including forests with several children of one parent.

The terminal boundary is bounded by $w_0^{|\mathrm{OLD}|}$. The suffix induction, followed by the initial boundary sum, yields

$$|F_p| \le P_0 a^N \delta^{r(p)}. \tag{F11}$$

No independence of the conditioned history is asserted in this argument: (F10) is uniform in the already exposed state and prescribed values. Combining (F8) and (F11) proves (M15), for the identical coefficients F_p defined in (F3)–(F4). The rank split and removal of the OLD cutoff then proceed as in the history restoration.

8 From complete-cell discrepancy to additive arc energy

We supply the local-energy inequality and the separation of variables used in (M26). The term with factor $(H/Y)^2$ is the error in replacing a linear window by a logarithmic window. The separate error in peeling a SMALL prime is supported on its square multiples.

8.1 A finite local-energy inequality

Let (a_n) be a finite complex sequence supported on $Y/A \le n \le AY$, where $A \ge 1$, and assume $1 \le H \le Y/(8A)$. Choose a fixed smooth function K supported on $[-\delta, \delta]$, $0 < \delta \le 1$, with Lipschitz constant M_K and

$$|\widehat{K}(t)| \ge c_K > 0 \quad (|t| \le 1).$$

Such a function is obtained by shrinking a nonnegative smooth bump of integral one. Its Fourier transform stays close to one on $[-1, 1]$. Put

$$P(\beta) = \sum_n a_n \mathrm{e}(n\beta), \qquad P_a(t) = \sum_n a_n n^{it},$$
$$Q(v) = \sum_n a_n K((n-v)/H), \qquad R(v) = \sum_n a_n K(v\log(n/v)/H), \quad v > 0.$$

For every integer $j \ge 2$ there is a constant C_j, depending only on A, K, j, such that

$$c_K^2 \int_{-1/H}^{1/H} |P(\beta)|^2 \, d\beta \le \frac{C_j}{Y} \int_{\mathbb{R}} \frac{|P_a(t)|^2}{(1 + (H/Y)|t|)^j} \, dt + 2C_{\mathrm{lin}}(H/Y)^2 \sum_n |a_n|^2, \tag{E1}$$

where $C_{\mathrm{lin}} = 4\delta(4\delta+1)(16A\delta^2 M_K)^2$. The constant is uniform in the sequence, H and Y.

Indeed, $\widehat{Q}(-\beta) = H\widehat{K}(-H\beta)P(\beta)$. Plancherel bounds the left side by $H^{-2}\int |Q(v)|^2\,dv$. The support of Q lies in $V = [Y/(2A), 2AY]$. If either kernel in $Q(v) - R(v)$ is nonzero, then $|n - v| \le 2\delta H$. Taylor's estimate for $\log(1+u)$, using $v \ge Y/(2A)$ and $H \le Y/(8A)$, gives

$$|K((n-v)/H) - K(v\log(n/v)/H)| \le 16A\delta^2 M_K H/Y. \tag{E2}$$

At each v at most $(4\delta+1)H$ integers contribute; for each n the contributing v lie in an interval of length $4\delta H$. Cauchy–Schwarz and integration give

$$H^{-2}\int_V |Q(v) - R(v)|^2\,dv \le C_{\mathrm{lin}}(H/Y)^2 \sum_n |a_n|^2. \tag{E3}$$

To estimate R, choose a fixed smooth compactly supported χ equal to one on $\log(V/Y)$, set $h = H/Y$, and write

$$Q(u) = \int_{\mathbb{R}} \chi(u) e^{-u} \widehat{K}(he^{-u}t/(2\pi)) e^{-itu} Y^{-it} P_a(t)\,dt.$$

Fourier inversion with $t = 2\pi v\xi/H$ gives exactly

$$\chi(\log(v/Y))R(v) = \frac{H}{2\pi Y} Q(\log(v/Y)). \tag{E4}$$

The amplitude is smooth and compactly supported in u. Its first two u-derivatives, multiplied by any fixed power of $1 + h|t|$, are bounded uniformly in h, t, by Schwartz decay of $\widehat{K}$. Two integrations by parts bound its Fourier transform by $C(1+h|t|)^{-j}(1+|s|)^{-2}$. Including e^{-itu} replaces the last factor by $(1 + |s + t/(2\pi)|)^{-2}$. Its row and column integrals are bounded constants. Weighted Schur and Plancherel yield

$$\int |Q(u)|^2\,du \le C_j' \int (1+h|t|)^{-j}|P_a(t)|^2\,dt.$$

The Jacobian of $v = Ye^u$ is $Ye^u = O(Y)$ on the support of χ. Together with (E4) and division by H^2, this leaves precisely Y^{-1}. Now $|Q|^2 \le 2|R|^2 + 2|Q-R|^2$ proves (E1).

8.2 Exact separation of the arithmetic coefficient

Use (M21)–(M23) at rational center h_0/d. Denote the distinguished SMALL prime set by S and its harmonic mass by V_s. After peeling its distinguished mark, the main coefficient is

$$a_n = \psi(n/Y)n^{iv_1}\mathrm{e}(h_0 n/d)\varepsilon(n)\,\mathrm{Core}(n_*)\frac{1}{V_s}\sum_{s\in S,\ s\mid n} W_{\mathrm{lo}}(n/s). \tag{E5}$$

The weight W_{lo} has one fewer mark in S and keeps all other groups and the bounded list coefficient b. Expanding the core writes $n = mspr$, where m is rough, $s \in S$ is the peeled prime, and p lies in the complete ordinary prime slot. The integers m, p have no marked prime factors, and all of m, s, p are coprime to d. Thus the remaining marked weight is exactly $W_{\mathrm{lo}}(r)$, while the parity factor is $\varepsilon(s)\varepsilon(r)$. The main term does not require s to occur only to the first power.

For $d_0 = \gcd(r, d)$ put $q = d/d_0$. On units modulo q expand

$$\mathrm{e}(h_0 u/q) = \sum_{\xi \bmod q} c_{q,\xi}\xi(u), \qquad c_{q,\xi} = \frac{1}{\varphi(q)}\sum_{u\in(\mathbb{Z}/q\mathbb{Z})^\times} \mathrm{e}(h_0 u/q)\overline{\xi(u)}. \tag{E6}$$

All characters, including principal and imprimitive ones, occur. We have $|c_{q,\xi}| \le 1$; their total number over $d_0 \mid d$ is $\sum_{d_0\mid d}\varphi(d/d_0) = d$. Define

$$d_{d_0,\xi}(r) = 1_{\gcd(r,d)=d_0}\,c(r_*)W_{\mathrm{lo}}(r)\varepsilon(r)\xi(r/d_0). \tag{E7}$$

This coefficient is independent of m, s, p. In particular $c(n_*/mp)$ has become $c(r_*)$; it is not absorbed into α. The marked-weight bound gives $|d_{d_0,\xi}(r)| \le CK^{g+1}$, where $g = O(\log L)$ is the group count and K depends only on the fixed mark counts, damping and lower harmonic-mass bound. This is a fixed polynomial in L.

Partition each of m, s, p, r into canonical dyads. On one box put

$$
\begin{aligned}
S_{\xi}(t) &= \sum_{s\in S \text{ in the dyad}} \varepsilon(s)\xi(s)s^{-1+it},\\
P_{\xi}(t) &= \sum_{p \text{ in the slot dyad}} \xi(p)p^{-1+it},\\
A_{\xi}(t) &= \sum_{m \text{ in the factor interval and dyad}} \alpha(m)\chi_0(m)\xi(m)m^{-1+it},\\
D_{d_0,\xi}(t) &= \sum_{r \text{ in the dyad}} d_{d_0,\xi}(r)r^{-1+it},\\
F_{d_0,\xi}(t) &= S_{\xi}(t)P_{\xi}(t+\sigma_1)A_{\xi}(t+\sigma_0)D_{d_0,\xi}(t).
\end{aligned}
\tag{E8}
$$

With $f(u) = e^u\psi(e^u)$, Fourier inversion of the product cutoff is

$$
\frac{P_a(t)}{Y} = \frac{1}{V_s}\sum_{d_0\mid d}\sum_{\xi \bmod d/d_0} c_{d/d_0,\xi}\sum_{\text{boxes}}\int_{\mathbb{R}}\hat{f}(\nu)Y^{-2\pi i\nu}F_{d_0,\xi}(t+v_1+2\pi\nu)\,d\nu. \tag{E9}
$$

All integer sums are finite. The cutoff restricts the product to $[Y/A, AY] \subset [1, N_{\text{cap}}]$, so the independent four dyads neither omit nor add a nonzero term. The reciprocal factors in (E8) multiply to $1/n$; the factor $e^u = n/Y$ in f explains P_a/Y. There are at most $(1+\log_2 N_{\text{cap}})^4$ boxes.

For $T_0 + |\sigma_0| \le L^B$, (M24) bounds $A_{\xi}(t+\sigma_0)$ on $|t| \le T_0$. Its joint character has modulus at most $q_{\text{base}}d$, within the requested budget. If the other coefficient bounds are C_s, C_p, C_r, their reciprocal sums on dyads are at most $2C_s, 2C_p, 2C_r$. Hence

$$
\int_{-T_0}^{T_0}|F_{d_0,\xi}(t)|^2\,dt \le 2T_0\big[(2C_s)(2C_p)(2C_r)L^{-B}\big]^2. \tag{E10}
$$

This applies discrepancy only after exact separation, with no m-dependent coefficient left in the other factors.

8.3 The rest of the Mellin window

The window extends beyond polynomial frequencies, so (E10) alone does not suffice. Put $T = 2LY/H$. Discard boxes whose product interval misses the cutoff support: their contribution before Fourier separation is zero. On each remaining box with small-prime scale P, long-prime scale N, and other scales M, R, the actual comparison cells imply

$$
T \le MNR, \qquad Y \asymp PMNR, \qquad \lceil Y\rceil^{3/5} \le MR \le \lceil Y\rceil. \tag{E11}
$$

Here $\log P \in [L^{a_s}, L^{b_s}]$ up to fixed factors, $a_s < b_s < 3/10$, and $x^{\tau}/2 \le N \le x^{\eta}$, $\eta < 1/4$. The two nonempty BIG probe products ensure $H' \gg LPL^{C_{\text{arc}}}$, giving the first inequality. The lower bound for MR follows from $Y \ge e^{3L/4}$ and $PN \le e^{o(L)}x^{\eta}$; the upper bound follows from the product support. Fixed comparison constants are absorbed by the strict exponent margins and a larger threshold.

Choose $\delta = L^{-A_s}$. Where $|S_{\xi}(t)| \le \delta$, collapse the other three factors to a Dirichlet polynomial of length at most $8MNR$. Its mean square on $[-T, T]$ is bounded by

$$
[2T + 128MNR(1+\log(8MNR))]\frac{(C_mC_pC_r)^2}{MNR}(1+\log(8MNR))^{16}.
$$

Integrating the off-diagonal terms uses $|\log(n/m)|^{-1} \le \max(m,n)/|n-m|$; the divisor moment bounds the collapsed coefficients. Multiplication by δ^2 and a sufficient choice of A_s give any required logarithmic saving.

Cover the remaining times by occupied unit intervals. The factorial moment estimate for the small-prime polynomial implies that their number in each class of indices modulo three is at most $\lceil Y\rceil^{1/20}$, eventually. Raise the polynomial to a power of order $\log Y/\log P$. Unique factorization bounds ordered representations by the factorial of that power. The separated sampling estimate then gives an exceptional count $\exp(o(\log Y))$: the factorial and threshold losses have logarithm $O((\log Y/\log P)\log L) = o(\log Y)$, and $\log P = o(\log Y)$. This proves the stated exponent for every fixed A_s.

On the occupied intervals the residual satisfies the explicit bound

$$\sum_{j\in\mathcal{J}}\int_j^{j+1} |A_{\tilde\xi}(t+\sigma_0)D_{d_0,\tilde\xi}(t)|^2\,dt \le 10752(C_mC_r)^2(1+\log(4\lceil Y\rceil))^4. \tag{E12}$$

Indeed, its collapsed coefficients on $[U,4U]$, $U = MR$, have squared norm at most $(C_mC_r/U)^2\,4U(1+\log(4U))^3$. For s one-separated sample times in $[-\lceil Y\rceil,\lceil Y\rceil]$, the complete Gram row is at most

$$448U(1+\log(2\lceil Y\rceil)) + 448s\sqrt{\lceil Y\rceil}.$$

The first-derivative bound for $\sum_{U\le n\le 4U} n^{it}$ is $O(U/|t|)$ for $1 \le |t| \le U$; the second-derivative bound is $O(\sqrt{\lceil Y\rceil})$ for $U < |t| \le 2\lceil Y\rceil$. Their sum at separated times gives this row estimate. Since $s \le \lceil Y\rceil^{1/20}$ and $U \ge \lceil Y\rceil^{3/5}$, the second term is absorbed by $448U$. Schur's inequality gives $3584(C_mC_r)^2(1+\log(4\lceil Y\rceil))^4$. Choose independently in each interval a residual maximizer and a point witnessing the large small-prime sum. Indices in one class modulo three make both families one-separated. Summing the three classes proves (E12).

On occupied intervals above a sufficiently large fixed power of L, the long-prime estimate (U1), proved below, gives $|P_{\tilde\xi}(t+\sigma_1)| \le K_DL^{-D}$ for every fixed D, uniformly for the polynomial conductor budget and $|t+\sigma_1| \le x^2$. Multiply it by (E12), using $|S_{\tilde\xi}| \le 2$. The other occupied intervals lie at polynomial frequencies and are covered by (E10). Thus the entire window has arbitrary logarithmic saving for each four-factor box.

Its constants, and those of the factorial-moment and Gram bounds, precede the cutoff and box endpoints. To use (E9), restrict $|\nu|$ to a fixed polynomial range, enlarge the phase budgets, and apply Minkowski. Bound the omitted Fourier integral by repeated integration by parts. The logarithmic derivative budget for ψ supplies the required norms and tails. Cauchy–Schwarz over the d character terms and four dyads then gives, for any prescribed D after requesting sufficient precision in (M24),

$$\int_{-2LY/H}^{2LY/H} |P_a(t)/Y|^2\,dt \le V_s^{-2}d^2(1+\log_2 N_{\mathrm{cap}})^8L^{-D}. \tag{E13}$$

8.4 The high-frequency prime input

Long-prime estimate. Fix $0 < \tau < \eta < 1/4$ and $C, A > 0$. There are $B_0, K > 0$ and $x_0 \ge 3$ such that, for $x \ge x_0$, uniformly for $x^\tau/2 \le N \le x^\eta$, $1 \le q \le (\log x)^C$, every Dirichlet character χ mod q, every interval $I \subset [N,2N]$, and $(\log x)^{B_0} \le |u| \le x^2$,

$$\left|\sum_{\substack{p\in I\\ p \text{ prime}}} \chi(p)p^{-1+iu}\right| \le K(\log x)^{-A}. \tag{U1}$$

Principal and imprimitive characters are included. In particular, the estimate holds at $u = t + \sigma_1$ when $L^{B_0} + |\sigma_1| \le |t|$ and $|t + \sigma_1| \le x^2$. This lower bound on $|t|$ explains the polynomial-frequency region handled separately by (E10).

The analytic input is the Vinogradov–Korobov zero-free region

$$L(s,\chi) \neq 0 \quad \text{for} \quad \Re s \ge 1 - \frac{1}{10.5\log q + 61.5(\log|\Im s|)^{2/3}(\log\log|\Im s|)^{1/3}}, \quad q \ge 3, \quad |\Im s| \ge 10. \tag{U2}$$

This is Khale, Theorem 1.1, with the displayed conductor and height range.

Put $L = \log x$ and choose

$$\epsilon = L^{-(A+5)}, \quad H = L^{A+7}, \quad c = L^{-1}, \quad Q = 3L^C, \quad V = 3e^{3L},$$

$$D = 10.5\log Q + 61.5(\log V)^{2/3}(\log\log V)^{1/3}, \qquad \delta = (100D)^{-1}, \qquad B_0 = A + 10. \tag{U3}$$

For moduli one and two induce to modulus $3q$. The extra Euler factor is nonzero in the positive half-plane, so (U2) gives the needed region for the original character as well. Smooth the interval I with relative boundary width ϵ. The Mellin transform of this cutoff has decay constant $B_j \ll_j \epsilon^{-j} \le C_j L^{(j+1)(A+5)}$ at order j. Mellin inversion expresses the smoothed Mangoldt sum with coefficient $\Lambda(n)\chi(n)n^{-1+iu}$ using $-L'/L(1 - iu + z, \chi)$ on $\Re z = c$. Shift the central segment $|\Im z| \le H$ to $\Re z = -\delta$. Here $|u| \ge L^{A+10}$ keeps the rectangle away from the principal pole and low heights. The denominator D dominates the zero-free widths for all its conductors and heights. The logarithmic-derivative bound in this smaller rectangle, the cutoff decay, and $N^c \le e^\eta$ bound the left side, the horizontal sides, and the remaining right-line tails by

$$\mathcal{M}_j \ll B_j\left(HLN^{-\delta} + (c+\delta)e^\eta LH^{-j} + e^\eta L^2 H^{1-j}\right). \tag{U4}$$

These bounds are uniform in the interval endpoints. Since $D \ll_C L^{3/4}$ and $\log N \ge \tau L - \log 2$, $N^{-\delta}$ is smaller than every fixed inverse power of L. Choose a fixed integer $j > 2A + 30$. The other terms in (U4) are also $O(L^{-A-2})$: substituting (U3) gives exponents at most $-2j + 2A + 14$. Removing the smoothing, deleting higher prime powers, and partial summation to remove $\log p$ give

$$\left|\sum_{p\in I}\chi(p)p^{-1+iu}\right| \le \frac{1}{\log N}\left[\frac{\mathcal{M}_j}{2\pi} + \frac{2\log(3N)}{N}(2\epsilon N + 2) + \frac{4\log(4N)}{\sqrt{N}}\right]. \tag{U5}$$

The last two terms are respectively the boundary collars and the prime powers. The exponential lower bound on N makes (U5) $O(L^{-A})$, proving (U1). Finally $|t+\sigma_1| \ge |t| - |\sigma_1|$ proves the shifted assertion.

8.5 The four terms in (M26)

For the peeled coefficient $|a_n| \le B_{\mathrm{amp}} d_2(n)^3$. With $N = \lceil AY \rceil$ this gives

$$\sum_n |a_n|^2 \le B_{\mathrm{amp}}^2 N (1 + \log N)^{63}, \tag{E14}$$

because $d_2(n)^6 \le d_{64}(n)$ and $\sum_{n \le N} d_{64}(n) \le N(1 + \log N)^{63}$. The finite Dirichlet-polynomial mean-square estimate, applied on dyadic time intervals and multiplied by $(1 + (H/Y)|t|)^{-j}$, bounds the part $|t| > LY/H$ by

$$4L^{-(j-2)}[LY/H + 16N(1 + \log N)] B_{\mathrm{amp}}^2 N (1 + \log N)^{63}. \tag{E15}$$

In (E1), (E13) contributes Y times its right side, and (E15) contributes Y^{-1} times its right side. Formula (E3), with (E14), gives the third line of (M26).

The difference between the original coefficient and (E5) vanishes unless $s^2 \mid n$ for a peeled prime $s \in S$. The marked-weight bound and the fourth divisor moment on square multiples bound its squared coefficient mass by

$$3LB_s^2 N (1 + \log N)^{16} e^{-L^{a_s}}. \tag{E16}$$

The factor $3L$ pays for the dyadic SMALL-prime ranges, and $e^{-L^{a_s}}$ comes from summing s^{-2} for $s \ge e^{L^{a_s}}$. Parseval bounds its arc energy by this full coefficient mass. Combining (E13)–(E16) with (E1) and $|u + v|^2 \le 2|u|^2 + 2|v|^2$ gives the four terms in (M26), up to constants fixed before L.

9 Distribution in residue classes

We prove the negative-parity estimate used in the sieve argument. All weights, prime bands, and normalizing factors are those of the main proof. Put $x = e^L$, $Y = \lfloor 2x \rfloor$, and $X = X_{-1} = \sum_{u \le Y} w_{-1}(u)$. For odd squarefree d, set

$$r(d) = \sum_{\substack{u \le Y \\ u \equiv -2 \pmod d}} w_{-1}(u) - \frac{X}{\varphi(d)}.$$

For fixed $0 < \vartheta < 1/2$ and every $D > 0$, the result is

$$\sum_{\substack{d \le x^\vartheta \\ d \text{ odd, squarefree}}} |r(d)| \ll_{\Psi,\vartheta,D} xL^{-D} + XL^{-18}. \tag{G1}$$

The proof below concerns w_{-1}. No distribution assertion for the other parity sign is needed here.

9.1 Orthogonality and the nonunit correction

Extend Dirichlet characters by zero on nonunits, and write

$$S_d(\chi) = \sum_{u \le Y} w_{-1}(u)\chi(u), \qquad N(d) = \sum_{\substack{u \le Y \\ (u,d)>1}} w_{-1}(u).$$

Since -2 is a unit modulo an odd d, character orthogonality gives

$$\varphi(d) r(d) = \sum_{\chi \ne \chi_0} \chi((-2)^{-1}) S_d(\chi) - N(d). \tag{G2}$$

The correction $N(d)$ occurs because the principal character sums to $X - N(d)$, rather than X. In particular,

$$|r(d)| \le \frac{1}{\varphi(d)} \left(\sum_{\chi \ne \chi_0} |S_d(\chi)| + N(d) \right). \tag{G3}$$

Let

$$H(N) = \sum_{\substack{d \le N \\ d \text{ squarefree}}} \frac{1}{\varphi(d)}.$$

For a fixed u, assign a noncoprime squarefree d to any prime $p \mid (u,d)$ and write $d = pe$. Dropping the conditions $(p,e) = 1$ and $e \le N/p$ gives

$$\sum_{\substack{d \le N \\ d \text{ squarefree} \\ (u,d)>1}} \frac{1}{\varphi(d)} \le H(N) \sum_{p \mid u} \frac{1}{p-1}. \tag{G4}$$

The squarefree totient sum satisfies $H(\lfloor x^{\vartheta} \rfloor) \le eL$ eventually. Every prime divisor of a supported $u \le 2x$ is at least L^{20}. Also $2^{\omega(u)} \le u$, where ω counts distinct prime divisors. Therefore, eventually,

$$\sum_{p \mid u} \frac{1}{p-1} \le \frac{2\omega(u)}{L^{20}} \le \frac{4}{\log 2} L^{-19}.$$

Multiply (G4) by $w_{-1}(u) \ge 0$ and interchange the finite sums. This proves

$$\sum_{\substack{d \le x^{\vartheta} \\ d \text{ odd, squarefree}}} \frac{N(d)}{\varphi(d)} \le \frac{4e}{\log 2} X L^{-18}. \tag{G5}$$

9.2 Exposing one prime variable

Reserve one labelled slot in Q_0, whose prime values satisfy

$$x^{\alpha} \le q \le x^{6\alpha/5}, \qquad \alpha = 1/44.$$

The ordered-slot expansion of the ordinary coefficient gives an exact convolution

$$w_{-1}(u) = \Psi(u/x) \sum_{\substack{q \in Q_0 \\ q \mid u}} b_L(u/q). \tag{G6}$$

Here b_L is the sum over the remaining ordinary slots and all marked group factors, with the original factorial normalization and negative parity projection. Deleting the slot changes the number of slot variables, not that normalization. If a prime occurs several times, its different slot assignments remain in this finite expansion. After factorial normalization the ordinary-slot coefficient is at most one, and the marked factors are bounded by a fixed power of L. Hence, for a fixed $c \ge 0$,

$$0 \le b_L(R) \le L^c. \tag{G7}$$

If χ modulo d is induced by a primitive character χ^* of conductor $f \mid d$, put $h = d/f$. Inserting (G6) gives the exact sum

$$S_d(\chi) = \sum_{\substack{q \in Q_0,\ R \ge 1 \\ (q,h)=(R,h)=1}} b_L(R) \Psi(qR/x) \chi^*(q) \chi^*(R). \tag{G8}$$

The two coprimality masks are retained. Since d is squarefree, $(f,h) = 1$ and $\varphi(d) = \varphi(f)\varphi(h)$. Summing the complementary level h costs at most $H(\lfloor x^{\vartheta} \rfloor) = O(L)$.

Partition the two variables into binary dyadic intervals $q \asymp P$, $R \asymp U$. The support $x \le qR \le 2x$ leaves $O(L)$ pairs of scales, and they satisfy

$$P \le x^{6\alpha/5}, \qquad U \le 4x^{1-\alpha}, \qquad x/4 < PU \le 2x. \tag{G9}$$

It remains to bound the primitive-character sums on each such rectangle.

9.3 Small and large conductors

The two analytic inputs are the prime Siegel–Walfisz estimate and the multiplicative large sieve. In the forms used here, Siegel–Walfisz gives arbitrary logarithmic saving for prime partial sums twisted by a nonprincipal character of modulus at most a fixed power of L. The large sieve bounds the primitive-character quadratic mean of a sequence on an interval of length N by its squared ℓ^2 norm times $Q^2 + \pi N$, with the usual $f/\varphi(f)$ conductor weight for $f \le Q$.

Choose a conductor cutoff $2^K \le L^{C_*}$. For $f \le 2^K$, apply Siegel–Walfisz to the q variable in (G8). Abel summation incorporates $\Psi(qR/x)$; its supremum and total variation are fixed constants. The excluded primes $q \mid h$ number at most ϑ/α, since $q \ge x^\alpha$ and $h \le x^\vartheta$. Thus the small-conductor contribution on one retained rectangle is at most

$$L^{C_*+c}\left(2xB_\Psi L^{-A}V_\Psi + 4x^{1-\alpha}\frac{\vartheta}{\alpha}S_\Psi\right), \tag{G10}$$

where B_Ψ, S_Ψ, V_Ψ are fixed nonnegative constants from the prime partial-sum estimate and the cutoff. The exponent A may be chosen arbitrarily large once C_* is fixed.

For $F < f \le 2F$, use Fourier inversion in the logarithmic variable to separate $\Psi(qR/x)$. If $\psi(t) = \Psi(e^t)$, write $M_\Psi = \int_{\mathbb{R}} |\widehat{\psi}(s)|\,ds < \infty$. At each frequency, Cauchy–Schwarz in the primitive characters, followed by the large sieve for the two factors, yields

$$\mathcal{B}_F \le M_\Psi L^c \sqrt{\frac{(4F^2 + \pi P)P}{F}}\sqrt{\frac{(4F^2 + \pi U)U}{F}}. \tag{G11}$$

Here $\mathcal{B}_F$ is the sum of the absolute bilinear character sums weighted by $1/\varphi(f)$, for the fixed complementary level h and rectangle. The masks in (G8) only decrease the coefficient norms. Expanding the right side with $\sqrt{a+b} \le \sqrt{a} + \sqrt{b}$, using (G9), and summing over $2^K < F \le 2^T$ gives

$$\sum_F \mathcal{B}_F \le M_\Psi L^c\left(8\cdot 2^T x^{1/2} + 12\sqrt{\pi}\,Tx^{1-\alpha/2} + \frac{4\pi x}{2^K}\right). \tag{G12}$$

For a requested exponent D, take

$$k = c + D + 2, \qquad C_* = k + 1, \qquad A = C_* + c + D + 2,$$

and choose integer K, T with

$$L^k \le 2^K \le L^{k+1}, \qquad x^\vartheta \le 2^T \le 2x^\vartheta.$$

The two remaining costs, the complementary levels and the rectangle count, are each $O(L)$. They turn the first term of (G10) and the last term of (G12) into $O(xL^{-D})$. All other terms contain a fixed negative power of x after division by x: the first term in (G12) uses exactly $\vartheta < 1/2$. These powers absorb every fixed logarithmic factor. We conclude that

$$\sum_{\substack{d \le x^\vartheta \\ d \text{ odd, squarefree}}} \frac{1}{\varphi(d)} \sum_{\chi \ne \chi_0} |S_d(\chi)| \ll xL^{-D}. \tag{G13}$$

Equations (G3), (G5), and (G13) prove (G1). At $D = 230$, the lower mass bound $X \gg xL^{-221}$ makes its right side $o(X/L)$.

10 The quantitative finite block sieve

We prove the finite estimate used for the rectangle upper bound and the one-dimensional sieves. Its constants are independent of the block count.

10.1 Statement

Let $z > 1$, let $h \geq 2$ be even, and let P_j, $0 \leq j < J$, be finite sets of primes in $(z^{2^{-(j+1)}}, z^{2^{-j}}]$. These blocks are disjoint; put $P = \bigcup_j P_j$. Let $\mathcal{X}$ be a finite set of objects with weights $w(x) \geq 0$ and bad-prime sets $B(x)$. Suppose $X \geq 0$, $0 \leq g(p) \leq 1 - \eta$ for $p \in P$, where $\eta > 0$, $C \geq 0$, and $\sum_{p \in P_j} g(p) \leq C$. For every $s \subseteq P$, assume

$$\sum_{x \in \mathcal{X}} w(x) \prod_{p \in s} 1_{p \in B(x)} = X \prod_{p \in s} g(p) + r\left(\prod_{p \in s} p\right). \tag{BS1}$$

Write

$$V = \prod_{p \in P} (1 - g(p)), \qquad D = z^{4h+2}, \qquad \mathcal{E} = \sum_{\substack{d \mid \prod_{p \in P} p \\ d \leq D}} |r(d)|,$$

$$K(C, \eta) = \frac{\exp(eC + C/\eta)}{1 - e^{-1}}, \qquad \rho = K(C, \eta) e^{-h}, \qquad S = \sum_{x \in \mathcal{X}} w(x) 1_{B(x) \cap P = \emptyset}.$$

Then

$$XV(1 - \rho e^{\rho}) - \mathcal{E} \leq S \leq XVe^{\rho} + \mathcal{E}. \tag{BS2}$$

Every divisor in $\mathcal{E}$ is squarefree. Both J and h may vary.

10.2 Proof

Set $k_j = (j+1)h$. For variables $t_p \in [0, 1]$, define

$$a_j(t) = \prod_{p \in P_j} (1 - t_p),$$

$$u_j(t) = \sum_{n=0}^{k_j} (-1)^n \sum_{\substack{s \subseteq P_j \\ |s| = n}} \prod_{p \in s} t_p,$$

$$e_j(t) = \sum_{\substack{s \subseteq P_j \\ |s| = k_j + 1}} \prod_{p \in s} t_p.$$

Even Bonferroni truncation gives $a_j \leq u_j$ and $0 \leq u_j - a_j \leq e_j$. To verify it, take independent Bernoulli variables of means t_p and average $\sum_{n=0}^{k} (-1)^n \binom{q}{n} = (-1)^k \binom{q-1}{k}$ for $q > 0$, handling $q = 0$ separately. A telescoping expansion now gives

$$\prod_j u_j - \sum_j e_j \prod_{i \neq j} u_i \leq \prod_j a_j \leq \prod_j u_j. \tag{BS3}$$

Apply (BS3) with $t_p = 1_{p\in B(x)}$, multiply by $w(x)$ and sum. Both bounding polynomials have monomial coefficients of absolute value at most one. A subset contributes to the upper polynomial only when its degree in every block is at most k_j. It contributes to the subtracted sum only when exactly one block has degree $k_j + 1$ and all other degrees are at most their cutoffs. These cases are disjoint, and the exceptional block is unique.

Every surviving monomial has prime product at most D: its exponent of z is at most $h\sum_{j\ge 0}(j+1)2^{-j} + 1 = 4h + 1 < 4h + 2$. Distinct prime subsets have distinct products. Inserting (BS1) therefore costs at most $\mathcal{E}$ for each bounding polynomial, with no extra divisor coefficients or factor J.

Evaluate next at $t_p = g(p)$. Abbreviate $a_j = a_j(g) > 0$. Since $\log(1-t) \ge -t/\eta$ for $0 \le t \le 1-\eta$, $a_j \ge e^{-C/\eta}$. Moreover

$$e_j(g) \le \frac{C^{k_j+1}}{(k_j+1)!} \le \exp(eC - k_j - 1),$$

by retaining one term in the series for $\exp(eC)$. Thus $\delta_j = e_j(g)/a_j$ satisfies $\sum_j \delta_j \le \rho$. From $a_j \le u_j(g) \le a_j(1+\delta_j)$ we obtain

$$V \le \prod_j u_j(g) \le Ve^{\rho}, \qquad \sum_j e_j(g)\prod_{i\ne j} u_i(g) \le V\rho e^{\rho}.$$

Together with (BS3) and the remainder bound, these inequalities prove (BS2).

10.3 Constants in the applications

For rectangles, the local estimates give $C = 12$, $\eta = 2/9$, and $h = 2$. Hence $\rho = e^{12e+52}/(1-e^{-1})$ and $D = z^{10}$, exactly as in (R7). The block count may grow.

For the preliminary sieve on $u + 2$, the density is zero at 2 and $1/(p-1)$ at odd primes. The reciprocal-prime block bounds give $C = 8$, and $\eta = 1/2$. Take $h = 2\lfloor 1/(40b)\rfloor$. If $0 < b \le 1/1000$, then $h \ge 1/(40b)$, $h \ge 50$ and $\rho \le 1$. Consequently

$$\rho e^{\rho} \le C_s e^{-1/(40b)}, \qquad C_s = 3K(8, 1/2).$$

This is the constant in (S7). The level exponent satisfies $b(4h+2) \le 1/5 + 2b$. For the conditioned sieve in (S8b), use the same C, η and let an even depth tend to infinity while keeping W^{4h+2} below the remaining level after conditioning on the prime. Estimate (BS2) is uniform in this depth and the block count.

For ordinary rough integers the density is $1/p$, with block bound $C = 4$ and gap $\eta = 1/2$. Put $K_0 = K(4, 1/2)$ and $C_{\text{sieve}} = K_0 e^{K_0}$. Since $\rho = K_0 e^{-h} \le K_0$, both relative errors in (BS2) are at most $C_{\text{sieve}} e^{-h}$. Counting multiples of each squarefree divisor has error at most one, so the total error is at most $2D$, as used in (S8d). Both C_s and C_{sieve} are fixed before b is chosen.

11 A positive estimate on factor rectangles

We prove the finite estimate underlying (S2) in the sieve argument. The notation for the ordinary bands Q_j, marked groups P_g, and harmonic factors H_Q, H_P is fixed in the weight chapter. Write $x = e^L$ and

$$\theta_0 = \frac{3}{2}\frac{1}{44}2^{-20}, \qquad z = x^{\theta_0}, \qquad Z = \lfloor z^{10} \rfloor.$$

Let $K \subset [A, B)$ and $N \subset [C, E)$ be finite sets of positive integers, with natural endpoints $A \le B$, $C \le E$. Set

$$\mathcal{A} = (B - A)(E - C), \qquad T = (B - A) + (E - C).$$

For $2 \le W \le z$, use the indicator R_W from (C1), which vanishes at 0 and 1. The finite rough-factor sum is

$$\mathcal{R}_\rho(K, N; W) = \sum_{u \le Y} w_\rho(u) \sum_{\substack{m \in K,\ n \in N \\ mn = u+2}} R_W(m) R_W(n).$$

The argument only uses $|\rho| \le 1$, $0 \le \Psi \le 1$, and $\Psi(0) = 0$; the outer cutoff Y may be any natural number. Uniformly in these data, for sufficiently large L,

$$\begin{aligned} \mathcal{R}_\rho(K, N; W) \le{}& K_0 \mathcal{A} H_Q H_P V(W)^2 V(z) \\ &+ 200^{400} H_Q \left(2TZx^{1/20} + 4Z^2 x^{1/10}\right)(1 + \log Z)^3, \end{aligned} \tag{R1}$$

where $V(y) = \prod_{p \le y}(1 - 1/p)$ and K_0 is an absolute constant. The proof keeps the additional factor $V(z)$ in the main term.

11.1 Assignments and random permissions

Since $A_\rho \le A_0$, it suffices to majorize the unsigned weight. Keep the twenty ordinary slots in each band $j \ge 20$, and choose one marked prime $p_g \in P_g$ from each group. For an assignment a, put

$$D_a = \prod_{j=20}^{d} \prod_{i=1}^{20} q_{j,i} \prod_g p_g, \qquad \eta = \frac{1}{(20!)^{d+1-20}} \prod_g \frac{1}{M_g}.$$

The geometric bound on the ordinary scales and the marked-band bounds give $D_a \le x^{1/20}$ eventually. Every prime divisor of D_a is at least L^{20}, so D_a is odd.

Let $\mathcal{P} = \bigcup_g P_g$ and $U_a = \{p \in \mathcal{P} : p \nmid D_a\}$. For each subset $S \subset U_a$, give S weight $2^{-|U_a|}$ and call its primes permitted. Ban every odd prime at most z that neither divides D_a nor belongs to S. For a fixed u, independent choices of permission give the exact identity

$$2^{-|U_a|} \sum_{S \subset U_a} \prod_{p \in U_a \setminus S} 1_{p \nmid u} = \prod_{p \in U_a} \begin{cases} 1/2, & p \mid u, \\ 1, & p \nmid u. \end{cases} \tag{R2}$$

When the marked prime is summed over the distinct divisors from a group, (R2) recovers its weight $\omega_g(u) 2^{-(\omega_g(u)-1)}/M_g$. Prime multiplicities remain in u; the permissions concern distinct primes.

The omitted ordinary bands $j < 20$ consist of primes larger than z. Thus, while the full ordinary factorization is still retained, every prime at most z in the ordinary part must divide the tail assignment. This proves the deterministic exclusions needed for the mask majorant. We may then drop the remaining nonnegative factorial coefficients and enlarge the tail-assignment sum. Consequently the weighted rectangle sum is bounded by the sum, over a and masks S, of $\eta 2^{-|U_a|}$ times the number of pairs in the rectangle satisfying

$$mn \equiv 2 \pmod{D_a}, \qquad R_W(m)R_W(n) = 1, \qquad p \nmid mn - 2 \text{ for every banned } p. \tag{R3}$$

The assignment moments used to sum this bound are

$$\begin{aligned} \sum_a \frac{\eta}{D_a} &= \prod_{j=20}^{d} \frac{V_j^{20}}{20!} \le 200^{400} H_Q, \\ \sum_a \eta &\le 200^{400} x^{1/20} H_Q, \\ \sum_a \eta D_a &\le 200^{400} x^{1/10} H_Q. \end{aligned} \tag{R4}$$

In the first identity the marked reciprocal sums cancel M_g exactly. The inequality uses $V_j^{20}/20! \ge 200^{-20}$ for the twenty omitted bands. The other two inequalities follow from $D_a \le x^{1/20}$.

11.2 Local densities and the finite sieve

Fix an assignment and mask. Write $D = D_a$. The base congruence $mn \equiv 2 \pmod{D}$ has exactly $\varphi(D)$ residue pairs modulo D: choose the unit m, then $n = 2m^{-1}$. At an odd prime $p \nmid D$, the bad residue pairs are $mn = 0$ when $p \le W$, and $mn = 2$ when p is banned. These events are disjoint and have $2p - 1$ and $p - 1$ solutions, respectively. Thus their combined density is

$$g(p) = \frac{(2p-1)1_{p \le W} + (p-1)1_{p \text{ banned}}}{p^2}, \qquad 0 \le g(p) \le 7/9, \quad g(p) \le 3/p. \tag{R5}$$

For squarefree d coprime to D, the Chinese remainder theorem makes the densities multiplicative. Counting each residue class in the two integer intervals gives

$$\#\{\text{base pairs bad at every } p \mid d\} = \frac{\varphi(D)\mathcal{A}}{D^2} \prod_{p \mid d} g(p) + r_a(d),$$

with the explicit boundary estimate

$$|r_a(d)| \le \tau(d)^2 (2T + 4Dd). \tag{R6}$$

Indeed, the count in a residue class differs from its interval length divided by the modulus by at most one in each coordinate; expansion of the product leaves the two side terms and a corner term. Summing over the allowed CRT classes and using the local bounds in (R5) gives (R6).

Apply the finite block sieve with parameter $h = 2$. The reciprocal-prime block bounds give block mass at most 12, and (R5) gives the uniform gap $1 - g(p) \geq 2/9$. The finite sieve is obtained by upper Bonferroni truncation in each block; it bounds the sifted count by

$$e^{\mathfrak{r}_2} \frac{\varphi(D)\mathcal{A}}{D^2} \prod_{\substack{3 \leq p \leq z \\ p \nmid D}} (1 - g(p)) + \sum_{d \leq Z} |r_a(d)|, \qquad \mathfrak{r}_2 = \frac{e^{12e+52}}{1 - e^{-1}}. \tag{R7}$$

Here the sieve level is z^{10}, so the integer cutoff is Z. The explicit constant follows from (BS2) with $C = 12$, $\eta = 2/9$. Its block count may grow with L; the displayed bound is uniform in that count by the finite proof given there.

The elementary divisor estimate $\sum_{d \leq Z} \tau(d)^2 \leq Z(1 + \log Z)^3$ now gives

$$\sum_{d \leq Z} |r_a(d)| \leq (2TZ + 4DZ^2)(1 + \log Z)^3. \tag{R8}$$

The mask weights sum to one. Averaging (R8) over masks and then applying the last two moments in (R4) gives exactly the error in (R1).

11.3 Averaging the Euler product

Put $v_p = 1 - 1/p$. Direct comparison of the four possibilities in (R5) gives

$$1 - g(p) \leq v_p^{2\,1_{p \leq W} + 1_{p \text{ banned}}} \left(1 + \frac{1}{p(p-1)}\right). \tag{R9}$$

The product of the correction factors over odd primes is at most $e^{1/2}$, since $\log(1 + t) \leq t$ and $\sum_{n \geq 3} 1/(n(n-1)) = 1/2$. Before deleting assignment primes or permitting marked primes, the remaining product is

$$\prod_{3 \leq p \leq z} v_p^{1 + 2\,1_{p \leq W}} = 8V(W)^2 V(z). \tag{R10}$$

The factor eight restores the three omitted factors at the prime two. Permitting $p \in S$ removes one power of v_p. Averaging these removals over the masks gives

$$2^{-|U_a|} \sum_{S \subseteq U_a} \prod_{p \in S} v_p^{-1} = \prod_{p \in U_a} \left(1 + \frac{1}{2(p-1)}\right) \leq \prod_g Z_g(1/2) \leq H_P. \tag{R11}$$

Deleting primes dividing D_a removes at most three powers per prime. Their roughness and $D_a \leq x^{1/20}$ imply, eventually,

$$\prod_{p \mid D_a} v_p^{-3} \leq e. \tag{R12}$$

For example, $-\log(1 - 1/p) \leq 2/p$, and the number of distinct prime divisors is at most $\log D_a / \log 2$; hence the logarithm of the left side tends to zero uniformly.

Finally $\varphi(D_a)/D_a^2 \leq 1/D_a$. Combining (R4) and (R7)–(R12) proves the main term in (R1), with the explicit possible value

$$K_0 = 200^{400}\, 8e^{3/2 + \mathfrak{r}_2}.$$

11.4 Normalization by the actual mass

Assume $\mathcal{A} \le C_{\mathrm{area}} x$ and $T \le C_{\mathrm{side}} x^{27/50}$, with fixed constants. Since $10\theta_0 \le 1/20$, we have $Z \le x^{1/20}$ and $(1+\log Z)^3 \le 8L^3$ for $L \ge 1$. The error in (R1) is therefore at most a fixed constant times

$$H_Q L^3 (x^{16/25} + x^{1/5}), \tag{R13}$$

where $27/50 + 1/20 + 1/20 = 16/25$ and $2/20 + 1/10 = 1/5$. The mass bound $X_\rho \gg x H_Q H_P / L$, Mertens' estimate $V(z) \asymp L^{-1}$, and $H_P \ge 1$ show that

$$X_\rho V(W)^2 \gg x H_Q / L^3 \qquad (2 \le W \le z).$$

The main term of (R1) is $O(X_\rho V(W)^2)$, and the ratio of (R13) to this scale is at most a constant times $L^6(e^{-9L/25} + e^{-4L/5}) \to 0$. Thus there is $C_{\mathrm{box}} > 0$ such that

$$\mathcal{R}_\rho(K, N; W) \le C_{\mathrm{box}} X_\rho V(W)^2 \qquad (\rho = \pm 1).$$

Both the constant and the eventual threshold precede the two signs, the admissible threshold W, the finite carriers, and their endpoints. In particular this applies to $W = e^{\sqrt{L}}$.

12 Sieve estimates and the remaining composites

We use the definitions of w_ρ, X_ρ, and $\mathfrak{S}$ in the main proof. The common outer cutoff is $Y = \lfloor 2e^L \rfloor$, because Ψ is supported in $[1,2]$.

12.1 Local densities

Write

$$V(y) = \prod_{p \le y} (1 - 1/p), \qquad V_2(y) = \prod_{3 \le p \le y} (1 - 1/(p-1)).$$

Every prime factor of a supported integer u is at least L^{20}. For large L this makes u, and therefore $u+2$, odd. At an odd prime the excluded unit residue is -2, and the density of the divisibility condition is $g(p) = 1/(p-1)$. The identity

$$1 - \frac{1}{p-1} = (1 - 1/p)\left(1 - \frac{1}{(p-1)^2}\right)$$

gives $V_2(y)/V(y) \to \mathfrak{S}$. Consequently $V_2(x^b) \sim \mathfrak{S} e^{-\gamma_E}/(bL)$ for fixed $b > 0$. The product defining $\mathfrak{S}$ converges because $\sum_{p \ge 3} (p-1)^{-2} < \infty$. It is at least one: each factor lies in $(0,1)$, so comparison with all integer indices gives

$$\prod_{p \ge 3} \left(1 - \frac{1}{(p-1)^2}\right) \ge \prod_{n=2}^{\infty} (1 - n^{-2}) = \frac{1}{2}.$$

The congruence estimate used here is specifically for the negative parity weight. Put

$$r(d) = \sum_{d \mid u+2} w_{-1}(u) - \frac{X_{-1}}{\varphi(d)}.$$

For every fixed $0 < \vartheta < 1/2$, some $C \ge 0$ satisfies, eventually,

$$\sum_{\substack{d \le x^\vartheta \\ d \text{ odd and squarefree}}} |r(d)| \le C(xL^{-230} + X_{-1}L^{-18}). \tag{S1}$$

The residue-class chapter proves the stronger version with any fixed accuracy in place of 230. Character orthogonality separates nonprincipal characters from the nonunit correction. A reserved ordinary prime slot, Siegel–Walfisz, and the multiplicative large sieve control the former; the L^{20}-rough support controls the latter. By $X_{-1} \ge c_0 x L^{-221}$, the right side of (S1) is $o(X_{-1}/L)$: the two ratios are bounded by constants times L^{-8} and L^{-17}.

12.2 Positive rectangles

Let A, B, C, E be natural numbers with $A \le B$ and $C \le E$, and let $K \subset [A, B)$ and $N \subset [C, E)$ be finite sets of positive integers. For a roughness threshold W, define the nonnegative sum

$$\mathcal{R}_\rho(K, N; W) = \sum_{u \le Y} w_\rho(u) \sum_{\substack{m \in K,\ n \in N \\ mn = u+2}} R_W(m) R_W(n).$$

Fix positive constants $C_{\mathrm{area}}, C_{\mathrm{side}}$. If

$$(B-A)(E-C) \le C_{\mathrm{area}} x, \qquad (B-A)+(E-C) \le C_{\mathrm{side}} x^{27/50},$$

then, for $W = e^{\sqrt{L}}$, there is a constant $C_{\mathrm{box}} > 0$ such that

$$\mathcal{R}_{\rho}(K, N; W) \le C_{\mathrm{box}} X_{\rho} V(W)^2. \tag{S2}$$

The constant and the sufficiently large threshold for L precede both parity signs and all the finite carriers and endpoints.

This is the specialization $W = e^{\sqrt{L}}$ of the normalized rectangle estimate. Its extra sieve factor $V(x^{\theta_0})$ controls prime divisors of $mn-2$, while the two factors $V(W)$ impose roughness on m, n. That normalization proves the stated uniformity before any balanced width is chosen.

12.3 Counting balanced rectangles

Put

$$\mathcal{B}_{\kappa} = [x^{1/2-2\kappa}, x^{1/2+2\kappa}], \qquad 0 \le \kappa \le 1/50.$$

Partition each positive factor by its unique binary index $k = \lfloor \log_2 m \rfloor$. Only rectangles containing a pair with $mn = u+2$ and $\Psi(u/x) \neq 0$ need to be retained. If (k, j) is such a pair of indices, its geometry implies

$$x/4 < 2^k 2^j \le 4x, \qquad x^{1/2-2\kappa} < 2^{k+1}, \quad 2^k \le x^{1/2+2\kappa}.$$

The number of first indices is at most $4\kappa L/\log 2 + 2$. For each first index there are at most five second indices, because their powers of two lie in an interval with ratio sixteen. Thus the number of active rectangles is at most

$$C_{\mathrm{dyad}}(\kappa L + 1), \qquad C_{\mathrm{dyad}} = 5(4/\log 2 + 2). \tag{S5}$$

The active rectangles partition the supported factor pairs. Each has area at most $4x$ and side sum at most $2x^{1/2+2\kappa} \le 2x^{27/50}$, so (S2) applies with fixed constants.

Summing (S2) and applying (S5) gives

$$\sum_{\substack{mn=u+2 \\ m,n \in \mathcal{B}_{\kappa} \\ P^-(m), P^-(n) > W}} w_{\rho}(u) \le C_1(\kappa L + 1) X_{\rho} V(W)^2. \tag{S6}$$

The constant C_1 is chosen before κ.

12.4 Removing the unbalanced composites

The approximation estimate in Section 3 permits replacement of the roughness test on the cofactor of a least prime divisor. The number of bins is fixed before the approximation parameters; the parameter theorem remains uniform over the subsequent choice of endpoints.

For small fixed $b > 0$, define the rough-number density by

$$D_b(w) = \frac{1}{w} + \sum_{k\ge 2} \frac{1}{k!} \int_{\substack{t_1,\dots,t_{k-1}\ge b \\ t_1+\cdots+t_{k-1}\le w-b}} \frac{dt_1 \cdots dt_{k-1}}{t_1 \cdots t_{k-1}(w - t_1 - \cdots - t_{k-1})}.$$

Only finitely many terms can be nonzero, since the domain is empty when $kb > w$. The finite block sieve gives the preliminary-sieve constant $C_s = 3\exp(8e+16)/(1-e^{-1})$, independent of b. Set

$$A_b = \frac{e^{-\gamma_E}}{b}\left(1 - C_s e^{-1/(40b)}\right).$$

The preliminary sieve and (S1) give, eventually,

$$\sum_{P^-(u+2)>x^b} w_{-1}(u) \ge \left(\mathfrak{S}A_b - \frac{\mathfrak{S}}{10}\right)\frac{X_{-1}}{L}. \tag{S7}$$

Choose a finite monotone partition of $(b, 1/2-\kappa]$. For a bin $(\gamma, \gamma']$, the cofactor test is $1_{P^-(m)>x^\gamma}$ and the other factor is a prime $n \in (x^\gamma, x^{\gamma'}]$. The sum over all such ordered pairs dominates the composites whose least prime factor is in that bin: choose $n = P^-(u+2)$ and $m = (u+2)/n$. No uniqueness of the other representations is needed because their weights are nonnegative.

For a fixed bin $(\gamma, \eta]$, put

$$C_{\gamma,\eta} = \sup_{\gamma\le t\le\eta} D_\gamma(1-t), \qquad F_{\gamma,\eta} = \sum_{x^\gamma < p \le x^\eta} \sum_{\substack{p \mid u+2 \\ P^-(u+2)>W}} w_{-1}(u), \quad W = e^{\sqrt{L}}.$$

The cofactor $m = (u+2)/p$ has logarithmic exponent in $[1-\eta, 1-\gamma+o(1)]$. The full-cell roughness approximation satisfies, for every fixed $\delta > 0$, eventually on these pairs,

$$0 \le B_{R_{x^\gamma}}(m) \le \frac{C_{\gamma,\eta}+\delta}{LV(W)}. \tag{S8a}$$

To obtain (S8a), count products of a fixed number of primes in each complete logarithmic cell. The prime number theorem and partial summation give the reciprocal-simplex integrals defining D_γ. There are only finitely many possible numbers of prime factors on the fixed compact exponent range. The denominator in the cell ratio is its length times $V(W)(1+o(1))$, by the finite rough sieve. Joint continuity of $D_\gamma(w)$ on this compact range absorbs the $o(1)$ overshoot at $1-\gamma$ and gives the displayed supremum bound.

If the cofactor approximation is nonzero, then m is W-rough. Since the prime $p > x^\gamma$ also exceeds W, the product $u+2 = mp$ is W-rough, even when $p \mid m$. Reindexing the ordered pair sum by p and u therefore bounds its approximated value by $(C_{\gamma,\eta}+\delta)F_{\gamma,\eta}/(LV(W))$.

The conditioned sieve gives

$$\frac{F_{\gamma,\eta}}{X_{-1}V(W)} \longrightarrow \mathfrak{S}\log(\eta/\gamma). \tag{S8b}$$

Here one chooses $\eta < \vartheta < 1/2$ in (S1). For a fixed $p \in (x^\gamma, x^\eta]$, sieve the condition $p \mid u+2$ by primes at most W. Since $p > W$, its local density separates as $g(pd) = g(p)g(d)$ for each sieve divisor d. The main term is $X_{-1}g(p)V_2(W)$. Summing over p gives $\sum g(p) \to \log(\eta/\gamma)$ by the prime reciprocal-sum estimate. The sieve level can be chosen below x^ϑ/p;

its logarithm divided by $\log W$ tends to infinity uniformly in this bin. Thus its relative sieve error tends to zero. In the remainder sum, each integer $pd \le x^{\vartheta}$ has at most ϑ/γ possible prime factors from the bin, a fixed bound. Estimate (S1) consequently controls the summed remainders. Finally $V_2(W)/V(W) \to \mathfrak{S}$, proving (S8b).

Combining (S8a) and (S8b), and taking δ sufficiently small, gives for any $\varepsilon > 0$ the per-bin bound

$$U_{\gamma,\eta}^{\text{proxy}} \le \left(\mathfrak{S}C_{\gamma,\eta}\log(\eta/\gamma) + \varepsilon\right)\frac{X_{-1}}{L}. \tag{S8c}$$

For completeness, the choice of partition in this sum is independent of all subsequent approximation parameters. Set $c = 1/2 - \kappa$. The function $F(s,t) = D_s(1-t)$ is continuous on $[b,c]^2$. Given $\varepsilon > 0$, uniform continuity permits an affine grid $t_i = b + (c-b)i/N$, with $N > 0$ fixed, for which

$$\sum_{i<N} C_{t_i,t_{i+1}}\log(t_{i+1}/t_i) \le \int_b^c D_t(1-t)\,\frac{dt}{t} + \varepsilon.$$

Indeed, on each cell the supremum differs from $F(t,t)$ by at most $\varepsilon/\log(c/b)$; integration against dt/t makes these errors sum to at most ε.

Partitioning each prime-factor simplex by its least coordinate gives

$$\int_b^{1/2} D_t(1-t)\,\frac{dt}{t} = D_b(1) - 1.$$

The one-factor term is 1. In a k-factor term, choose its least coordinate t and integrate the remaining $k-1$ coordinates with lower bound t; the k choices change $1/k!$ to $1/(k-1)!$. The sets where two coordinates tie have measure zero. All terms are nonnegative and only finitely many can be nonzero, which justifies the partition and integration. Restricting the integral to $[b,c]$ only reduces it. Choose the grid above with $\varepsilon = 1/10$. Sum (S8c) over this fixed grid and use the approximation error $o(X_{-1}/L)$ from Section 3. Allocating the finite per-bin errors and the replacement error gives

$$U \le \left(\mathfrak{S}(D_b(1) - 1 + 1/10) + \frac{\mathfrak{S}}{10}\right)\frac{X_{-1}}{L}. \tag{S8}$$

The needed small-b estimate is the one-sided bound

$$D_b(1) \le \frac{e^{-\gamma_E}}{b}\left(1 + C_{\text{sieve}}e^{-1/(40b)}\right), \qquad 0 < b \le 1/100. \tag{S8d}$$

To prove this, let $R_b(z)$ count the integers in $(0,z]$ with no prime factor at most z^b, and take the even sieve depth $h = 2\lfloor 1/(40b)\rfloor$. The finite rough sieve gives

$$R_b(z) \le zV(z^b)(1 + C_{\text{sieve}}e^{-h}) + 2z^{b(4h+2)}.$$

For $b \le 1/100$, $h \ge 1/(40b)$ and $b(4h+2) \le 1/5 + 2b < 1$. The last term disappears after multiplication by $\log z/z$. Mertens' formula gives $(\log z)V(z^b) \to e^{-\gamma_E}/b$. On the other hand, counting products of distinct primes in compact subsets of the simplices by the prime number theorem gives

$$D_b(1) \le \liminf_{z\to\infty}\frac{\log z}{z}R_b(z).$$

To see the factorial, count ordered k-tuples and divide by $k!$; distinct tuples have exactly $k!$ orderings. Repeated-prime tuples may be discarded in this lower bound. Refine the compact grids and let their boundaries tend to the simplex boundary; positivity gives the complete integral. Again $k \le 1/b$ is bounded for fixed b. Combining the lower comparison with the upper sieve bound proves (S8d).

Now put $b = 1/(40T)$, with $T \ge 25$ chosen so large that

$$40e^{-\gamma_E}(C_s + C_{\text{sieve}})Te^{-T} \le 1/10.$$

Such a T exists because $Te^{-T} \to 0$. Then $0 < b \le 1/1000$, and (S8d) gives

$$A_b - (D_b(1) - 1 + 1/10) \ge \frac{9}{10} - \frac{e^{-\gamma_E}}{b}(C_s + C_{\text{sieve}})e^{-1/(40b)} \ge \frac{4}{5}.$$

Subtracting (S8) from (S7) leaves at least $(4/5 - 1/10 - 1/10)\mathfrak{S}X_{-1}/L$, which is at least $(\mathfrak{S}/2)X_{-1}/L$.

Every composite surviving that subtraction has least prime factor greater than $x^{1/2-\kappa}$. The finite partition theorem accounts for the strict lower and closed upper endpoints of every bin. Hence

$$\mathcal{M}_{-1} + \mathcal{C}_{-1}(L, \kappa) \ge \frac{\mathfrak{S}}{2}\frac{X_{-1}}{L}. \tag{S9}$$

12.5 Balanced composites

For sufficiently large x, a composite $u + 2 \le 2x + 2$ whose least prime factor exceeds $x^{1/2-\kappa}$ has exactly two prime factors, counted with multiplicity: three factors would give $u + 2 > x^{3(1/2-\kappa)} > 2x + 2$. Both factors lie in $\mathcal{B}_\kappa$. The argument includes a square of a prime. Let $I(m) = 1_{m \text{ prime}}$, and let $B(m)$ be its full-cell approximation. The complete-cell prime estimate gives $0 \le B(m) \le 3/(LV(W))$ on the balanced range, with rough support. Therefore (S6) bounds the pair sum with two approximations by

$$9C_1(\kappa + 1/L)\frac{X_\rho}{L}.$$

For fixed $\kappa > 0$, eventually $1/L \le \kappa$. Thus the constant $C_{\text{bal}} = 18C_1$ works for the approximated sum, independently of κ and of the cell precision.

There are exactly two replacement steps:

$$I(m)I(n) - B(m)B(n) = (I(m) - B(m))I(n) + B(m)(I(n) - B(n)).$$

The companion in the second step is bounded but need not be smooth. Their total is bounded by $\mathcal{E}(L) = o(X_\rho/L)$ from (C9)–(C12).

It follows that for every fixed $0 < \kappa < 1/50$ and $\varepsilon > 0$, eventually for both signs,

$$\mathcal{C}_\rho(L, \kappa) \le (C_{\text{bal}}\kappa + \varepsilon)\frac{X_\rho}{L}. \tag{S10}$$

The bound holds first for the ordered prime-pair sum and hence for the composite mass it covers. For (S10), the auxiliary partition may be fixed to one admissible bin with lower endpoint $1/200$. Equations (S9) and (S10) are the two estimates used in the main proof.

A Reading the Lean proof

The public theorem is `Green64.green_64` in `Main.lean`. Its statement is the direct proposition

```
{p : ℕ | p.Prime ∧ Odd (Ω (p - 2))}.Infinite
```

The proof first constructs a prime above every natural bound. `Green64.primes_unbounded` records positivity of $p - 2$ as well as its odd prime-factor count. `scripts/StatementTests.lean` checks both exact propositions, including the multiplicity convention, and `scripts/AxiomAudit.lean` prints their dependency axioms. Mathlib's `Ω` is the length of `Nat.primeFactorsList`, so it counts prime factors with multiplicity.

The project separates a short final argument from the supporting analytic library. Module imports form an acyclic graph. Supporting declarations remain source code in `Green64/Library`; they are not external packages or precompiled proof objects. The final argument uses the following modules.

Mathematical step	Module	Main declaration
Infinitely many primes	`Main`	`Green64.green_64`
Witness above any bound	`Infinitude`	`prime_witnesses_unbounded`
Positive mass gives a prime of the required parity	`PrimeExtraction`	`negative_parity_prime_candidates`
A fixed smooth cutoff	`Localization`	`localizingBump`
Subtract balanced composites	`Sieve/PrimeMass`	`negative_prime_mass_lower_bound`
Cover the composite remainder by prime pairs	`Sieve/CompositeMass`	`balanced_composite_mass_bound`
Transfer the proxy bound to sharp prime indicators	`Sieve/BalancedMass`	`balanced_prime_pair_and_composite_bound`
Bound the two prime proxies	`Sieve/BalancedApproximation`	`balanced_proxy_bound_uniform`
Sum positive bounds over balanced rectangles	`Sieve/BalancedBoxes`	`balanced_rough_pair_bound`
Uniform weighted rectangle bound	`Sieve/WeightedBoxes`	`weighted_box_bound`
Linear count of compatible dyadic rectangles	`Sieve/DyadicCounting`	`compatible_dyad_labels_card`
Exact finite partition	`Sieve/DyadicPartition`	the active-rectangle partition identities
Support and rectangle geometry	`Sieve/DyadicGeometry`	the bounds for attained dyadic scales
Normalize the finite main term	`Sieve/BoxNormalization`	`box_main_Xrho_eventually`
Bound the finite remainder	`Sieve/BoxRemainder`	`box_finite_error_exp_bound`
Make the remainder negligible	`Sieve/BoxError`	`box_error_uniform_little_o`
Euler-product estimates	`Sieve/EulerProduct`	`box_euler_bounds`
Account for both replacements	`Sieve/ApproximationError`	`balanced_sharp_le_proxy_error`

Mathematical step	Module	Main declaration
Close the analytic parameter choices	Sieve/Parameters	proxy_parameters_exist

Unless a row says otherwise, the public analytic declarations are in namespace Green64Analytic.Parity. Lower-level declarations retain their specific namespaces. Search a declaration with rg if using an editor without Lean's go-to-definition support.

A.1 Correspondence with the mathematical text

The following map records the exact source results used in the exposition. The probability notation in the weight chapter rewrites normalized sums; it does not introduce a probabilistic assumption.

Text	Lean declarations	Source
Ordinary bands and moving depth	sourceOriginalQ0Scale, sourceOriginalQActualBands, sourceOriginalLiteralDepth	Correlation/Part065, Weights/Part097
Every admissible marked band	sourceOriginalPGroupIndex, source_original_P_all_groups_mem_iff, sourceOriginalPAllActualBands	group geometry, Weights/Part019, Correlation/Part032
Factorial coefficient and marked weight	sourceOriginalOrdinaryCoefficient, sourceOriginalW1, sourceOriginalA0	TrueCellPrimeFibre
Parity projection	parityWeight	Foundations/Part044
Exact multiplicity sign	b628_parity_liouville_equals_P_phase434, b628_parity_negative_candidate_exact_target434	ParityLiteralParityTarget
Ordinary harmonic lower bounds	b635_actual_moving_Q_twelfth_floor434, b635_parity_admissible_band_count434, b635_parity_band_factor_numeric434, b635_parity_actual_HQ_lower_220434	Weights/Part102
Ordinary harmonic upper bound	source_original_Q_actual_band_harmonic_le_four	Weights/Part105
Marked harmonic bounds	source_original_P_all_actual_reciprocal_le_four434, source_original_P_all_reciprocal_floor_eventually	Correlation/Part032, Weights/Part107
Euler normalization and Rankin tail	source_original_P_group_harmonic_tsum_eq_Hg, source_original_P_actual_polylog_rankin_tail	Weights/Part103

Text	Lean declarations	Source
Product law and both parity probabilities	source_original_global_actual_P_normalized_harmonic_law, parity_original_P_global_actual_uniform_positive_parity434	Weights/Part104
Exact localized weight identity and corridor	parity_original_XA_QP_literal_weights_exact_parity434, source_original_QP_actual_good_rest_corridor	Weights/Part106
Strong mass lower bound	parity_original_XA_actual_HQ_HP_floor_eventually_parity434, parity_original_XA_literal_depth_lower_eventually_parity434	Foundations/Part005
Shared $X_\rho \geq c_0 e^L L^{-221}$	localizedMass_lower_bound	Foundations/Part050
Full logarithmic cells and roughness proxy	sourceD20LogCell, sourceD20FullProxy, sourceD20FullDelta	Weights/Part012
Prime proxy and character estimate	b627PrimeFullProxy434, b627_actual_fine_rough_all_character_saving434	PrimeFineProxy
Norm of the second proxy	parity_original_actual_second_proxy_beta_half_norm_le_one434	Foundations/Part045
Actual signed source and common analytic precision	parity_eventually_original_signed_quarter_b629_fine_one_slot_Psi_Mellin_source_anylog_common_precision434	Foundations/Part050
Whole rectangles and both replacement orders	parity_eventually_original_both_parity_full_rectangles_proxy_steps434	Foundations/Part052
All finite error terms and normalization	requiredProxyError, allOrdersProxyError, parity_original_littleO_exp222_to_Xrho434	Foundations/Part053
Singular product	singularSeries_converges_and_ge_one	Foundations/Part006
Slot allocation, diagonal estimate, Fourier tails	Detailed source map in Section B.2	Foundations/Part050
Alternating resampling, physical moments, and comparison subtraction	Derivation in Section 5; source map in Section B.3	Weights/Part092, Foundations/Part008
Negative-parity congruence estimate	parity_original_negative_congruence_D230_bound434	Foundations/Part053
Character decomposition, deleted-slot convolution, and conductor estimates	Detailed source map in Section B.7	Weights/Part107, Part108

Text	Lean declarations	Source
Assignment masks, CRT, and the finite rectangle sieve	Detailed source map in Section B.9	Foundations/Part056, Part057
Rough-number density	sourceSimplexTerm, sourceDensity	Analysis/Estimates/Part003
Remove unbalanced composites	parity_original_exists_small_b_unbalanced_margin434, negative_prime_or_composite_mass_lower_bound	Foundations/Part055

A.2 The analytic library

Library/Weights develops labelled prime assignments, multiplicities, harmonic normalization, and the finite combinatorial operators used by the correlation argument. Library/Correlation contains their signed correlations and reindexing identities. Library/Sieve/Foundations connects those identities to the sieve and the finite factor sums. Library/Analysis, Library/OAI, Library/ChenTheorem, and the other retained source directories provide the analytic and arithmetic estimates used in those proofs. See THIRD_PARTY.md for their origins.

The bilinear chapter follows the exact Cauchy square, the diagonal divisor bound, and the integration over the Fourier ball and its tail. The marked-correlation chapter supplies the cancellation used inside that ball, including the separately bounded comparison term. The final analytic parameter construction discharges the hypotheses of these lower-level estimates.

The PartNNN files contain supporting results; the named files in Green64/Sieve expose the final argument.

A.3 Checking what is proved

A successful build alone is not the axiom audit. Run scripts/verify.py and inspect its VALIDATION.json receipt for the public theorem names, axioms, source hashes, dependency revisions, and command exit codes. The audit checks the infinite set proposition itself, together with the stronger unbounded-witness theorem. It does not substitute a conditional statement or a wrapper with an assumed analytic conclusion.

B Further source correspondence

B.1 Replacing prime and roughness tests

Text	Definitions and proved identities
(C1)–(C3), complete-cell discrepancies	`Weights/Part012`: `sourceD20Rough`, `sourceD20LogCell`, `sourceD20FullProxy`, `sourceD20FullDelta`, `sourceD20_full_delta_zero`; `Estimates/Part005`: `sourceD20_principal_interval_bound`, `sourceD20_mesh_scalar`, `sourceD21_full_nonprincipal_saving`; `PrimeFineProxy`: `b627PrimeFullProxy434`, `b627_actual_fine_rough_all_character_saving434`, `b627_prime_full_delta_eq_actual_rough434`
(C5)–(C6), common bilinear precision	`Foundations/Part050`: `parity_original_TypeII_target_K0_precision_compatible434`, `parity_eventually_original_signed_quarter_b629_fine_one_slot_Psi_Mellin_source_anylog_common_precision434`; the exponents are defined in `Foundations/Part013`: `parityOriginalW1PolylogExponent434`, `Part011`: `parityOriginalSelectedECoefficientPolylogExponent434`, and `Part010`: `parityOriginalB590FixedFourierPolynomialExponent434`
(C7)–(C8), two replacements and dyadic partition	`Foundations/Part045`: `parity_original_actual_fullcell_proxy_beta_norm_le_two434`, `parity_original_actual_second_proxy_beta_half_norm_le_one434`; `Foundations/Part050`: `parity_original_parity_positive_Icc_eq_quarter_nested_sources434`
(C9)–(C12), total error and normalization	`Foundations/Part053`: `requiredProxyError`, `allOrdersProxyError`, `parity_eventually_original_sieve_all_proxy_errors434`, `parity_original_littleO_exp222_to_Xrho434`; `Parameters`: `proxy_parameters_exist`

B.2 The bilinear reduction

Argument	Lean source
Common-precision rectangle statement	Foundations/Part050: parity_eventually_original_signed_quarter_b629_fine_one_slot_Psi_Mellin_source_anylog_common_precision434
Smooth allocation functions	Weights/Part003: sourceBufferedGateReal
Slot ordering and frequency scales	Correlation/Part065: sourceOrderedLabels, sourceActualSortedFrequencyScales
Prefix bounds, partition, and the exponent 88	Weights/Part117
Fourier phases and endpoint sums	Weights/Part020, Foundations/Part009, Part011
Intrinsic H and a_v	Foundations/Part017: parityOriginalIntrinsicH434, parityOriginalIntrinsicSignedAV434
Windowed a_e and source comparison	Foundations/Part018: parityOriginalHistoricalWindowAE434
Finite $\mathcal{B}'$ and divisor-moment diagonal	Foundations/Part032: parityOriginalHistoricalUngatedCauchyAtDeltaNoUCap434
Smooth square and exact off-diagonal	Foundations/Part022, Part025, Part028
Shift coordinates and residue mask	Foundations/Part019, Part029, Weights/Part005
Exact two-endpoint cofactor identity	Weights/Part102: b633_parity_full_physical_pair434
Physical correlation saving (B18)	Foundations/Part008: parity_eventually_actual_original_Parity417_physical_cofactor_saving_common_precision434
Actual Fourier moments (B19a) and the extra precision in (B19b)	Foundations/Part026: parity_original_historical_full_odd_3d_fourier_moment_uniform_polylog434; Part032: parity_original_historical_box_source_BH_uniform_polylog434; Part037: parity_original_actual_capfree_Bprime_M0_uniform_polylog434, parity_original_actual_capfree_Bprime_primary_inverse_log_from_M0434
Fourier ball, tails, and common precision	Foundations/Part046, Part049, Part050

B.3 Cancellation in the marked correlation

The transfer formulas have the following more detailed correspondence. Here `k` in a local operator declaration means $s = 2k$ in (B14), `rho` in the memory declarations means ρ_0 (not parity), and `mu` in a physical-state declaration is the tilted law (T2).

Text and mathematical operation	Definitions and proved identities
(M0), arbitrary joint endpoint tests and permutation averaging	`Weights/Part090`: `sourceOriginalUnfilteredIdealEndpointMean434`; `Part091`: `source_original_unfiltered_ideal_ENDPOINT_PAIR_MEAN434`
(M0a), the phase and the asymmetric splice in (T7)	`Weights/Part024`: `originalEdgeScalar`; `Part091`: `sourceOriginalUnfilteredIdealHeadPhase434`, `source_original_unfiltered_ideal_raw_splice_phase434`
(M4a), product matching and the resampling reference	`Weights/Part091`: `source_original_unfiltered_ideal_raw_splice_phase434`; `Part092`: `source_original_unfiltered_ideal_RAW_FORM_RESIDUAL434` (the displayed matching kernel explains the finite product-law substitution in these identities)
(T1)–(T3), actual state spaces and measures	`Estimates/Part003`: `SmallState`, `stateMass`; `Weights/Part002`: `FockState`, `fockMass`, `BigState`, `bigStateMass`; `Correlation/Part006`: `physicalSourceBaseline`, `physicalSourceMuTilde`
(T4), quotient map, mass and norm	`Correlation/Part005`: `physicalFockProjection`, `physicalOrbitMass`; `Part006`: `fullPhysicalOrbitAverage`; `Weights/Part093`: `sourceOriginalUnfilteredIdealOrbitEnergy434`
(T5), legal transitions, fresh births and the two OLD gates	`Weights/Part002`: `PhysicalEdgeLegal`, `physicalBigBranchCoefficient`; `Part024`: `sourceOrientedPhysicalCoefficient`; `Part093`: `sourceOriginalUnfilteredEdgeLiteralKeep434`, `sourceOriginalUnfilteredEdgeSeenErasedPhase434`
(T6)–(T7), frozen form and raw/comparison/ban identity	`B26FrozenCleanPhysicalFourier`: `sourceOriginalB26FrozenFreshLaw434`, `sourceOriginalB26FrozenBigWeight434`; `Weights/Part092`: `source_original_unfiltered_ideal_FULL_FROZEN_PAIR434`, `eventually_source_original_unfiltered_ideal_FULL_UNIT434`

Text and mathematical operation	Definitions and proved identities
(T8)–(T12), common density, fresh collisions and exact three-term identity	Weights/Part090: source_original_unfiltered_ideal_COMMON_ENERGY434, source_original_unfiltered_ideal_PHYSICAL_COMMON_DENSITY434; Part092: source_original_physical_FROZEN_BAD_FAMILY434; Part094: source_original_physical_EXACT_THREE_TERM434, eventually_source_original_physical_FULL_TILTED_NODUP_BIG434
(T13)–(T15), SMALL fibres and Fourier transform of the actual translation	B26CleanSmallCoherentPairing: sourceOriginalB26SmallLift434, source_original_B26_MOVING_SMALL_CLEAN_FRAME_PAIR434, source_original_B26_ORIGINAL_CLEAN_SMALL_FOURIER434
(T16)–(T17), both endpoint energies and orbit transfer	Estimates/Part013: source_original_B26_SMALL_OLD_TRANSFER434; Weights/Part091: source_original_unfiltered_ideal_LITERAL_SECTOR_TRANSFER434; Correlation/Part063: eventually_source_original_physical_FULL_SEEN_ERASED_CLEAN434
(M11)–(M13), exceptional transitions; (T18)–(T19), full local estimate	Correlation/Part062: source_original_edge_LITERAL_DIRTY_FINITE434; Weights/Part094: eventually_source_original_edge_SOURCE_PHASED_BOUNDS434; Correlation/Part065: eventually_source_original_edge_FULL_NEAR_CONTRACTION434, eventually_source_original_root_SOURCE_LOCAL_BANKS434
(T20), background births, signs and factorials	Weights/Part002: sourceGhostTerm; Correlation/Part005: physicalVisitGKernel; Weights/Part094: source_original_evolution_FINAL_G_ENERGY434
(T21)–(T23), all moment constants	Correlation/Part005: memorySmallRowConstant, memoryPromotionDampingConstant; Part062: sourceOriginalRootPinnedRowBudget434; Weights/Part072: sourceOriginalActualSourceSignedInitialSmallMass434; Part093: sourceOriginalUnfilteredEdgeExponent434; Part094: source_original_global_root_PREFACTOR434

Text and mathematical operation	Definitions and proved identities
(T24)–(T25), prime lifespans and the exact residue model	Correlation/Part009: memoryAbsentTerm, memoryRepresentedTerm; Part010: memory_frozen_prime_all_lifespans, memory_frozen_prime_supported_normal_form; Part025: memory_source_root_actual_prime_product_exact; Part027: memory_source_original_physical_root_eq_frozen_Haar
(M16a)–(M19), same-integer densities and signed transfer	WeightedActualPaths: b645ActualMarkedPathModel434, b645_eventually_actual_weighted_paths_CRT434; Weights/Part073: b645_eventually_actual_parity_physical_root_transfer434, b654_actual_parity_moment_eq_return434; Correlation/Part042: b650_physical_root_moment_array434; Weights/Part048: b654ParityIntervalWindow434, b654ParitySeedPositions434
(M20), spectral interpolation and position blocks	Weights/Part005: b652_mark_independent_position_endpoint434
(M20a), the full path window and fixed endpoint bounds	ParityStepCap: b651ParityDConstant434, b651ParityMaxStep434; ParityLocalizationWindow: b651ParityWindowSize434; Weights/Part109: b660_eventually_actual_parity_dyadic_saving434 (uses norm exponent 9 and supremum exponent 5)
(M23a)–(M23b), rank budgets and the coupled auxiliary sum	Correlation/Part072: parity_canonical_rank_card434, parity_eventually_actual_rank_family_budget434; Weights/Part112: parity_actual_rank_aux_coupled_reindex434; Weights/Part111: parity_fixed_rank_cutoff_pair_budget434
(M23c)–(M23f), exact endpoints, shared restoration and the free-auxiliary error	Weights/Part046: parityFirstRankMark434, paritySecondRankMark434; Correlation/Part072: parityActualRankEndpoint434, parity_eventually_comparison_weighted_restored434; WeightedStrippedRank: parity_actual_weighted_stripped_rank_difference434; Weights/Part113: parity_eventually_actual_free_collision_pair_bound434; FreeAuxCollisionJoin: parity_actual_stripped_family_rank_main_difference434

Text and mathematical operation	Definitions and proved identities
(M23g)–(M23i), Mellin-circle pairing with the full auxiliary product	RankFourier: parity_actual_rank_pair_fourier434; Estimates/Part016: parity_actual_weighted_cell_rank_main_mellin434, parity_actual_cell_rank_family_circle434, parity_actual_cell_coupled_kernel_norm434

The chapter follows the finite constructions and their estimates in the following files. These references identify the proof being explained; the mathematical argument above uses the displayed identities and inequalities.

Step	Included Lean source
Hoeffding components and (M1)	Weights/Part017, source_original_hoeffding_physical_all_degree_saving434
Actual BIG block sizes	Weights/Part100
Alternating identity and its squared bound	Weights/Part092, source_original_unfiltered_ideal_RAW_FORM_RESIDUAL434
Character Gram matrices	Correlation/Part037
Global kernel norm	ActualGlobalKernelL2
Rational phase, product fibres, and minor arcs	Correlation/Part039, Weights/Part047, Part053
Arithmetic mesh choices	Correlation/Part041
SMALL omission uniqueness	Correlation/Part043
OLD, density, and collision corrections	Correlation/Part062, Weights/Part094
Forest rank split and restoration	GlobalRootScalar, Correlation/Part065
Removal of the OLD cutoff	Weights/Part064, Part090, Correlation/Part050, Part052
Bonferroni–CRT interval estimate	Correlation/Part035
Signed path transfer	WeightedActualPaths, Foundations/Part005
Physical states, matrix entries, and returning paths	Correlation/Part021, Part041, Part042, Weights/Part073
Spectral endpoint estimate	Analysis/Estimates/Part015, b655_eventually_actual_parity_endpoint_saving434
Literal core and ordered-mark weights	Weights/Part037, Part038, Part044
Marked arc estimate (M25)–(M26)	Weights/Part045, eventually_source_original_arc_conclusion434
Comparison main term and its subtraction	Foundations/Part007, Part008
Cap and shared-divisor errors	Weights/Part111, Part115

B.4 From one mark to the physical matrix

The raw slice is `parityRawPhysicalSlice434` in `Correlation/Part072`. The clean identity and (L4) are `parity_raw_clean_slice_exact434` and `parity_raw_shared_restoration_bound434` in `RawSharedRestoration`. The carrier reindexing is in `RawSharedCarrier`, and the arithmetic cancellation and original-source identification are in `Weights/Part115`. The endpoint vectors, removed-product fourth moment, and exact Fourier pairing are in `Weights/Part109`.

B.5 The forest expansion of the history sum

The list identity is `source_original_root_LIST_FOREST_EXPANSION434` in `Weights/Part093`. The actual linear functional, exact forest sum, sign-field average, owner count and energy iteration are in `Correlation/Part064`, especially `sourceOriginalRootFunctionalRoot434`, `source_original_root_ORIGINAL_FOREST_SUM434` and `source_original_root_FOREST_FINITE_BOUND434`. The pinned row estimate is `source_original_forest_E_PINNED_FULL_ROW434` in `Weights/Part090`. The full-rank suffix induction and the initial distinctness case are in `Correlation/Part064`; the resulting bound (F11) is `eventually_source_original_global_root_SOURCE_HIGH434` in `Correlation/Part065`.

B.6 From complete-cell discrepancy to additive arc energy

The local-energy transfer is `source_linear_to_log_arc434` in `Weights/Part012`, using `source_physical_collar_replacement_error434` in `Analysis/PrimeNumberTheorem/Part003`. The weighted Schur estimate is `source_actual_physical_normalized_L2_schur434` in `Weights/Part023`. The arithmetic identities and (E13) are in `Analysis/Estimates/Part009`, especially `source_original_global_physical_mellin_exact434` and `eventually_source_original_global_physical_from_literal_cells434`. The direct low-frequency estimate is in `SourceLowFrequencyFourFactorIntegral`. The three-region argument and retained Fourier frequencies are in `Weights/Part040`. The exceptional residual estimates are in `Weights/Part035` and `Weights/Part039`. The long-prime statement (U1) and its shift are `sourceCorrelationLongPrime_shifted_U1` in `Correlation/Part002`. The contour bound (U4), its closure (U3)–(U5), and the matching zero-free interface are `p1p2_canonical_source_prime_bound`, `sourceLongPrimeU1_of_sourceKhaleZeroFree`, and `KhalePublished11` in `Estimates/Part003`. The unconditional declaration there, `sourceLongPrimeU1_unconditional`, obtains (U2) from the included stronger theorem `OAI.DirichletCharacter.LFunction_ne_zero_of_seven_eighths_lt_re` in `Detector/Part016`. The text invokes the weaker, independently published statement (U2); no stronger zero-free conclusion is needed for the contour proof. The use of (U1) on occupied intervals is in `Analysis/Estimates/Part006`. The final four-term estimate is `eventually_source_original_full_additive434` in `Correlation/Part033`.

B.7 Distribution in residue classes

Argument	Lean source and declaration
Orthogonality with the nonunit correction	`Weights/Part108`: `source_unit_class_remainder_identity`, `source_shift_two_genuine_remainder_bound`
Squarefree totient sum	`Weights/Part107`: `source_squarefree_totient_sum_harmonic`
Rough-support correction	`Weights/Part108`: `shift_two_nonunit_reciprocal_harmonic`, `shift_two_prime_reciprocal_rough_bound`
Deleted-slot convolution and both masks	`Foundations/Part005`: `source_original_moving_deleted_prime_convolution`, `source_original_conductor_masked_convolution`
Small conductors and final parameter choice	`Weights/Part108`: `source_original_SW_finite_Echi_eventually`, `source_shift_two_congruence_input_unconditional`
Large-conductor bilinear estimate	`Weights/Part107`, with its dyadic expansion in `Part108`
Prime Siegel–Walfisz interface	`Analysis/Estimates/Part005`: `sourcePrimeSiegelWalfisz_unconditional`
Multiplicative large sieve	`Foundation/LargeSieve/Part001`
Specialization used in the final argument	`Foundations/Part053`: `parity_original_negative_congruence_D230_bound434`

B.8 The quantitative finite block sieve

The statement and proof are `weighted_block_sieve` in `Foundations/Part001`. The shifted constant and depth estimates are in `Foundations/Part053` and `Part054`. The ordinary-rough constant is `sieveC` in `Analysis/Estimates/Part003`.

B.9 A positive estimate on factor rectangles

Argument	Lean source and declaration
Assignment-mask majorant	`Foundations/Part056`: `parity_original_A0_assignment_mask_majorant434`
Assignment moments, weighted-fibre reduction, sieve application	`Foundations/Part056`
CRT density and boundary count	`Foundations/Part002`, `Weights/Part035`
Finite Brun–Hooley block sieve	`Foundations/Part001`
Euler-product comparison and mask average	`Foundations/Part057`
Complete finite bound (R1)	`Foundations/Part057`: `parity_original_literal_weighted_box_normalized434`
Error exponents and uniform normalization	`BoxRemainder`, `BoxError`, `BoxNormalization`

Argument	Lean source and declaration
Uniform final statement	`WeightedBoxes`: `weighted_box_bound_uniform`

B.10 Sieve estimates and the remaining composites

Text	Definitions and proved identities
Singular product and (S1)	`Foundations/Part006`: `singularSeries_converges_and_ge_one`; the congruence source map is in Section B.7
(S6), balanced rectangle count	`BalancedBoxes`: `balanced_rough_pair_bound`
(S7)–(S8d), rough density and the full-cell estimate	`Estimates/Part003`: `sourceDensity` (the displayed index is $k = n + 2$); `Weights/Part037`: `sourceD29_original_full_proxy_pointwise_compact`, `literal_density_rough_unconditional`, `literal_small_b_exp_bound_unconditional`
Conditional Euler limit and (S9)	`Foundations/Part055`: `parity_original_negative_conditional_natural_euler_limit434`, `negative_prime_or_composite_mass_lower_bound`
(S10), composite cover and two proxy errors	`Foundations/Part052`: `parity_eventually_original_balanced_composite_prime_pair434`; `ApproximationError`: `balancedTwoStepError`; `BalancedMass`

The one-bin specialization for (S10) is in `CompositeMass`.

References

[1] Tanmay Khale, *An explicit Vinogradov–Korobov zero-free region for Dirichlet L-functions*, Theorem 1.1, https://arxiv.org/abs/2210.06457. Published in *The Quarterly Journal of Mathematics* 75 (2024), no. 1, https://doi.org/10.1093/qmath/haae010.

[2] Ben Green, *100 Open Problems*, December 2025 revision, Problem 64, p. 31. https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf.

[3] Jizhou Guo, *Shifted primes with a prescribed parity of the number of prime factors*, October 2026. Retained TeX source from https://github.com/aster2024/openai-math-followups, revision 563cece082b628dfe3ba0290c9e4e30b92ffeca6, shifted-prime-parity/source/main.tex. The source identity and availability are recorded in the accompanying ACKNOWLEDGEMENTS.md.

[4] OpenAI, *Prime Predecessors with an Even Number of Prime Factors*, manuscript dated 17 September 2026, https://github.com/openai/math, revision adc7f1241b42e322a6451854ab7e4b4c146bf78a.

[5] OpenAI, *Weighted dilation graphs, smooth shifted primes and totient fibers*, manuscript dated 24 September 2026, https://github.com/openai/math, revision adc7f1241b42e322a6451854ab7e4b4c146bf78a.